Financial Statement Simulator

End-of-Internship Report

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Contents

1 Introduction

Financial statement analysis is the process of evaluating a firm's financial statements, including the balance sheet, income statement, and cash flow statement, to gain insights into its financial health, performance, and future prospects. It serves as a crucial tool for managers, investors, creditors, and other stakeholders, enabling them to make informed decisions. Through careful analysis, stakeholders can identify strengths and weaknesses in a company's financial position, evaluate its capacity to generate profits and cash flow, and assess the implications of alternative strategies.

The breadth of its applications reflects this importance:

- **Firm managers:** evaluate performance and guide strategic resource allocation
- **Investors:** assess financial health to inform buy, hold, or sell decisions
- **Creditors:** evaluate creditworthiness for lending decisions
- **Regulators:** monitor compliance and identify systemic risks
- **Banks:** assess financial position for credit and transaction decisions

Despite its importance, financial statement analysis can be a complex and time-consuming undertaking. It demands a deep understanding of accounting principles, financial ratios, and industry dynamics, along with the ability to interpret large volumes of financial data.

A key source of this complexity lies in the structure of the statements themselves. The three core financial statements are tightly interconnected through double-entry accounting: every transaction touches at least two accounts, and a change in one assumption can cascade through dozens of line items in non-obvious ways. For instance, a revision to the depreciation schedule simultaneously affects operating expenses on the income statement, the carrying value of fixed assets on the balance sheet, and the depreciation add-back in operating cash flow. Understanding any single assumption in isolation is therefore insufficient; the analyst must trace its consequences across all three statements simultaneously.

This interconnectedness raises two key challenges:

- **Counterfactual reasoning:** Questions such as "what if capex accelerates?" or "what if payout policy changes?" require not just a forecast, but a coherent simulation respecting all accounting identities across the three statements.
- **Ad-hoc plug entries:** Traditional spreadsheet approaches force balance-sheet closure using arbitrary "plug" entries that absorb imbalances into residual accounts, sacrificing transparency and making it difficult to audit assumptions or attribute outcomes.

These limitations point to the need for a more systematic and principled approach. This project presents an IFRS-grounded financial statement simulator that treats the accounting system as a stock-and-flow dynamic system, where balance-sheet stocks evolve through income-statement and cash-flow transactions under double-entry constraints. Macroeconomic conditions enter as exogenous inputs, while the firm's own decisions on investment, financing, working-capital management, and payout are modeled as explicit policies whose consequences propagate through the full set of statements. Concretely, the simulator is designed to deliver the following capabilities:

- **Complete statements:** generate internally consistent IAS 1 financial statements by construction
- **Counterfactual analysis:** enable scenario analysis by modifying policies and observing multi-period propagation
- **Stochastic outcomes:** produce outcome distributions separating deterministic mechanics from stochastic drivers
- **Auditability:** ensure transparency through reconciliation checks, execution ordering, and localized error attribution

The remainder of this report is organized as follows. Section 2 covers accounting foundations and financial statement structure. Section 3 reviews prior work and related research. Section 4 presents our modeling framework and methodology. Section 8 details the simulator's architecture and implementation. Finally, Section 9 outlines directions for future work and the potential impact of the project.

Note: The in-depth accounting rules and the first prototype implementation of the work are listed in Hongxiao's report.

2 Foundations

2.1 Accounting

2.2 Financial Statements

A financial statement is a formal record of the financial activities and position of a business, person, or other entity. The three main financial statements are:

- **Balance Sheet**: Shows the company's assets, liabilities, and equity at a specific point in time.
- **Income Statement**: Reports the company's financial performance over a period, including revenues, expenses, and profits.
- **Cash Flow Statement**: Provides a summary of cash inflows and outflows over a period, highlighting operating, investing, and financing activities.

These three statements are interconnected, providing a comprehensive view of a company's financial health.

Table 1: Example Balance Sheet Structure

Assets	**Liabilities and Equity**
Current Assets	*Current Liabilities*
Cash and Cash Equivalents	Accounts Payable
Accounts Receivable	Short-term Debt
Inventory	Accrued Expenses
Prepaid Expenses	
Total Current Assets	**Total Current Liabilities**
Non-current Assets	*Non-current Liabilities*
Property, Plant, and Equipment	Long-term Debt
Intangible Assets	Deferred Tax Liabilities
Investments	Pension Liabilities
Total Non-current Assets	**Total Non-current Liabilities**
	Equity
	Common Stock
	Retained Earnings
	Total Equity
Total Assets	**Total Liabilities + Total Equity**

Table 2: Example Income Statement Structure

Item	**Amount ($)**
Revenue	xxx
Cost of Goods Sold (COGS)	(xxx)
Gross Profit	**xxx**
Operating Expenses	
Selling, General & Administrative	xxx
Research & Development	xxx
Depreciation & Amortization	xxx
Total Operating Expenses	**(xxx)**
Operating Income (EBIT)	**xxx**
Non-operating Items	
Interest Income	xxx
Interest Expense	(xxx)
Other Income / (Expense)	xxx
Total Non-operating Items	**xxx**
Income Before Tax	**xxx**
Income Tax Expense	(xxx)
Net Income	**xxx**

2.2.1 Connection and Interdependence

The three financial statements are interconnected through the double-entry accounting system. For example, with indirect approach to cash flow statement, net income from the income statement is the starting point for the cash flow statement, and adjustments are made for non-cash expenses (like depreciation) and changes in working capital (which are reflected in the balance sheet). The ending cash balance from the cash flow statement must match the cash and cash equivalents line item on the balance sheet. Additionally, net income from the income statement contributes to retained earnings in the equity section of the balance sheet. This interconnectedness means that changes in one statement will have ripple effects across the others, making it essential to analyze them together for a comprehensive understanding of a company's financial health.

2.2.2 Accounting System

A standardized system for recording, measuring, and reporting a firm's economic activities. This is the rule book that defines

- **How** transaction are recognized

Table 3: Example Cash Flow Statement Structure

Item	**Amount ($)**
Cash Flows from Operating Activities	
Net Income	xxx
Depreciation & Amortization	xxx
Changes in Working Capital	xxx
Net Cash from Operations	**xxx**
Cash Flows from Investing Activities	
Capital Expenditures	(xxx)
Sale / Purchase of Investments	xxx
Net Cash from Investing	**(xxx)**
Cash Flows from Financing Activities	
Net Debt Issued / (Repaid)	xxx
Dividends Paid	(xxx)
Share Buybacks	(xxx)
Net Cash from Financing	**(xxx)**
Net Change in Cash	**xxx**
Beginning Cash Balance	xxx
Ending Cash Balance	**xxx**

- **When** they are recognized
- **Where** they are recognized (which statement)
- **What** value they are recognized at (e.g., historical cost, fair value)

There are two main accounting-standards regimes Different accounting standards can lead to

Table 4: Comparison of IFRS and US GAAP

	IFRS	**US GAAP**
Standard setter	IASB	FASB
Adoption	140+ jurisdictions worldwide	Primarily the United States
Philosophy	Principles-based	Rules-based

different financial statement presentations and measurements for the same underlying economic events. For instance, IFRS allows revaluation of certain assets to fair value, while US GAAP generally requires historical cost. This can affect the comparability of financial statements across firms and jurisdictions.

Although JPMorganChase is a US-based firm, we choose to implement an IFRS-grounded simulator for broader applicability and to avoid US GAAP's idiosyncratic complexities.

2.2.3 Double Entry System

The double entry system is a fundamental concept in accounting where every transaction affects at least two accounts, maintaining the accounting equation

$$Assets = Liabilities + Equity. \tag{1}$$

This system ensures that the balance sheet always balances, as every debit entry must have a corresponding credit entry. For example, if a company takes out a loan, it would debit cash (an asset) and credit a loan payable account (a liability), keeping the equation in balance.

2.3 Industry Classification

One of the key in financial statement analysis is interprecting financial ratio and metrics, like profit margin, return on assets, or debt-to-equity ratio. However, these ratios can vary significantly across industries due to differences in business models, capital intensity, and risk profiles. For instance, a high debt-to-equity ratio might be normal for a utility company but could indicate financial distress for a technology firm. Therefore, identifying the key data and structure of financial statements across different industries is crucial for accurate analysis and for our simulation model.

2.3.1 Classification System

Multiple industry classification systems are in use (see table below). Each has a distinct purpose, grouping logic, and geographic scope, which can influence how industries are analyzed and compared. Although the SASB Industry Classification System (SICS) is maintained by the IFRS Foundation, it groups industries by sustainability and ESG risk profiles, distinguishing it from traditional, financially oriented taxonomies such as GICS and ICB. For our analysis, we adopt Global Industry Classification Standard (GICS) as the primary taxonomy due to its broad acceptance in financial markets and investment analysis.

System	Primary Purpose	Key Factor for Grouping	Geographical Scope
SICS	Sustainability reporting	ESG risks & resource intensity	Global
GICS	Stock market analysis	Revenue & market perception	Global
ICB	Investment analysis	Audited account & Business activity	Global
NAICS	Economic statistics	Production processes	North America
SIC	Gov. reporting / legacy	Primary business activity	United States

Table 5: Comparison of Industry Classification Systems

The Global Industry Classification Standard (GICS)[1] is a widely used taxonomy developed by MSCI and S&P Dow Jones Indices. It categorizes companies into 11 sectors, 25 industry groups, 74 industries, and 163 sub-industries based on their primary business activities and revenue sources. GICS is designed to facilitate investment analysis and portfolio management by grouping companies with similar financial characteristics and market perceptions.

[1]https://www.msci.com/indexes/index-resources/gics (retrieved 11 Feb 2026)

	Sector name	Industries	Sub industries
1	Energy	2	7
2	Materials	5	17
3	Industrials	14	27
4	Consumer Discretionary	10	27
5	Consumer Staples	6	12
6	Health Care	6	10
7	Financials	6	18
8	Information Technology	6	12
9	Communication Services	5	10
10	Utilities	5	6
11	Real Estate	9	17

Table 6: GICS Sectors, Industries, and Sub-industries

2.3.2 Case Study

We selected some firms across different GICS sectors and analyzed their financial statements to identify key industry-specific characteristics. The table below summarizes our findings, highlighting unique line items, structural features, and common financial metrics that are particularly relevant for each sector. This analysis informs the design of our financial statement simulator, ensuring that it captures the nuances of different industries and allows for accurate scenario analysis. (See Table 7 for details.)

We also analyzed the financial ratios and metrics across different industries to identify key drivers and patterns. These insights are crucial for designing our simulator to accurately reflect industry-specific dynamics and allow for meaningful scenario analysis. (See Figures 1 to 5 for details.)

Table 7: Industry-specific IFRS financial statement characteristics

Sector	**Reference**	**Income statement**	**Balance sheet**	**Cash flow statement**
Energy	UK Oil and Gas PLC	Impairment charges salient	Property-intensive; limited financial assets	
Materials	James Cropper	Raw materials and energy cost lines	Impairment provision; translation reserve	Capex; intangible capex
Health Care	AstraZeneca	Collaboration revenue; R&D; distribution costs		
Financials	HSBC	No revenue/COGS; interest, fees, insurance; acquisition effects	No inventory; extensive financial assets/liabilities; equity includes called-up share capital and share premium	
Real Estate	British Land	Investment property; disposal gains/losses	Explicit fixed assets; debtors under assets; creditors under liabilities; revaluation reserve	

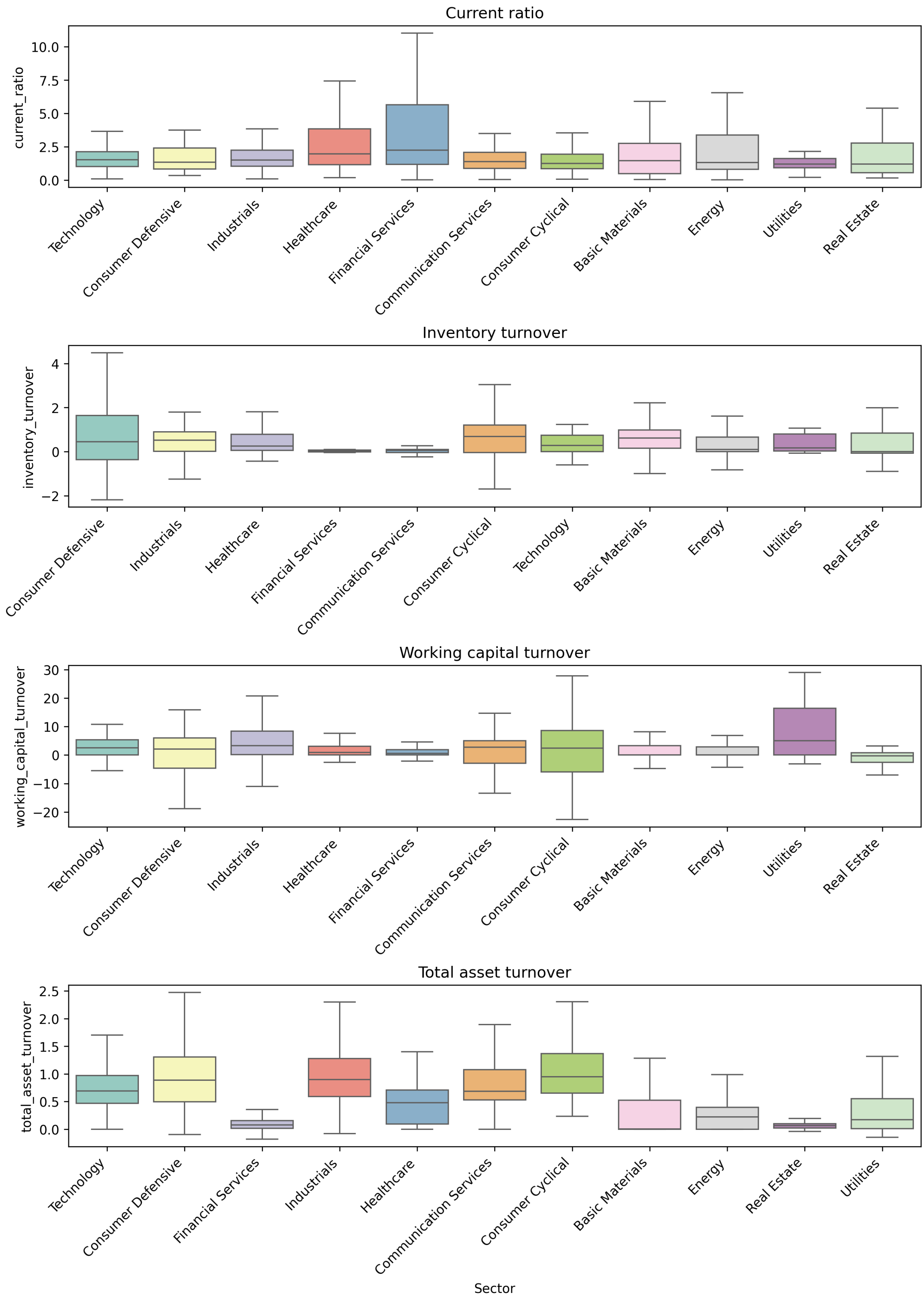


Figure 1: Financial Ratios Across Industries Part 1

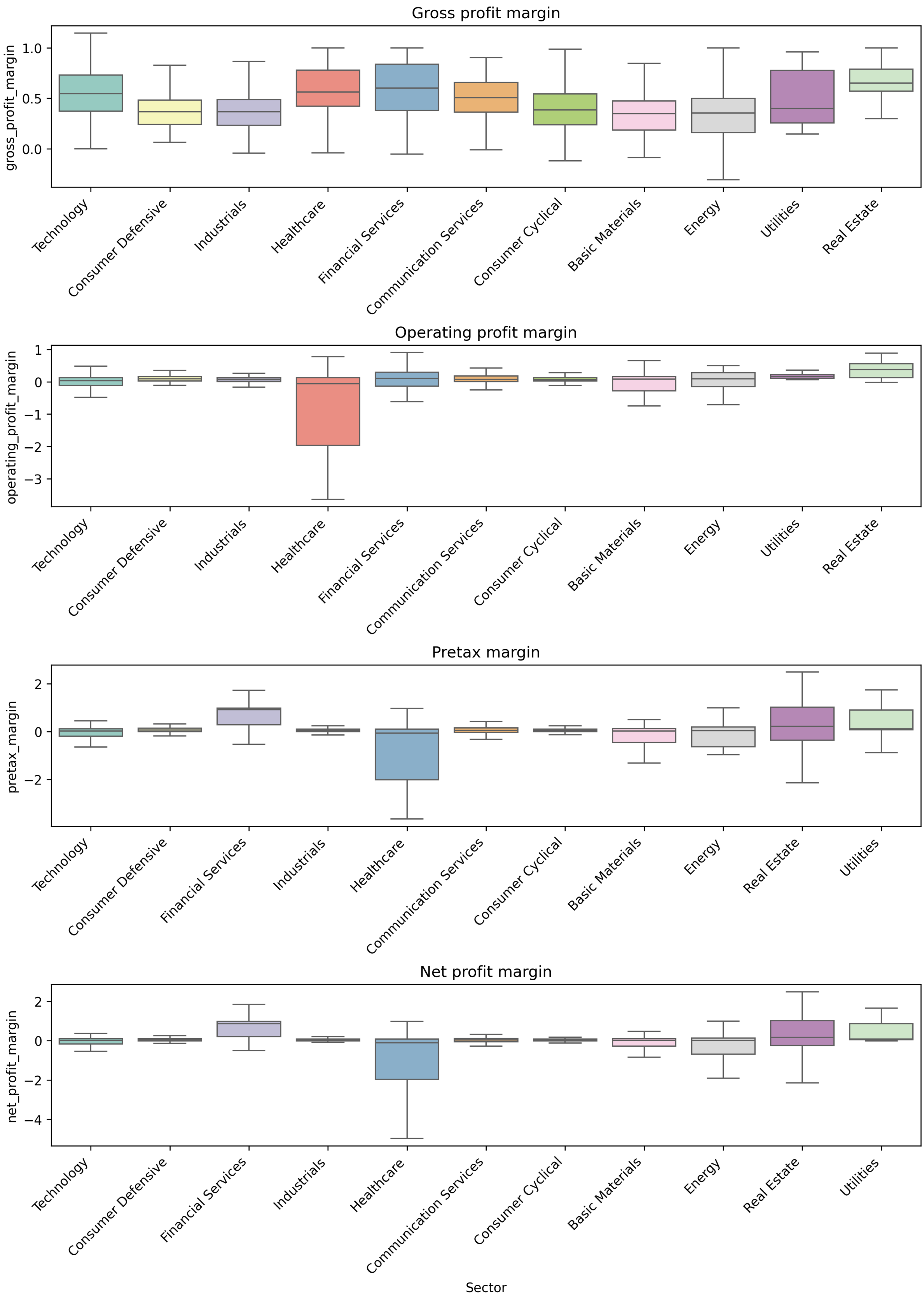


Figure 2: Financial Ratios Across Industries Part 2

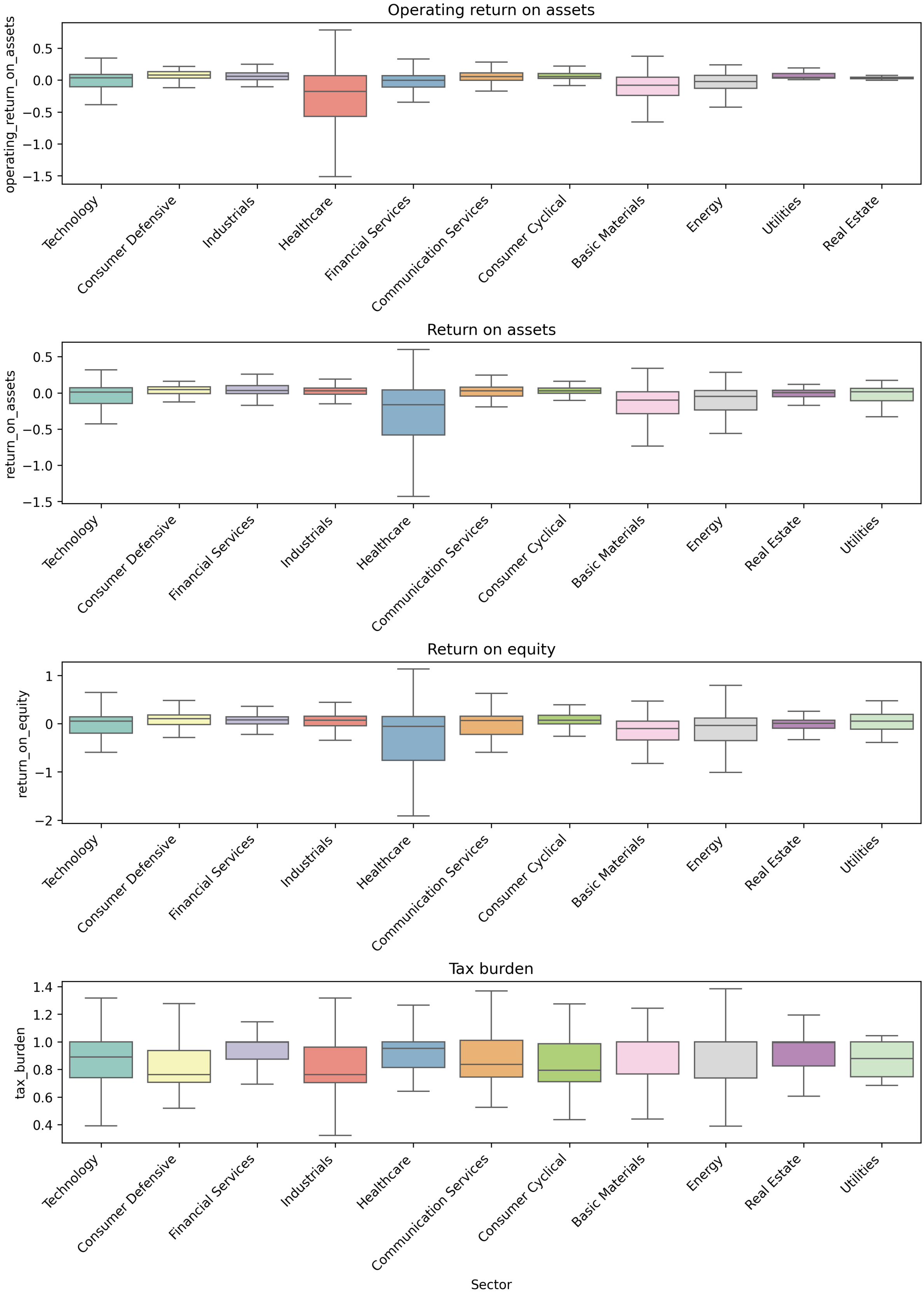


Figure 3: Financial Ratios Across Industries Part 3

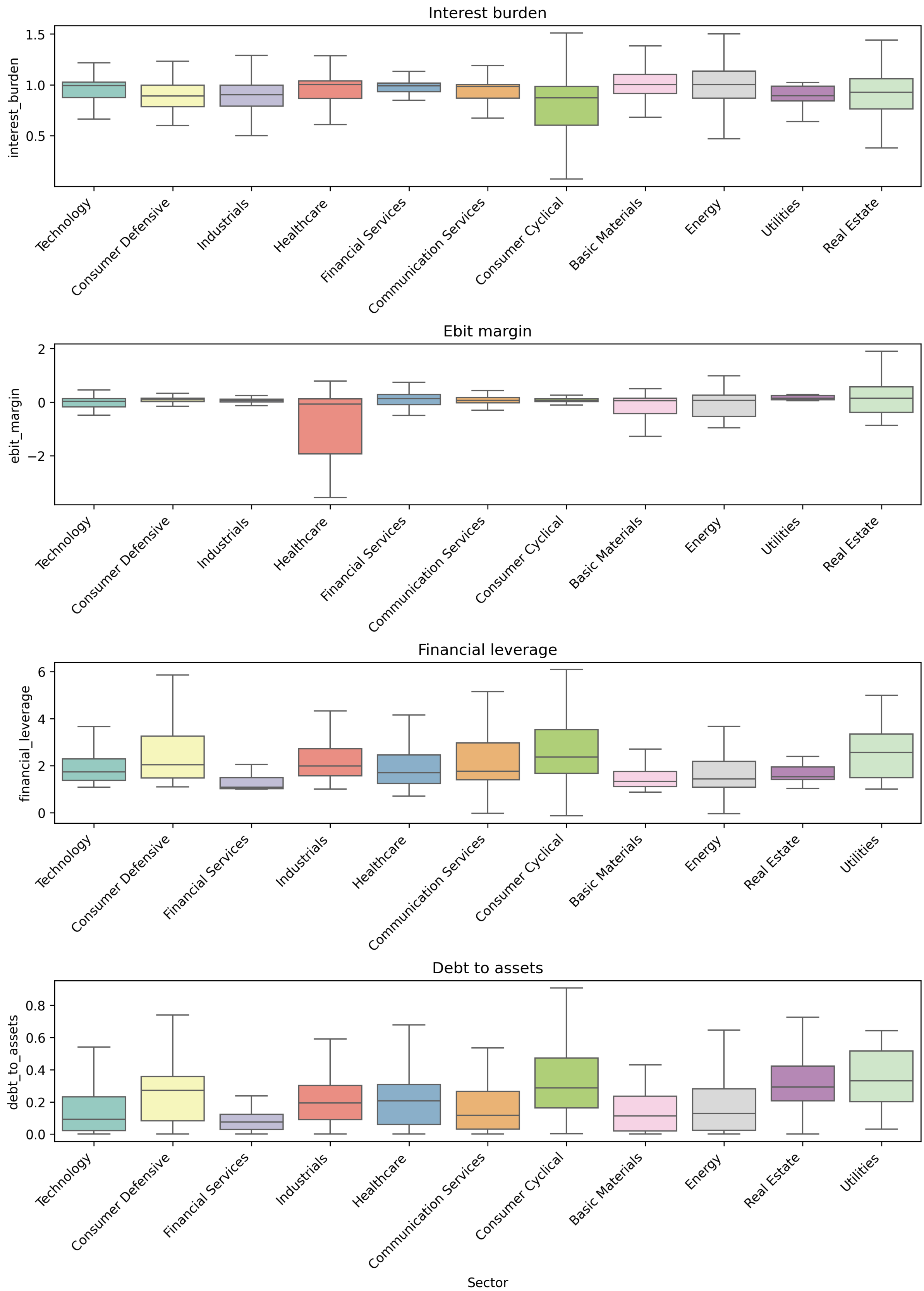


Figure 4: Financial Ratios Across Industries Part 4

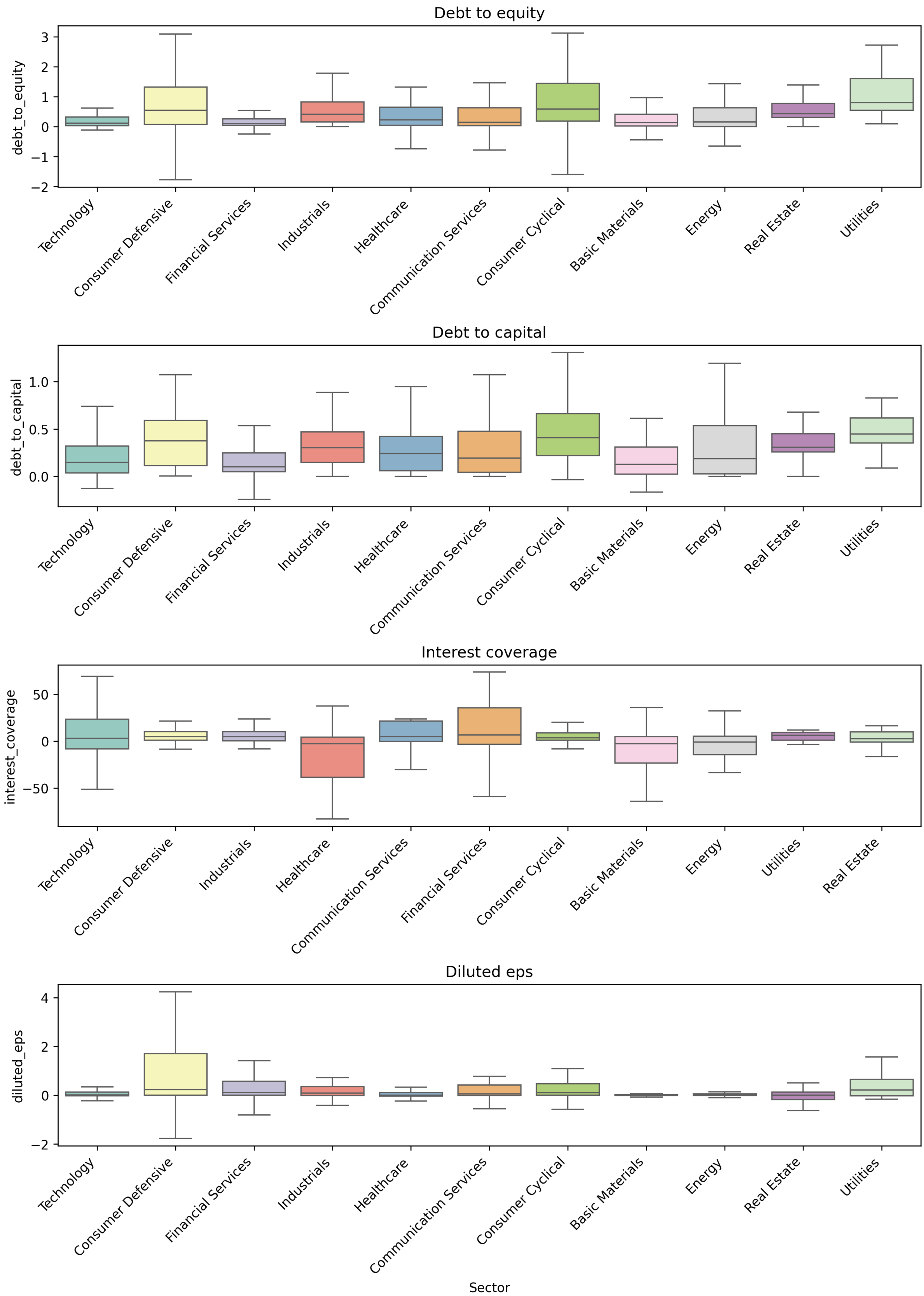


Figure 5: Financial Ratios Across Industries Part 5

3 Literature Review

3.1 Accounting Choices

3.1.1 Choice and Market Imperfection

The study of accounting choice addresses the fundamental question of whether, and why, accounting decisions matter. If markets were complete and perfect, where information is free and transparent, financial disclosures and regulations would be unnecessary. However, in a world characterized by market imperfections, accounting becomes a critical mechanism for addressing information gaps and facilitating contracts. For the purposes of this review, "accounting choice" is defined broadly. It encompasses any decision intended to influence the output of the accounting system, including the selection of accounting methods (e.g., LIFO vs. FIFO inventory valuation), the timing of adoption for new standards, and decisions regarding tax returns and regulatory filings. Notably, this definition extends to real decisions and operational actions taken primarily to affect accounting numbers, such as accelerating production to reduce per-unit fixed costs or cutting research and development (R&D) expenditures to inflate current earnings. The central premise is that rational managers engage in earnings management because they perceive market frictions that prevent users from completely unraveling these choices.

3.1.2 Rationale for Discretion

A rigid, rule-based accounting system might seem desirable to eliminate manipulation, but it is practically infeasible. A system that prescribes fixed rules for all scenarios, such as mandating a 10% allowance for uncollectibles regardless of actual credit risk, cannot accommodate the complexity of the economic environment or the emergence of new financial instruments. Consequently, accounting flexibility is necessary. This discretion is a double-edged sword, where it allows managers to signal private information about the firm's future prospects, but it also provides an opportunity for opportunistic behavior, such as inflating earnings to maximize personal compensation. The persistence of accounting choice suggests it may be part of an optimal solution to agency problems, mitigating the need for managers to make costly real economic decisions solely to meet rigid accounting targets.

3.1.3 Taxonomy

Research in the 1990s is generally classified into three categories based on the market imperfection driving the manager's decision: agency costs (contracting), information asymmetries (asset pricing), and externalities (influence on third parties).

Contractual Motivations

This category focuses on the contracts that define the firm, specifically the conflicts between shareholders and managers (internal agency) and between the firm and debtholders (external agency).

Executive Compensation

Managers often have incentives to manipulate earnings to maximize payouts from bonus plans. The "Healy Hypothesis" posits that managers will manage earnings upward when they are close to meeting bonus targets. Conversely, if earnings are so low that a bonus is unreachable, managers may engage in a "big bath," recognizing large write-offs in the current period to lower the bar for future years. However, later research has challenged this, suggesting that managers may smooth income even when performance is poor to demonstrate stability. A significant challenge in this literature is endogeneity; for instance, a reduction in R&D expenses prior to a CEO's departure may be a result of the firm's poor cash position rather than an opportunistic attempt by the CEO to inflate final earnings.

Debt Covenants

Lending agreements often include covenants based on accounting numbers, such as limits on leverage ratios. To minimize monitoring costs, these contracts often use "floating GAAP," meaning changes in accounting rules affect covenant calculations. Research investigates whether managers manipulate accounts to avoid technical default. While early studies relied on crude proxies like debt-to-equity ratios, 1990s research moved toward analyzing actual covenant slack. The evidence is mixed, while some firms approaching default shift to income-increasing methods, others cut dividends or reflect true financial distress without manipulation.

Asset Pricing Motivations

This category examines attempts by managers to influence stock prices by exploiting their superior knowledge of the firm.

Corporate Control and Valuation

Accounting choices are often heightened during major corporate events. For example, managers planning a Management Buyout (MBO) have been found to depress earnings via discretionary accruals to lower the stock price and purchase the company at a discount. Conversely, when a firm acquires another company using stock, managers may inflate earnings to boost their stock price and reduce the dilution of ownership. Similarly, managers may manage earnings to meet analyst forecasts to avoid the negative stock price reaction associated with missing targets.

Disclosure Policy

Managers also decide how much information to voluntarily disclose. While increased disclosure can theoretically lower the cost of capital by reducing investor uncertainty, it carries the risk of revealing proprietary information to competitors. For instance, firms in competitive industries may aggregate segment data to hide the profitability of specific product lines.

3.1.4 Externalities

The final category addresses accounting choices motivated by the desire to influence parties other than investors, such as tax authorities and regulators.

Taxation

Managers face a trade-off between reporting high earnings to shareholders and low earnings to tax authorities. This is evident in the LIFO (Last-In, First-Out) inventory decision; while LIFO saves taxes during inflation, it lowers reported earnings. The requirement for book-tax conformity often leads firms to forgo tax savings to maintain higher reported earnings. Furthermore, studies show that firms shift income across years or jurisdictions (transfer pricing) in response to changes in statutory tax rates.

Regulation and Political Costs

In regulated industries like banking, managers manage accruals to meet capital adequacy requirements imposed by the government. In other sectors, high profitability can attract unwanted political attention ("political costs"). For example, during the Persian Gulf crisis, oil companies managed earnings downward to avoid public backlash and potential windfall profit taxes. Similarly, firms seeking import relief protection may deflate earnings to appear more damaged by foreign competition.

3.1.5 Impediments to Progress in Research

Despite the volume of research, progress in understanding the economic implications of accounting choice has been modest. Several methodological hurdles persist:

- **Multiple Method Choices:** Managers often use a portfolio of accounting choices (e.g., depreciation, bad debts, and timing of expenses) to achieve a goal. Research that focuses on a single method in isolation fails to capture the aggregate effect.

- **Multiple Motivations:** Managers rarely face a single incentive. A decision to maximize earnings for a bonus might conflict with a desire to minimize taxes. Ignoring these trade-offs leads to correlated omitted variable problems.

- **Endogeneity:** It is difficult to distinguish whether an accounting choice causes an economic outcome or is merely a symptom of underlying firm characteristics (e.g., risky firms choosing both high leverage and aggressive accounting).

- **Lack of Theory:** There is currently no comprehensive theory that integrates these conflicting goals and choices, leaving researchers to rely on fragmented hypotheses.

3.1.6 Summary

The literature confirms that accounting choice is a pervasive feature of financial reporting, driven by complex interactions of contractual incentives, asset pricing goals, and regulatory pressures. However, the inability to consistently document the economic consequences, such as whether earnings management actually results in higher total compensation or sustained stock price increases, remains a gap. Future research must move beyond replicating existing studies in new settings. It requires broader research designs that account for multiple simultaneous motivations and the use of more sophisticated econometric techniques to address endogeneity.

3.2 Managerial Accounting

3.2.1 Foundational Concept

Managerial accounting provides financial and non-financial information to an organization's internal managers to aid in three core responsibilities of the management process:

- **Planning:** Setting strategic and operational goals. Managerial accounting provides the data to formulate these goals (e.g., cost data for budgets).
- **Directing:** Overseeing day-to-day operations. Accounting provides reports to monitor these operations (e.g., daily scrap reports).
- **Controlling:** Comparing actual results to the plan and taking corrective action. This involves a feedback loop where performance reports highlight variances that require attention.

Unlike financial accounting, which is bound by GAAP/IFRS and serves external stakeholders, managerial accounting is forward-looking, not governed by fixed rules, and focuses on reporting for specific segments of the business as needed.

Cost

A "cost" can be classified in several ways depending on the context.

- **By Traceability:**
  - *Direct Costs:* Costs that can be easily and economically traced to a cost object (e.g., wood for a guitar).
  - *Indirect Costs (Overhead):* Costs that cannot be easily traced and must be allocated (e.g., factory electricity).
- **For Financial Statements:**
  - *Product Costs:* Costs to manufacture a product (Direct Materials, Direct Labor, Factory Overhead). These are held as an asset (Inventory) until the product is sold.
  - *Period Costs:* Non-manufacturing costs expensed in the period incurred (e.g., selling and administrative expenses).
- **By Behavior:**

  - *Variable Costs:* Costs that change in total in direct proportion to changes in activity level.

  - *Fixed Costs:* Costs that remain constant in total regardless of activity level within a relevant range.

These foundational concepts are the absolute bedrock of our simulator. The classification of costs into product/period and variable/fixed categories defines the core structure of the income statement and balance sheet models. We could program the simulator to understand these distinctions to ensure that costs flow correctly through inventory and onto the income statement.

Costing System

Job Order Costing

Costing system is used by companies that produce unique, custom, or distinct products or services, where costs can be traced to individual "jobs." Examples include construction companies, consulting firms, and custom furniture makers.

Mechanism

- **Job Cost Sheet:** The central document used to accumulate the costs for a specific job.

- **Cost Flow:**

  1. *Direct Materials* and *Direct Labor* costs are traced directly to individual job cost sheets as they are incurred.

  2. *Factory Overhead* costs (indirect costs) cannot be traced directly. Instead, they are applied to jobs using a predetermined overhead rate.

Predetermined Overhead Rate

This rate is calculated at the beginning of the period to ensure that overhead can be applied to jobs throughout the period, rather than waiting until the end when actual overhead is known.

$$\text{Predetermined OH Rate} = \frac{\text{Estimated Total Factory Overhead Cost}}{\text{Estimated Activity Base (e.g., Direct Labor Hours)}}$$

The overhead applied to a specific job is then:

$$\text{Overhead Applied} = \text{Predetermined OH Rate} \times \text{Actual Activity Base Used by the Job}$$

For our simulator, incorporating different costing systems like job order costing could add a layer of sophistication. We could model a "project-based" company where revenue and costs are accumulated on a job-by-job basis. This would be a practical approach for simulating firms in industries like construction or defense contracting. For a general-purpose financial statement forecaster, a process costing model (covered next) is often more representative of typical manufacturing and might be a better starting point due to its relative simplicity.

Process Cost Systems

A process cost system is used by companies that mass-produce identical or very similar products in a continuous production process. Examples include food processing, chemicals, and oil refining.

Mechanism Instead of tracking costs for individual jobs, a process system accumulates costs by department or process for a given period. The total cost is then averaged over all units produced.

- **Cost of Production Report:** This is the key document that summarizes the costs and units for a department. It calculates the cost per unit and assigns costs to units transferred out and to those remaining in ending work-in-process inventory.
- **Equivalent Units of Production:** A core concept used to measure the amount of work done during a period, expressed in terms of fully completed units. It is necessary because some units are only partially completed at the end of a period. For example, 100 units that are 70% complete represent 70 equivalent units.

$$\text{Cost per Equivalent Unit} = \frac{\text{Total Costs for the Period}}{\text{Total Equivalent Units of Work Done}}$$

The process costing model is highly suitable for our simulator, especially for modeling manufacturing firms. Its logic, accumulating costs by department and averaging them over production, is systematic and can be programmed effectively. This approach would form a robust engine for the simulator's cost of goods sold and inventory calculations. It is a practical and common system. A potential challenge in a dynamic simulation would be managing the complexity of multi-department production flows and the calculation of equivalent units, but it is a necessary complexity for realistic forecasting.

Activity-Based Costing

Refined method for allocating factory overhead. Traditional systems often use a single, volume-based allocator (like direct labor hours), which can distort product costs. ABC recognizes that overhead costs are driven by many different activities, not just production volume.

Mechanism ABC involves a two-stage allocation process:

1. **Identify Activities and Pool Costs:** Identify the major activities that drive overhead costs (e.g., machine setups, quality inspections, material handling) and create a "cost pool" for each.
2. **Identify Cost Drivers and Allocate Costs:** For each activity cost pool, identify a "cost driver", measure of the activity that causes the cost. The overhead rate is then

calculated for each activity.

$$\text{Activity Rate} = \frac{\text{Estimated Cost for the Activity Pool}}{\text{Estimated Cost Driver Volume}}$$

Overhead is then assigned to products based on their actual consumption of each activity.

Implementing ABC in our simulator would be an advanced feature. It would allow us to model how a company's understanding of its cost structure can lead to better strategic decisions (e.g., changing pricing or dropping an unprofitable product that a traditional costing system failed to identify). We might consider this a "policy choice" within the simulator; the user could select between a traditional costing system and an ABC system to see how it impacts the reported product costs and overall profitability. It is a practical and powerful technique, though its implementation would be more complex than a traditional system.

Support Department Allocation

Support (or service) departments do not engage in production directly but provide necessary services to the production departments (e.g., maintenance, human resources, IT). Their costs are part of factory overhead and must be allocated to the production departments before being applied to products.

- **Allocation Methods:**
  - *Direct Method:* Allocates each support department's costs directly to production departments only. It is the simplest but least accurate method.
  - *Sequential Method:* Allocates support department costs in a specific order, allowing for some (but not all) inter-departmental services.

Joint Cost Allocation

A **joint cost** is a cost incurred in a single process that yields multiple products simultaneously (e.g., processing crude oil to get gasoline, kerosene, etc.). These costs must be allocated to the different **joint products**.

- **Allocation Methods:** Typically based on the relative sales value of the products at the "split-off point" (the point where they become separately identifiable).

These allocation concepts add further realism to the simulator's overhead model. For a sophisticated simulation of a large, integrated company, modeling support departments is a practical necessity. We could implement the direct method for simplicity. Joint cost allocation is more specialized and might be less critical for a general-purpose simulator unless we intend to model specific industries like natural resources or chemicals. We should probably consider support department allocation as a potential feature, while joint cost allocation could be a lower priority.

3.2.2 Managerial Decision Making

Cost-Volume-Profit Analysis

Cost-Volume-Profit (CVP) analysis is a foundational managerial tool used to study the relationships between sales volume, costs (both variable and fixed), and profit. It provides a model for answering critical planning questions, such as determining a break-even point or forecasting the profit effects of changes in business operations.

Contribution Margin

The cornerstone of CVP analysis is the **contribution margin**, which represents the revenue left over to cover fixed costs after variable costs have been met. Any remaining margin contributes directly to operating income.

- **Contribution Margin per Unit:** The amount each unit sold contributes towards covering fixed costs and generating profit.

$$\text{CM per Unit} = \text{Selling Price per Unit} - \text{Variable Cost per Unit}$$

- **Contribution Margin Ratio (%):** The percentage of each sales dollar that is available to cover fixed costs and generate profit. This is particularly useful when analyzing overall profitability rather than per-unit results.

$$\text{CM Ratio} = \frac{\text{Contribution Margin per Unit}}{\text{Selling Price per Unit}} \quad \text{or} \quad \frac{\text{Total Contribution Margin}}{\text{Total Sales}}$$

Break-Even Point (BEP)

The break-even point is the level of sales at which total revenues equal total costs, resulting in zero operating income. A business must exceed this sales level to become profitable.

- **BEP in Sales Units:** To find the number of units that must be sold to break even.

$$\text{BEP (Units)} = \frac{\text{Total Fixed Costs}}{\text{Contribution Margin per Unit}}$$

- **BEP in Sales Dollars:** To find the total sales revenue required to break even.

$$\text{BEP (Dollars)} = \frac{\text{Total Fixed Costs}}{\text{Contribution Margin Ratio}}$$

Target Profit Analysis

CVP analysis can be easily adapted to determine the sales volume required to achieve a specific, desired level of operating income (target profit). The formulas are modified to include the target profit alongside fixed costs.

- **Sales Units for Target Profit:**

$$\text{Sales (Units)} = \frac{\text{Total Fixed Costs} + \text{Target Profit}}{\text{Contribution Margin per Unit}}$$

- **Sales Dollars for Target Profit:**

$$\text{Sales (Dollars)} = \frac{\text{Total Fixed Costs} + \text{Target Profit}}{\text{Contribution Margin Ratio}}$$

Sensitivity Analysis ("What-If" Analysis)

CVP's primary strength lies in its ability to model the financial impact of changing business conditions. Managers can use it to perform sensitivity analysis by altering key variables:

- *Change in Selling Price:* A price change directly affects the contribution margin per unit and the slope of the total revenue line.
- *Change in Variable Costs:* A change in variable costs (e.g., raw materials) alters the contribution margin and the slope of the total cost line.
- *Change in Fixed Costs:* A change in fixed costs (e.g., adding an advertising campaign) shifts the total cost line up or down, thereby changing the break-even point.

Margin of Safety

The margin of safety measures the "cushion" between current or budgeted sales and the break-even point. It indicates how much sales can decline before the company begins to incur a loss.

- **Formula in Dollars:**

$$\text{Margin of Safety (\$)} = \text{Current Sales (\$)} - \text{Break-Even Sales (\$)}$$

- **Formula as a Percentage:**

$$\text{Margin of Safety (\%)} = \frac{\text{Margin of Safety (\$)}}{\text{Current Sales (\$)}} \times 100\%$$

Assumptions and Limitations of CVP Analysis

The CVP model relies on several key assumptions:

1. Costs can be accurately classified as either fixed or variable.
2. Cost and revenue relationships are linear within a "relevant range" of activity.

3. The unit selling price, unit variable cost, and total fixed costs are constant.
4. In multi-product companies, the sales mix remains constant.
5. For manufacturers, production volume equals sales volume (i.e., no significant change in inventory levels).

For our simulator, we may consider these variable as part of the input.

Variable Costing for Management Analysis

This subsection introduces a different approach to income statement presentation known as **Variable Costing**. It is contrasted with the traditional method, **Absorption Costing**, which is required for external financial reporting under GAAP and IFRS. Variable costing is primarily used for internal analysis and decision-making because it provides a clearer view of cost behavior.

Absorption Costing vs. Variable Costing

The fundamental difference between the two methods lies in the treatment of **fixed manufacturing overhead**.

- **Absorption Costing (Full Costing):** Treats fixed manufacturing overhead as a **product cost**. This means the fixed overhead is "absorbed" into the cost of inventory and is only expensed (as part of Cost of Goods Sold) when the product is sold. This is the method used for external reporting.

$$\text{Product Cost}_{\text{Absorption}} = \text{Direct Materials} + \text{Direct Labor} + \text{Variable OH} + \text{Fixed OH}$$

- **Variable Costing (Direct Costing):** Treats fixed manufacturing overhead as a **period cost**. This means fixed overhead is expensed in the period it is incurred, regardless of how many units are produced or sold. Only variable manufacturing costs are included in the product cost.

$$\text{Product Cost}_{\text{Variable}} = \text{Direct Materials} + \text{Direct Labor} + \text{Variable OH}$$

The Impact on the Income Statement

This difference in cost treatment leads to a different income statement format and can result in different operating income figures if production and sales volumes are not equal.

- **Absorption Costing Income Statement:**

$$\text{Sales} - \text{COGS} = \text{Gross Margin}$$
$$\text{Gross Margin} - \text{Selling \& Admin Expenses} = \text{Operating Income}$$

- **Variable Costing Income Statement (Contribution Margin Format):**

$$\text{Sales} - \text{Variable Costs} = \text{Contribution Margin}$$
$$\text{Contribution Margin} - \text{Fixed Costs} = \text{Operating Income}$$

Reconciliation of Operating Income

The operating income between the two methods will differ whenever the number of units produced does not equal the number of units sold. This difference is due to the fixed manufacturing overhead that is either deferred in (or released from) inventory under absorption costing.

- If **Production** > **Sales**: Inventory increases. Operating income will be **higher** under absorption costing because some of the period's fixed overhead is capitalized into the ending inventory asset, rather than being expensed.
- If **Production** < **Sales**: Inventory decreases. Operating income will be **lower** under absorption costing because fixed overhead from a previous period (stored in beginning inventory) is now released and expensed.
- If **Production** = **Sales**: Inventory is constant. Operating income will be the **same** under both methods.

The difference can be calculated as:

$$\text{Difference in Income} = (\text{Ending Inventory Units} - \text{Beginning Inventory Units}) \times \text{Fixed OH per Unit}$$

Managerial Use of Variable Costing

Variable costing is preferred for many internal decisions because:

1. **Supports CVP Analysis:** The contribution margin income statement format is essential for break-even, target profit, and other CVP calculations.
2. **Explains Profit Changes:** It provides a clearer picture of profitability. Under variable costing, if sales increase, profit increases. Under absorption costing, profit can sometimes increase even if sales decline, simply by producing more units and building up inventory (which can be misleading).
3. **Aids in Decision-Making:** It helps managers make better decisions about pricing, special orders, and evaluating segment profitability by focusing on the contribution margin that each product or segment generates.

Differential Analysis and Product Pricing

Differential analysis is a decision-making approach that focuses only on the changes in revenues and costs between alternative courses of action. It is also known as *incremental analysis*. The key is to ignore irrelevant information, specifically **sunk costs** (costs already incurred that cannot be changed) and costs that do not differ between alternatives.

Differential Analysis

The analysis typically involves comparing the differential revenue, differential cost, and resulting differential income (or loss) of two or more options. Some common applications of differential analysis are as follows:

- **Leasing or Selling Equipment:** The analysis compares the differential revenue from selling the equipment against the differential revenue from leasing it. The book value of the equipment is a sunk cost and therefore irrelevant.
- **Discontinuing a Segment or Product:** A segment or product should only be discontinued if the decrease in total revenue is less than the decrease in total costs. Crucially, this involves identifying which fixed costs are *avoidable* (will disappear if the segment is dropped) and which are *unavoidable* (will remain and must be absorbed by other segments).
- **Make-or-Buy Decisions:** The company compares the cost of internally manufacturing a component (the relevant variable costs plus any avoidable fixed costs) with the cost of purchasing it from an external supplier. Opportunity cost (e.g., the profit that could be earned if the manufacturing capacity were used for something else) is also a key factor.
- **Accepting Business at a Special Price (Special Orders):** A special, one-time order should be accepted if the special price offered is greater than the differential costs of producing the order. These costs are typically the variable manufacturing costs, as fixed costs will not change if the company has excess production capacity.

Product Pricing Methods

Managerial accounting provides structured methods for setting product prices.

- **Cost-Plus Methods:** The price is determined by adding a desired markup to a cost base. Common approaches include:
  - *Total Cost Concept:* Markup is added to the total cost (all manufacturing, selling, and admin costs).
  - *Product Cost Concept:* Markup is added to the total manufacturing cost.
  - *Variable Cost Concept:* Markup is added to the total variable cost.
- **Target Costing:** This is a market-driven approach. The company starts with the price customers are willing to pay and then designs a product that can be manufactured at a low enough cost (the target cost) to generate a profit.

$$\text{Target Cost} = \text{Expected Selling Price} - \text{Desired Profit}$$

These approaches provide a set of powerful, logical frameworks for short-term decision-making. For the simulator, these concepts can be useful and practical. We can program "events" or "opportunities" for the simulated company, such as a special order offer or a make-or-buy dilemma. The simulator could then execute a differential analysis based on its current financial state (e.g., production capacity, variable cost structure) to make the most profitable choice. This is good for modeling tactical, operational decisions where the alternatives are clear and quantifiable. It is less suited for complex strategic choices involving many qualitative factors (like brand image or long-term supplier relationships) that are hard to model.

Capital Investment Analysis

Capital investment analysis is the process of evaluating and prioritizing long-term investment proposals, which typically involve large initial cash outlays and returns that are expected to span several years. These decisions are critical to a company's strategic direction.

Methods that Ignore the TVM

These methods are simpler to calculate but can be less accurate because they treat a dollar received in the future as being worth the same as a dollar today.

- **Average Rate of Return (ARR):** Measures the average annual income expected from the investment as a percentage of the average investment.

$$\text{ARR} = \frac{\text{Average Annual Income}}{\text{Average Investment}}$$

  A project is accepted if its ARR exceeds a minimum required rate of return.

- **Cash Payback Period:** Measures the time (in years) required to recoup the initial cash investment from the project's net cash inflows.

$$\text{Payback Period} = \frac{\text{Initial Investment}}{\text{Annual Net Cash Inflow}} \quad \text{(for equal annual inflows)}$$

  This method is often used as a screening tool to assess risk and liquidity.

Methods that Use the TVM

These methods are conceptually superior because they explicitly account for the fact that future cash flows are worth less than current ones. They rely on discounting future cash flows back to their present value.

- **Net Present Value (NPV):** Calculates the difference between the present value of a project's future net cash inflows and the initial investment amount. A discount rate (often the company's minimum desired rate of return or cost of capital) is used.

$$\text{NPV} = \text{PV of Future Cash Inflows} - \text{Initial Investment}$$

If **NPV is positive or zero**, the project is considered acceptable because its return meets or exceeds the minimum required rate. If negative, it is rejected.

- **Internal Rate of Return (IRR):** This is the discount rate at which the NPV of a project equals zero. It represents the project's true expected rate of return. The IRR is found through trial and error or with a financial calculator. A project is acceptable if its **IRR is greater than** the company's minimum desired rate of return.

Capital investment analysis provides the decision rules for a company's long-term evolution, making it highly relevant for the simulator. We could model a company's growth by having it evaluate potential "capital projects" (e.g., build a new factory, launch a major product line). The simulator could use an NPV or IRR calculation as the core decision rule to determine whether to commit capital. This is very practical for simulating strategic growth and resource allocation. These methods are good for situations involving large, distinct projects with predictable cash flows over multiple periods. They are less effective for investments where benefits are hard to quantify in cash terms (e.g., R&D, employee training) or in highly uncertain environments where cash flow forecasts are unreliable.

3.2.3 Planning, Control, and Performance Evaluation

Budgeting

Budgeting is the process of creating a detailed plan that outlines a company's financial and operational goals for a specific period. The comprehensive result of this process is the **Master Budget**, which serves as a crucial tool for planning, communication, coordination, and control. The master budget is typically prepared for a one-year period, often broken down into quarterly or monthly segments.

Master Budget Component

The master budget is composed of two primary categories: operating budgets and financial budgets. The process is sequential, with the output of one budget often serving as the input for the next. The **Sales Budget** is always the first step, as it drives the entire process.

A. The Operating Budgets

These budgets focus on the income-generating activities of the business and culminate in a budgeted income statement.

1. **Sales Budget:** The starting point. It is a detailed forecast of expected sales in units and dollars for the budget period.

$$\text{Budgeted Sales} = \text{Expected Sales Volume (Units)} \times \text{Expected Unit Selling Price}$$

2. **Production Budget:** This budget plans the number of units that must be produced to

meet sales needs and desired ending inventory levels.

$$\text{Units to Produce} = \text{Expected Sales} + \text{Desired Ending Inventory} - \text{Beginning Inventory}$$

3. **Direct Materials Purchases Budget:** This budget calculates the quantity and cost of direct materials that must be purchased to meet production needs.

$$\text{Materials to Purchase} = (\text{Units to Produce} \times \text{DM per Unit}) + \text{Desired End. DM Inv.} - \text{Beg. DM Inv.}$$

4. **Direct Labor Cost Budget:** This budget estimates the direct labor hours and cost required to meet the production budget.

$$\text{Total DL Cost} = \text{Units to Produce} \times \text{DL Hours per Unit} \times \text{DL Cost per Hour}$$

5. **Factory Overhead Cost Budget:** This budget provides a schedule of all expected manufacturing costs other than direct materials and direct labor. It separates costs into variable and fixed components.
6. **Cost of Goods Sold Budget:** This budget integrates the information from the direct materials, direct labor, and factory overhead budgets to estimate the cost of the goods that will be sold.
7. **Selling and Administrative Expenses Budget:** This budget outlines all planned non-manufacturing expenses, separating them into variable and fixed components.

B. The Financial Budgets

These budgets focus on the financial implications of the operating plan, specifically cash flows and the company's financial position.

1. **Cash Budget:** This is a critical budget for managing liquidity. It details the expected cash receipts and payments for the period and shows any anticipated cash excess or deficiency. Its main sections are:
   - Cash Receipts (from sales, etc.).
   - Cash Payments (for materials, labor, overhead, etc.).
   - Net Increase/Decrease in Cash.
   - Financing subsection (for borrowing or repaying debt).
2. **Budgeted Income Statement:** This is prepared using the information from the operating budgets (Sales, COGS, S&A Expenses) to project the company's profitability for the period.

3. **Budgeted Balance Sheet:** This is the final step. It projects the company's financial position at the end of the budget period, using data from the budgeted balance sheet of the previous period and integrating the effects of all the budgets in the current period.

The master budget is perhaps the most directly applicable concept for our simulator. It provides a complete, step-by-step causal logic for forecasting financial statements. We could structure the simulator's core forecasting engine to follow this exact sequence, starting with a sales forecast and then deriving all subsequent operational activities and financial outcomes. This approach is practical and forms the basis of corporate financial planning. Its primary limitation, in a standard form, is that it can be somewhat static and deterministic. For our simulator, we might consider this the "base case" and then introduce dynamic elements, like the decision rules from differential analysis or variance analysis feedback loops, to allow the simulated company to react to changing conditions rather than just follow a fixed plan.

Evaluating Variances from Standard Costs

After creating a budget, managers need a way to control operations and evaluate performance against that plan. A **standard cost system** provides this by setting detailed cost benchmarks for a single unit of product. These standards represent what costs should be under efficient operating conditions. The difference between the actual cost incurred and the standard cost is called a **variance**.

Standard Costs vs. Budgets

- A **budget** sets total cost goals for a given level of activity.
- A **standard** sets the cost goal for a single unit (*per-unit* budget). Standards are used to build flexible budgets that can be adjusted for the actual level of output.

Direct Materials Variances

The total variance for direct materials is broken down into two components to identify the underlying cause: price or quantity.

- **Price Variance:** Measures the difference between what was actually paid for materials and what should have been paid (the standard price). This variance is typically the responsibility of the purchasing department.

  $$\text{DM Price Variance} = (\text{Actual Price} - \text{Standard Price}) \times \text{Actual Quantity Purchased}$$

  An unfavorable variance means the actual price was higher than the standard.

- **Quantity Variance:** Measures the difference between the quantity of materials actually used in production and the quantity that *should have been* used for the actual output (the standard quantity). This variance is the responsibility of the production department.

$$\text{DM Quantity Variance} = (\text{Actual Quantity Used} - \text{Standard Quantity Allowed}) \times \text{Standard Price}$$

An unfavorable variance means more material was used (e.g., due to waste or spoilage) than the standard allowed.

Direct Labor Variances

Similar to materials, the total labor variance is split into two components: rate or efficiency.

- **Rate Variance:** Measures the difference between the actual hourly wage rate paid to workers and the standard hourly rate.

  $$\text{DL Rate Variance} = (\text{Actual Rate} - \text{Standard Rate}) \times \text{Actual Hours Worked}$$

  An unfavorable variance means the average wage rate was higher than planned, perhaps due to using more senior employees or unplanned overtime.

- **Efficiency Variance:** Measures the difference between the actual labor hours worked and the standard labor hours that *should have been* worked for the actual output. This is a measure of worker productivity and is the responsibility of the production supervisor.

  $$\text{DL Efficiency Variance} = (\text{Actual Hours Worked} - \text{Standard Hours Allowed}) \times \text{Standard Rate}$$

  An unfavorable variance means production took more time than the standard allowed.

Reporting and Investigating Variances

Variances are summarized in performance reports. The principle of **management by exception** is applied, meaning that managers should focus their attention only on variances that are significant or material in amount. Both unfavorable (costing more than planned) and favorable (costing less than planned) variances should be investigated if they are large, as a favorable variance might indicate an issue (e.g., purchasing cheap, low-quality materials).

This concept of variance analysis appears valuable for our simulation. It provides the essential "control" and "feedback loop" mechanism that follows the "planning" stage of the budget. We could model the simulator to generate "actual" results with some degree of randomness or in response to external shocks. A variance analysis module could then run automatically, comparing these actuals to the budget standards. This is a practical and widely used management technique. Its strength is in enabling a more dynamic simulation. A potential limitation is that modeling the causes and magnitude of variances realistically can be complex. Furthermore, an overemphasis on variance minimization in a simulation might lead to simplistic outcomes, whereas in reality, managers must balance cost control with quality, employee morale, and other qualitative factors.

Evaluating Decentralized Operations

As businesses grow, it is often impractical for a central authority to make all decisions. This leads to **decentralization**, where authority is delegated to managers of lower-level organizational units. Managerial accounting provides a framework called **responsibility accounting** to measure the performance of these units.

Responsibility Centers

In a decentralized company, units are structured as responsibility centers, and managers are evaluated based on the items they can directly control. There are three main types:

- **Cost Center:** The manager has responsibility only for controlling costs. The unit does not directly generate revenue, e.g. a factory's production department or the company's accounting department.
- **Profit Center:** The manager has responsibility for both revenues and costs, and therefore for profits, e.g. a specific retail store in a chain or a particular product line.
- **Investment Center:** The manager has responsibility for revenues, costs, and also for the investment of assets. They are accountable for using the center's assets to generate profit, e.g. a large, independent corporate division like Chevrolet within General Motors.

Performance Evaluation for Investment Centers

Evaluating investment centers is the most complex task. Two primary measures are used:

- **Return on Investment (ROI):** This is a widely used metric that measures the profitability of a division relative to the size of its assets. It is calculated as:

  $$\text{ROI} = \frac{\text{Operating Income}}{\text{Invested Assets}}$$

  The ROI formula can be expanded to provide more insight into its drivers:

  $$\text{ROI} = \left(\frac{\text{Operating Income}}{\text{Sales}}\right) \times \left(\frac{\text{Sales}}{\text{Invested Assets}}\right)$$

  This expanded formula shows that ROI can be improved by increasing profitability (margin), using assets more efficiently to generate sales (turnover), or both.

- **Residual Income (RI):** This metric measures the amount of operating income a division earns above a minimum required return on its assets. The minimum return is based on the company's cost of capital.

  $$\text{Residual Income} = \text{Operating Income} - (\text{Minimum Rate of Return} \times \text{Invested Assets})$$

  A positive residual income means the division is exceeding the company's minimum performance target. A key advantage of RI over ROI is that it encourages managers to accept any profitable investment that exceeds the minimum required return, avoiding the problem of a high-ROI division rejecting a good project simply because it would lower its average ROI.

Transfer Pricing

When divisions within the same company sell goods or services to each other, a **transfer price** must be set for the internal transaction. This price becomes revenue for the selling division and a cost for the buying division, directly affecting the performance metrics (like ROI and RI) of both. Common approaches to setting transfer prices include:

- **Market Price Approach:** Based on the price that would be charged to an external customer.
- **Cost-Based Approaches:** Based on the cost of producing the product (e.g., variable cost or full absorption cost), often with a markup.
- **Negotiated Price Approach:** The price is determined through negotiation between the division managers.

The concepts of decentralization and responsibility accounting introduce an interesting layer of complexity for our simulation. We could model the simulated company not as a single monolith, but as a collection of interacting divisions (investment centers). Each division could have its own manager AI with a specific goal: maximize its ROI or RI. This could lead to more realistic and emergent behavior. For instance, a division manager might reject a capital project that is good for the company as a whole if it slightly harms their division's ROI. This is a practical representation of real-world internal corporate dynamics. We should be conservative in our approach, as modeling the behavioral incentives and potential for conflict (e.g., disputes over transfer prices) could be quite complex. However, even a simplified implementation where divisional performance is tracked could significantly enhance the simulator's robustness by showing how internal structure affects overall financial results.

Lean Manufacturing and Activity Analysis

Lean manufacturing is a management philosophy focused on systematically eliminating waste in all aspects of the production process, with the goal of delivering value to the customer more efficiently. Rather than producing large batches of goods to be held in inventory, lean principles emphasize producing products only as they are needed.

Waste Elimination

Lean manufacturing identifies several forms of waste to be eliminated. These include:

- Producing defective products.
- Producing more goods than are needed (overproduction).

- Excess inventory.
- Unnecessary movement of employees.
- Unnecessary processing steps.
- Waiting time (e.g., for materials or the next production step).
- Inefficient transportation of materials.

Key Practices in Lean Manufacturing

- **Just-in-Time (JIT):** This is a core practice where materials are received from suppliers and products are manufactured "just in time" to meet customer demand. This dramatically reduces or eliminates inventory, which in turn reduces storage costs, waste, and the capital tied up in stock.
- **Cellular Manufacturing:** Workstations and equipment are arranged in a "cell" to support a smooth flow of production through a sequence of operations. This minimizes the distance materials have to travel and reduces setup and waiting times.
- **Employee Involvement:** Lean systems empower employees with the responsibility to identify and eliminate waste, continuously improve processes (a concept known as *kaizen*), and maintain quality at the source.

Impact on Managerial Accounting

The shift to a lean environment changes how performance is measured and how costs are tracked.

- **Performance Measurement:** Traditional metrics, such as direct labor efficiency variance, can be less relevant or even counterproductive in a lean system. Instead, lean accounting focuses on non-financial measures that reflect the core principles of lean, such as:
  - **Lead Time (or Cycle Time):** The total time from when an order is received to when it is shipped.
  - **Value-Added Ratio:** The proportion of lead time that is spent on activities that actually add value to the product.
  - **Setup Time:** The time required to switch production from one product to another.
  - **First-Time-Through Rate:** The percentage of products that are completed without any need for rework.
- **Costing Systems:** With minimal inventory, the distinction between variable and absorption costing becomes less significant. Some lean companies simplify their costing by combining direct labor and overhead into a single category called **conversion costs**.

Costs may be tracked by "value streams" (all the activities required to bring a product to the customer) rather than by traditional departments.

The principles of lean manufacturing could provide a fascinating dimension to our simulator. We might consider modeling different "types" of companies, such as a traditional batch-production company versus a lean, JIT-based company. We could then observe the differences in their financial statements, the lean company would likely show much lower inventory levels on its balance sheet and potentially higher asset turnover ratios. This seems like a practical way to simulate the impact of different operational strategies. We should be mindful that a lean model in the simulator might be more sensitive to "shocks," such as a sudden supply disruption, which would be a realistic trade-off to model. This concept is exceptionally well-suited for simulating manufacturing firms. It might be less directly applicable, though still conceptually useful, for service-based or purely financial firms where physical inventory is not a factor.

Balanced Scorecard and Corporate Social Responsibility

This chapter moves beyond purely financial metrics to discuss a more holistic framework for evaluating organizational performance: the **Balanced Scorecard**. It also addresses the increasing importance of reporting on a company's social and environmental impact.

Balanced Scorecard

It is a strategic performance management system that translates an organization's mission and strategy into a comprehensive set of performance measures. It provides a "balanced" view by supplementing traditional financial measures with metrics from three additional perspectives. The goal is to align day-to-day work with overall strategy and to monitor performance against strategic objectives.

The framework consists of four perspectives:

1. **The Financial Perspective:** This perspective answers the question, "How do we look to shareholders?" It focuses on traditional financial measures of success.

    - *Common Objectives:* Increase revenue, control costs, improve productivity, enhance shareholder value.

    - *Example Measures:* Operating income, return on investment (ROI), revenue growth, cash flow.

2. **The Customer Perspective:** This perspective answers the question, "How do customers see us?" It focuses on identifying the target customer segments and measuring the company's success in meeting their needs.

    - *Common Objectives:* Improve customer satisfaction, increase market share, retain existing customers.

- *Example Measures:* Customer satisfaction ratings, market share percentage, customer retention rate, number of new customers.

3. **The Internal Process Perspective:** This perspective answers the question, "What must we excel at?" It focuses on the internal business processes that are critical for satisfying customers and achieving financial objectives.

    - *Common Objectives:* Improve product quality, enhance operational efficiency, foster innovation.

    - *Example Measures:* Defect rates, order fulfillment time (lead time), cost of quality, number of new products launched.

4. **The Learning and Growth Perspective:** This perspective answers the question, "How can we continue to improve and create value?" It focuses on the infrastructure and intangible assets necessary for long-term growth, such as employee skills, information systems, and organizational culture.

    - *Common Objectives:* Enhance employee skills, improve access to information, foster a culture of innovation.

    - *Example Measures:* Employee training hours, employee satisfaction scores, percentage of revenue from new products.

A key feature of the scorecard is that the objectives across the four perspectives are linked in a **cause-and-effect chain**. For example, investing in employee training (Learning & Growth) should lead to improved internal processes, which in turn leads to higher customer satisfaction, and ultimately results in better financial performance.

Corporate Social Responsibility (CSR)

CSR is a management concept whereby companies integrate social and environmental concerns into their business operations and interactions with their stakeholders. It reflects a company's commitment to being a responsible corporate citizen. The reporting of a company's performance on these dimensions is often called **sustainability reporting**. This can include:

- **Environmental Performance:** Reporting on resource usage, emissions, and environmental stewardship (e.g., tons of CO2 emitted, water usage, recycling rates).

- **Social Performance:** Reporting on employee welfare, community involvement, and ethical practices (e.g., employee turnover, workplace safety incidents, charitable contributions).

- **Economic Performance:** The company's traditional financial performance, viewed in the context of its broader social and environmental impact.

The Balanced Scorecard provides a powerful conceptual framework for our simulator. While financial statement forecasting is the primary goal, we could use the scorecard logic to build more sophisticated causal relationships within the model. For example, a decision to invest in "Learning and Growth" (e.g., R&D spending) could be programmed to have a delayed, positive impact on "Internal Process" metrics (e.g., lower production costs), which in turn could boost customer metrics and finally, financial results. This would allow us to simulate how non-financial investments and strategies translate into long-term financial outcomes. We should be conservative with this approach, as quantifying these relationships is a significant challenge. However, even a simplified model could demonstrate that short-term cost-cutting might harm long-term value, adding a layer of strategic depth to the simulation. The CSR component is less of a direct fit for a financial forecasting model but could be included as a set of non-financial outputs that the simulator tracks and reports.

3.2.4 Financial Statement Analysis from a Managerial Perspective

This final part bridges the gap between generating internal managerial data and interpreting the primary financial statements that are the ultimate output of the accounting cycle. While these topics are also central to financial accounting, they are viewed here from the perspective of a manager using the information for internal decision-making, performance evaluation, and strategic planning.

The Statement of Cash Flows

The Statement of Cash Flows reports a company's cash inflows and outflows for a period, providing crucial insights into its liquidity and solvency. It reconciles the beginning and ending cash balances on the balance sheet and explains how cash was generated and used. The statement is divided into three main activities.

1. Cash Flows from Operating Activities

This subsection shows the cash effects of the transactions that create revenues and expenses, and thus enter into the determination of net income. It is a critical indicator of a company's ability to generate cash from its core business operations.

- **Cash Inflows:** Primarily cash received from customers.
- **Cash Outflows:** Primarily cash paid to suppliers for inventory, to employees for wages, and to others for operating expenses like rent and utilities.
- **Calculation Methods:**
  - *Direct Method:* Presents the cash inflows and outflows directly (e.g., "Cash received from customers").
  - *Indirect Method:* Starts with net income and adjusts it for non-cash items (like depreciation) and changes in working capital accounts (like Accounts Receivable and Accounts Payable) to arrive at net cash flow from operations. The indirect method is far more common.

2. Cash Flows from Investing Activities

This subsection reports the cash effects of purchasing and selling long-term assets and other investments. It provides insight into how the company is investing in its long-term future.

- **Cash Inflows:** Cash from selling property, plant, and equipment (PP&E); selling investments in other companies.
- **Cash Outflows:** Cash used to purchase PP&E; purchasing investments.

3. Cash Flows from Financing Activities

This subsection shows the cash effects of transactions with the company's owners (stockholders) and long-term creditors (bondholders). It illustrates how the company is financed.

- **Cash Inflows:** Cash from issuing stock; issuing bonds or notes.
- **Cash Outflows:** Cash paid to repurchase stock (treasury stock); repaying debt principal; paying dividends.

Understanding the Statement of Cash Flows is fundamental to our simulator. Since our goal is to forecast financial statements, generating a correct and complete Statement of Cash Flows is a required output. The structure of this statement provides a logical check on the simulator's integrity, the changes in the balance sheet and income statement must translate directly into the cash flows reported. We could use the indirect method as a core component of the simulator's reporting module, which would automatically reconcile the forecasted net income with the forecasted change in the cash balance. This is a practical and necessary step for ensuring the simulator's outputs are internally consistent and realistic.

Financial Statement Analysis

Financial statement analysis involves using various tools to evaluate a company's performance and financial condition. The goal is to identify strengths and weaknesses and to help predict future performance. The analysis is typically based on comparative data and industry benchmarks.

Methods of Analysis

- **Horizontal Analysis (Trend Analysis):** Compares financial data over a series of reporting periods. Each line item (e.g., Sales, Net Income) is expressed as a percentage of the base-year amount, making it easy to see the direction and magnitude of trends.

$$\text{Percent Change} = \frac{(\text{Current Year Amount} - \text{Base Year Amount})}{\text{Base Year Amount}} \times 100\%$$

- **Vertical Analysis (Common-Size Analysis):** Expresses each financial statement item as a percentage of a base amount for a single period.

- For the *Income Statement*, all items are usually expressed as a percentage of **Sales**.
- For the *Balance Sheet*, all items are usually expressed as a percentage of **Total Assets**.

This method is useful for comparing companies of different sizes or for identifying significant changes in the composition of assets or expenses.

Ratio Analysis

Financial ratios are used to assess different aspects of a company's performance. They are most useful when compared to historical trends or industry averages. Key categories include:

- **Liquidity Ratios:** Measure short-term ability to pay maturing obligations and meet unexpected needs for cash.
  - *Current Ratio* = Current Assets / Current Liabilities
  - *Quick Ratio (Acid-Test)* = (Cash + Marketable Securities + Accounts Receivable) / Current Liabilities
- **Solvency Ratios:** Measure the ability of a company to survive over a long period of time by assessing its level of debt.
  - *Debt to Assets Ratio* = Total Liabilities / Total Assets
- **Profitability Ratios:** Measure the income or operating success of a company for a given period.
  - *Profit Margin* = Net Income / Sales
  - *Return on Assets (ROA)* = Net Income / Average Total Assets
  - *Return on Equity (ROE)* = Net Income / Average Stockholders' Equity
- **Activity Ratios:** Measure how efficiently a company is using its assets.
  - *Asset Turnover* = Sales / Average Total Assets
  - *Inventory Turnover* = Cost of Goods Sold / Average Inventory

This analysis is directly relevant to the "output" side of our simulator. Once the simulator generates the forecasted financial statements, we could build an automated analysis module that calculates these key ratios and trends. This would allow us to not only see the raw forecasted numbers but also to interpret them through the lens of a real financial analyst. We could have the simulator track these ratios over time to see if the company's performance is improving or deteriorating based on the strategic decisions it makes. This is a very practical feature that would significantly increase the simulator's utility as a tool for understanding the consequences of business decisions. We might be conservative in how we interpret the ratios, as a simulation cannot capture all the qualitative factors that affect a real company, but the quantitative framework is essential.

3.3 Others

Here listed out some fields that are related to the simulator, but have not been detailed studied yet. We can consider incorporating them in future iteration.

- Financial Shenanigans, Schilit

4 Modelling Approach

Literature on no-plug no-circularity are in Hongxiao's report. This section presents the core modelling framework of our financial statement simulator, which is designed to generate internally consistent financial statements while enabling flexible scenario analysis. The key innovation is the adoption of a stock-flow approach that embeds the double-entry accounting system at its core, ensuring that all generated statements are balanced by construction without the need for ad-hoc plug entries. We also discuss the directed acyclic graph structure that underpins the computational pipeline, as well as the modelling choices for state dynamics and policy functions.

4.1 Stock-Flow Approach

A natural first attempt at financial statement simulation is to forecast each balance sheet line item independently using standard time-series models. Let $\hat{A}_t$, $\hat{L}_t$, and $\hat{E}_t$ denote the forecasts of total assets, liabilities, and equity at time t, each produced by its own model. However, there is no structural mechanism ensuring that the fundamental accounting identity

$$A_t = L_t + E_t \tag{2}$$

holds for the forecasted values. In practice, it almost never does. One could impose (2) as a post-hoc constraint (the plug approach), but this treats the identity as an inconvenient numerical artefact rather than what it truly is: a *law* inherited from the double-entry bookkeeping system.

Under double-entry bookkeeping, every economic transaction is recorded as a pair of equal and opposite entries. A cash purchase of equipment, for example, simultaneously decreases cash and increases PP&E by the same amount; borrowing simultaneously increases cash and increases debt. Consequently, the balance sheet is not balanced by coincidence, it is balanced by construction, because every flow of value into one account is matched by a flow out of another. Any simulation framework that aspires to produce internally consistent financial statements must embed this principle at its core.

The *stock-flow* approach does precisely this. Rather than modelling balance sheet levels, we model the flows, where the individual transactions and economic events occur, and let the levels emerge from deterministic accumulation rules. Stock variables, corresponding to balance sheet items, are quantities measured at a point in time. Flow variables, corresponding to income statement and cash flow statement items, are quantities measured over a period. Because every flow that enters one stock must exit another (or originate from an explicitly tracked external source), the double-entry principle is hard-wired into the dynamics, and (2) is guaranteed at every simulated time step without post-hoc correction.

A remaining modelling question is: what *generates* the flows? In our framework, flows are ultimately driven by the economic environment the firm faces and the decisions its managers make. We formalise this through a four-layer variable taxonomy: stock, state, policy, and flow, described in the following subsections.

4.1.1 Stock Variables

Stock variables represent quantities measured at a *point in time* and correspond to balance sheet items, e.g., cash and cash equivalents, property, plant and equipment (PP&E), total debt, retained earnings, and so on.

Let $S_t^{(i)}$ denote the i-th stock variable at time t. Consistent with the double-entry principle, its dynamics are governed by an accumulation equation that aggregates all flows entering and leaving the account:

$$S_{t+1}^{(i)} = S_t^{(i)} + \sum_{j \in \mathcal{I}^{(i)}} f_{t+1}^{(j)} - \sum_{k \in \mathcal{O}^{(i)}} f_{t+1}^{(k)}, \tag{3}$$

where $\mathcal{I}^{(i)}$ and $\mathcal{O}^{(i)}$ are the index sets of inflow and outflow variables for stock i. For example, PP&E evolves as

$$\mathrm{PPE}_{t+1} = \mathrm{PPE}_t + \mathrm{CAPEX}_{t+1} - \mathrm{Depreciation}_{t+1} - \mathrm{Disposal}_{t+1}. \tag{4}$$

Because these accumulation equations are *deterministic accounting identities*, no estimation is required at the stock layer. All stochasticity enters through the flow variables and propagates upward.

4.1.2 State Variables

State variables describe the economic environment in which the firm operates and constitute the information set upon which managerial decisions are conditioned. We partition them into two groups.

Exogenous (macro) states.

These include macroeconomic indicators such as GDP growth, inflation, and the risk-free rate. In the prototype, the vector of exogenous states $\mathbf{x}_t^{(\text{macro})}$ follows a VAR(1) process:

$$\mathbf{x}_{t+1}^{(\text{macro})} = \mathbf{c} + \mathbf{\Phi}\,\mathbf{x}_t^{(\text{macro})} + \boldsymbol{\varepsilon}_{t+1}, \quad \boldsymbol{\varepsilon}_{t+1} \sim \mathcal{N}(\mathbf{0},\, \mathbf{\Sigma}), \tag{5}$$

where $\mathbf{c}$ is a vector of intercepts, $\mathbf{\Phi}$ is the autoregressive coefficient matrix, and $\mathbf{\Sigma}$ is the innovation covariance matrix. Users may override these processes by supplying scenario-specific time series.

Note. The modelling for stationarity is deferred to the below section 4.3.

Endogenous (firm-specific) states.

These include variables such as the firm's current leverage ratio, the market price of its goods, or its realised return on invested capital. They are computed from lagged stocks and contemporaneous macro states:

$$x_t^{(\text{firm})} = g\big(\mathbf{S}_t,\ \mathbf{x}_t^{(\text{macro})}\big), \tag{6}$$

where $g(\cdot)$ is a known deterministic mapping (e.g. leverage $= D_t/(D_t + E_t)$). Because (6) depends only on quantities already realised at time t, no circularity is introduced.

4.1.3 Policy Variables

Policy variables encode *managerial decisions* made in response to the prevailing state. At each period, a manager decides, among others, how much cash to retain, how much debt to issue, what share of capital to allocate to risky versus safe assets, and whether to distribute dividends. These decisions embody economic trade-offs, e.g., retaining cash provides a precautionary buffer against future idiosyncratic shocks but forgoes current investment opportunities; issuing debt expands the investment set but raises future interest obligations and default risk.

In the prototype, each policy variable $p_t^{(i)}$ is modelled as a linear function of the full state vector $\mathbf{s}_t = \left[\mathbf{x}_t^{(\text{macro})};\ \mathbf{x}_t^{(\text{firm})}\right]$:

$$p_t^{(i)} = \alpha^{(i)} + \left(\boldsymbol{\gamma}^{(i)}\right)^\top \mathbf{s}_t + \eta_t^{(i)}, \quad \eta_t^{(i)} \sim \mathcal{D}^{(i)}, \tag{7}$$

where $(\alpha^{(i)}, \boldsymbol{\gamma}^{(i)})$ are policy parameters and $\mathcal{D}^{(i)}$ is a noise distribution that may be degenerate (yielding a deterministic policy), continuous (yielding a stochastic policy), or discrete (e.g. a Bernoulli gate for "issue debt or not").

Equation (7) provides the natural interface for scenario analysis: a user modifies $(\alpha^{(i)}, \boldsymbol{\gamma}^{(i)})$ or replaces the linear form entirely to explore alternative managerial strategies while the accounting backbone remains intact.

4.1.4 Flow Variables

Flow variables represent quantities measured over a *period* and correspond to income statement and cash flow statement items, e.g., revenue, cost of goods sold, capital expenditure, interest expense, depreciation, dividends, and so on. Given the state and policy variables, each flow is determined by

$$f_{t+1}^{(j)} = h^{(j)}\big(\mathbf{s}_{t+1},\ \mathbf{p}_{t+1}\big) + \xi_{t+1}^{(j)}, \quad \xi_{t+1}^{(j)} \sim \mathcal{N}\big(0, {\sigma^{(j)}}^2\big), \tag{8}$$

where $h^{(j)}(\cdot)$ encodes the economic and accounting logic linking decisions to financial outcomes. For instance, revenue is the product of the market price (a state) and the quantity produced (a policy):

$$\text{Revenue}_{t+1} = P_{t+1} \times Q_{t+1}, \tag{9}$$

and interest expense is the product of the prevailing rate (a state) and outstanding debt (a stock determined at the end of the prior period):

$$\text{InterestExpense}_{t+1} = r_{t+1} \times D_t. \tag{10}$$

Once all flows are evaluated, they enter the accumulation equations (3) to update each stock, completing one simulation step.

4.2 Directed Acyclic Graph Structure

Within each time step, the causal ordering of computation is

$$\mathbf{s}_t \xrightarrow{\text{policy}} \mathbf{p}_t \xrightarrow{\text{accounting}} \mathbf{f}_t \xrightarrow{\text{accumulation}} \mathbf{S}_{t+1}. \tag{11}$$

This pipeline forms a *Directed Acyclic Graph* (DAG): every variable at time $t+1$ can be computed in a single forward pass with no variable depending on its own updated value.

Why does acyclicity matter? Consider a plausible circular dependency: lower net income reduces operating cash flow; reduced cash flow triggers additional debt issuance; additional debt raises interest expense; higher interest expense further reduces net income. If these relationships are evaluated within the same period, the system becomes a set of simultaneous equations that must be solved iteratively, increasing computational cost per time step and precluding straightforward parallelisation across Monte Carlo paths.

By enforcing the DAG structure, specifically, by requiring that debt issuance at $t+1$ responds to the state observed at t, not to the endogenous outcomes at $t+1$. We break such cycles without loss of economic realism (the manager sets policy based on information available *before* the period's outcomes are realised). The result is a simulation step that is a pure feed-forward computation, naturally vectorised and parallelisable.

Inter-temporal feedback.

Although the within-period graph is acyclic, the model is not open-loop. Updated stocks $\mathbf{S}_{t+1}$ feed into the firm-specific state variables $\mathbf{x}_{t+1}^{(\text{firm})}$ via (6), closing the dynamic feedback loop across periods. The system therefore captures rich inter-temporal interactions, where leverage today affects borrowing cost tomorrow, which affects profitability, which affects future leverage, while retaining a clean feed-forward structure within each step.

Scalability.

As the number of accounts and flow equations grows, maintaining the DAG property and verifying accounting identities by manual inspection becomes impractical. We address this challenge through a symbolic computation framework detailed in Section 8.

Trainability.

The DAG structure, combined with the differentiable formulation of (7) and (8), implies that gradients of any simulated financial quantity with respect to the policy parameters can be obtained via automatic differentiation. This opens the door to gradient-based optimisation of corporate policies, effectively solving for the managerial strategy that optimises a user-specified objective. We discuss the computational realisation of this capability in Section 8.

4.3 State Modelling

4.3.1 VAR(1)

The exogenous macro-state dynamics were introduced in (5). We now detail the stationarity requirements, moment structure, and unconstrained parameterisation that together make the VAR(1) component both theoretically sound and amenable to gradient-based estimation.

Stationarity condition.

The process (5) is weakly stationary if its mean $\boldsymbol{\mu} = \mathbb{E}[\mathbf{x}_t^{(\text{macro})}]$ and autocovariance $\boldsymbol{\Gamma}(h) = \text{Cov}(\mathbf{x}_t^{(\text{macro})}, \mathbf{x}_{t-h}^{(\text{macro})})$ are time-invariant and all second moments are finite. The necessary and sufficient condition is that every eigenvalue λ of $\boldsymbol{\Phi}$ satisfies

$$|\lambda| < 1, \tag{12}$$

equivalently, the spectral radius $\rho(\boldsymbol{\Phi}) = \max_i |\lambda_i(\boldsymbol{\Phi})| < 1$. Because $\boldsymbol{\Phi}$ is real but may have complex eigenvalues $\lambda = a + bi$, the modulus $|\lambda| = \sqrt{a^2 + b^2}$ measures distance from the origin in the complex plane; condition (12) therefore requires all eigenvalues to lie strictly inside the unit circle.

An equivalent formulation uses the lag-polynomial determinant. Writing the model in lag-operator form $(I - \boldsymbol{\Phi}B)\,\mathbf{x}_t = \mathbf{c} + \boldsymbol{\varepsilon}_t$ and replacing B by a complex variable z, stationarity holds if and only if all roots of $|I - \boldsymbol{\Phi}z| = 0$ satisfy $|z| > 1$. The two views are linked by the identity

$$|\lambda I - \boldsymbol{\Phi}| = \lambda^K \left| I - \boldsymbol{\Phi}\,\lambda^{-1} \right|, \tag{13}$$

so that λ is an eigenvalue of $\boldsymbol{\Phi}$ if and only if $z = 1/\lambda$ is a root of the determinant polynomial, and $|\lambda| < 1 \Leftrightarrow |z| > 1$. In the scalar AR(1) case, this reduces to the familiar condition $|\phi| < 1$.

Why stationarity matters.

Condition (12) guarantees that the matrix powers $\boldsymbol{\Phi}^j$ decay to zero as $j \to \infty$. This is the discrete-time stability result: $\boldsymbol{\Phi}^j \to 0$ if and only if $\rho(\boldsymbol{\Phi}) < 1$. For the diagonalisable case, writing $\boldsymbol{\Phi} = P\Lambda P^{-1}$ with diagonal $\Lambda = \text{diag}(\lambda_1, \ldots, \lambda_K)$, the conclusion follows immediately from $\boldsymbol{\Phi}^j = P\Lambda^j P^{-1}$ and $\lambda_i^j \to 0$. The result extends to non-diagonalisable matrices via the Jordan normal form, where each Jordan block of size m satisfies

$$\begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}^j = \begin{pmatrix} \lambda^j & j\,\lambda^{j-1} \\ 0 & \lambda^j \end{pmatrix}, \tag{14}$$

and the polynomial prefactor $j\,\lambda^{j-1}$ is dominated by the geometric decay of $|\lambda|^{j-1}$ whenever $|\lambda| < 1$.

The decay of $\boldsymbol{\Phi}^j$ has two consequences for the simulation. First, it ensures convergence of the

MA(∞) representation

$$\tilde{\mathbf{x}}_t = \sum_{j=0}^{\infty} \boldsymbol{\Phi}^j \, \boldsymbol{\varepsilon}_{t-j}, \qquad \tilde{\mathbf{x}}_t := \mathbf{x}_t^{(\text{macro})} - \boldsymbol{\mu}, \tag{15}$$

so that the process has finite variance and the initial condition is eventually forgotten. Second, it ensures that the long-run mean

$$\boldsymbol{\mu} = (I - \boldsymbol{\Phi})^{-1} \, \mathbf{c} \tag{16}$$

is well defined, anchoring the simulated macro states to economically interpretable levels.

Moment structure.

From the MA(∞) expansion and serial independence of shocks, the contemporaneous covariance matrix $\boldsymbol{\Gamma}_0 = \operatorname{Cov}(\mathbf{x}_t^{(\text{macro})})$ admits the infinite-sum representation

$$\boldsymbol{\Gamma}_0 = \sum_{j=0}^{\infty} \boldsymbol{\Phi}^j \, \boldsymbol{\Sigma} \left(\boldsymbol{\Phi}^j\right)^{\top}, \tag{17}$$

which converges under (12). Equivalently, $\boldsymbol{\Gamma}_0$ is the unique solution of the discrete Lyapunov equation

$$\boldsymbol{\Gamma}_0 = \boldsymbol{\Phi} \, \boldsymbol{\Gamma}_0 \, \boldsymbol{\Phi}^{\top} + \boldsymbol{\Sigma}. \tag{18}$$

In practice, $\boldsymbol{\Gamma}_0$ is computed by solving (18) directly rather than truncating (17). Lagged covariances follow from $\boldsymbol{\Gamma}_h = \boldsymbol{\Phi}^h \, \boldsymbol{\Gamma}_0$ for $h \geq 0$.

Two useful moment identities that arise from the MA(∞) structure are

$$\mathbb{E}\left[\tilde{\mathbf{x}}_t \, \boldsymbol{\varepsilon}_t^{\top}\right] = \boldsymbol{\Sigma}, \qquad \mathbb{E}\left[\tilde{\mathbf{x}}_{t-j} \, \boldsymbol{\varepsilon}_t^{\top}\right] = 0 \quad \text{for } j > 0. \tag{19}$$

The first states that the current state is contemporaneously correlated with the current shock through the innovation covariance; the second states that past states carry no information about future shocks.

Unconstrained parameterisation of $\boldsymbol{\Phi}$.

During estimation and scenario optimisation the coefficient matrix $\boldsymbol{\Phi}$ is updated by gradient-based methods. A naïve unconstrained optimisation may produce iterates that violate (12), yielding an explosive process. We therefore adopt an unconstrained parameterisation that maps $\mathbb{R}^d$ to the set of stable matrices, ensuring stationarity by construction. The approach rests on the singular value decomposition, which in turn requires two matrix-algebraic ingredients that we briefly recall.

Orthogonal matrices. A square matrix $Q \in \mathbb{R}^{K \times K}$ is *orthogonal* if $Q^{\top} Q = Q Q^{\top} = I$, i.e. its transpose equals its inverse. Geometrically, multiplication by Q is a rotation or reflection: it preserves lengths and angles. Every eigenvalue of an orthogonal matrix has modulus exactly 1.

Skew-symmetric matrices. A square matrix $A \in \mathbb{R}^{K\times K}$ is *skew-symmetric* if $A^\top = -A$. This forces all diagonal entries to be zero and the off-diagonal entries to satisfy $a_{ij} = -a_{ji}$, so a $K\times K$ skew-symmetric matrix is determined by its $K(K-1)/2$ strictly upper-triangular entries. A fundamental property is that the matrix exponential of a skew-symmetric matrix is always orthogonal:

$$A^\top = -A \quad\Longrightarrow\quad \exp(A)^\top \exp(A) = \exp(-A)\exp(A) = I. \tag{20}$$

This provides a smooth, surjective map from $\mathbb{R}^{K(K-1)/2}$ (the free entries of A) onto the orthogonal group, which is the key ingredient for an unconstrained parameterisation.

Singular value decomposition (SVD). Any real matrix $\mathbf{\Phi} \in \mathbb{R}^{K\times K}$ admits a factorisation

$$\mathbf{\Phi} = U\,D\,V^\top, \tag{21}$$

where $U, V \in \mathbb{R}^{K\times K}$ are orthogonal and $D = \mathrm{diag}(\sigma_1, \ldots, \sigma_K)$ is a diagonal matrix of non-negative *singular values*. Geometrically, the SVD decomposes $\mathbf{\Phi}$ into a rotation ($V^\top$), a coordinate-wise scaling (D), and a second rotation (U). The orthogonal factors U and V preserve vector norms (since $\|Qx\| = \|x\|$ for any orthogonal Q), so the maximum amount by which $\mathbf{\Phi}$ can stretch any vector is governed entirely by D:

$$\|\mathbf{\Phi}\,x\| = \|UDV^\top x\| = \|DV^\top x\| \le \sigma_{\max}\,\|V^\top x\| = \sigma_{\max}\,\|x\|, \tag{22}$$

where $\sigma_{\max} = \max_i \sigma_i$ is the largest singular value. Now suppose λ is an eigenvalue of $\mathbf{\Phi}$ with eigenvector $v \neq 0$. Then $\mathbf{\Phi}v = \lambda v$, so $|\lambda|\,\|v\| = \|\mathbf{\Phi}v\| \le \sigma_{\max}\,\|v\|$, giving

$$|\lambda| \le \sigma_{\max}. \tag{23}$$

Consequently, if every singular value satisfies $\sigma_i < 1$, then $\sigma_{\max} < 1$, and (23) implies $|\lambda| < 1$ for every eigenvalue. This is a sufficient (though not necessary) condition for (12), and it is the reason the SVD parameterisation guarantees stability: bounding the diagonal scaling automatically bounds the eigenvalue moduli, regardless of the choice of orthogonal factors U and V.

Using these ingredients, we parameterise each component of the SVD as follows.

Singular values. Each singular value is expressed as

$$\sigma_i = \tanh(\theta_i), \qquad \theta_i \in \mathbb{R}, \tag{24}$$

where $\tanh : \mathbb{R} \to (-1, 1)$. Because the image of tanh is bounded by 1, we have $\sigma_i < 1$ for any unconstrained θ_i, and the stationarity constraint is automatically satisfied.

Orthogonal factors. The orthogonal matrices U and V are parameterised using (20):

$$U = \exp(A_U), \qquad V = \exp(A_V), \tag{25}$$

where A_U and A_V are skew-symmetric matrices, each constructed from $K(K-1)/2$ unconstrained real parameters that fill the strictly upper-triangular entries. By (20), U and V are guaranteed to be orthogonal for any choice of these parameters.

Total parameter count. The unconstrained parameter vector for $\mathbf{\Phi}$ has dimension

$$\underbrace{K}_{\text{singular values}} + \underbrace{2 \times \frac{K(K-1)}{2}}_{\text{two skew-symmetric matrices}} = K^2, \tag{26}$$

matching the number of free entries in a general $K \times K$ matrix. The full reconstruction is

$$\mathbf{\Phi}(\boldsymbol{\theta}) = \exp\big(A_U(\boldsymbol{\theta}_U)\big) \operatorname{diag}\big(\tanh(\theta_1), \ldots, \tanh(\theta_K)\big) \exp\big(A_V(\boldsymbol{\theta}_V)\big)^\top, \tag{27}$$

which is differentiable with respect to the full unconstrained parameter vector $\boldsymbol{\theta} = (\boldsymbol{\theta}_U, \boldsymbol{\theta}_\Sigma, \boldsymbol{\theta}_V)$. Every component, the matrix exponential, tanh, and matrix multiplication, admits well-defined gradients, so standard automatic-differentiation machinery applies directly.

Note. We acknowledge that several other approaches exist for enforcing VAR stationarity during unconstrained optimization, including eigendecomposition-based methods using the Cayley transform, Givens rotation constructions, and projection-based approaches. Each involves a different trade-off between coverage of the stable set, computational cost, and numerical robustness.

Unconstrained parameterisation of $\mathbf{\Sigma}$.

The innovation covariance $\mathbf{\Sigma}$ must be symmetric positive definite. We parameterise $\mathbf{\Sigma}$ through its Cholesky decomposition: every symmetric positive definite matrix admits a unique factorisation $\mathbf{\Sigma} = LL^\top$, where L is lower-triangular with strictly positive diagonal entries. To remove the positivity constraint on the diagonal, we write $L_{ii} = \exp(\ell_i)$ for unconstrained $\ell_i \in \mathbb{R}$, and leave the $K(K-1)/2$ strictly lower-triangular entries of L unconstrained. Because $\exp(\ell_i) > 0$ for all ℓ_i, the product $LL^\top$ is positive definite for any parameter value. Together with (27), this places the entire VAR(1) estimation problem in unconstrained parameter space.

4.4 Policy Modelling

Section 4.1.3 introduced policy variables as the modelling layer that translates the economic state into managerial decisions. The general form (7) is deliberately flexible, accommodating deterministic rules, stochastic policies, and discrete gates. In this subsection we describe the concrete policy specifications used in the simulator, progressing from simple univariate rules to structured multi-objective formulations.

The key design principle is that every policy must respect the DAG ordering of Section 4.2. A policy at time t may depend on the state $\mathbf{s}_t$ (which includes lagged stocks and contemporaneous macro variables) but not on any endogenous outcome realised at t. This ensures that each simulation step remains a feed-forward computation, regardless of the policy's internal complexity.

4.4.1 Basic Policies

The simplest policy specifications are univariate rules that map the state to a single decision variable. All basic policies share a common *linear core*: the affine-plus-noise form (7) produces an unconstrained real-valued score

$$z_t^{(i)} = \alpha^{(i)} + \big(\boldsymbol{\gamma}^{(i)}\big)^\top \mathbf{s}_t + \eta_t^{(i)}, \quad \eta_t^{(i)} \sim \mathcal{N}\big(0,\, \sigma_\eta^{(i)2}\big), \tag{28}$$

where $\alpha^{(i)}$ is a baseline intercept, $\boldsymbol{\gamma}^{(i)}$ is a coefficient vector encoding sensitivity to each state variable, and $\eta_t^{(i)}$ captures idiosyncratic variation not explained by the observable state. The parameters $(\alpha^{(i)}, \boldsymbol{\gamma}^{(i)}, \sigma_\eta^{(i)})$ are estimated from historical data or set by the user for scenario analysis.

Different economic decisions require different output domains. Rather than designing separate models, we compose the linear core with an output transformation chosen to match the domain constraint. This keeps the parameterisation uniform, every policy is ultimately driven by the same triple $(\alpha^{(i)}, \boldsymbol{\gamma}^{(i)}, \sigma_\eta^{(i)})$, while ensuring that simulated values are economically valid.

Linear policy ($p \in \mathbb{R}$).

When the decision variable is naturally unbounded, for example an absolute dollar change in a reserve account, no transformation is needed:

$$p_t^{(i)} = z_t^{(i)}. \tag{29}$$

Non-negative policy ($p \in \mathbb{R}_+$).

For quantities that cannot be negative, such as capital expenditure or gross debt issuance, we apply the exponential map $p_t^{(i)} = \exp(z_t^{(i)})$.

Ratio policy ($p \in [0, 1]$).

Decisions naturally expressed as a proportion of a reference quantity, such as the dividend payout ratio or the reinvestment rate, are obtained by applying the sigmoid function

$$\rho_t^{(i)} = \operatorname{sigmoid}\big(z_t^{(i)}\big) = \frac{1}{1 + e^{-z_t^{(i)}}}, \tag{30}$$

and multiplying by a reference quantity $R_t^{(i)}$ (a stock or flow already determined at time t) to give $p_t^{(i)} = \rho_t^{(i)} \times R_t^{(i)}$.

Bernoulli gate ($p \in \{0, 1\}$).

Discrete actions, such as whether to issue new debt or whether to pay a dividend, are modelled by drawing a Bernoulli sample whose activation probability is the sigmoid ratio from (30):

$$p_t^{(i)} \sim \operatorname{Bernoulli}\big(\rho_t^{(i)}\big), \qquad \rho_t^{(i)} = \operatorname{sigmoid}\big(z_t^{(i)}\big). \tag{31}$$

When the gate fires ($p_t^{(i)} = 1$) a downstream flow is triggered; otherwise the flow is zero. This captures lumpy corporate actions that cannot be meaningfully represented by a continuous policy.

4.4.2 Partial-Adjustment Policy

Motivation.

The basic policies above are purely *forward-looking* in that each decision is a function of the current state $\mathbf{s}_t$, with no explicit reference to where the firm "should" be in terms of financial structure. In practice, firms often operate with implicit or explicit targets for key financial ratios, such as a target leverage, a target interest coverage, or a target liquidity buffer, and adjust their decisions gradually toward these targets over time. The empirical corporate finance literature documents this behaviour extensively. Firms do not jump to their target capital structure instantaneously but instead close only a fraction of the gap each period, reflecting adjustment costs, market timing, and managerial inertia (Flannery and Rangan, 2006).

Embedding target-seeking behaviour directly in the policy layer is important for two reasons. First, it produces more realistic simulated trajectories. Without it, the simulator behaviours can be erratic and unrealistic. Second, it provides a natural scenario-analysis interface. A user can specify "the firm targets a debt-to-equity ratio of 0.5 with a half-life of 3 years" and observe the resulting financial statement dynamics, without manually engineering the flow-level decisions that implement this target.

Formulation.

Let r_t denote a financial ratio at time t and r_t^* the target value of that ratio. The partial-adjustment model specifies that the ratio evolves as

$$r_t - r_{t-1} = \lambda \left(r_t^* - r_{t-1}\right) + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \sigma^2), \tag{32}$$

where $\lambda \in (0, 1)$ is the *speed of adjustment*. Rearranging gives the equivalent form

$$r_t = \lambda\, r_t^* + (1 - \lambda)\, r_{t-1} + \varepsilon_t, \tag{33}$$

which is a convex combination of the target and the lagged ratio, perturbed by noise. The parameter λ controls the speed at which the firm closes the gap to its target. Setting $\lambda = 1$ implies immediate adjustment within one period, while λ close to 0 implies near-complete inertia. The stability condition $|1 - \lambda| < 1$ is satisfied for any $\lambda \in (0, 1)$, ensuring the ratio mean-reverts toward r_t^* rather than diverging.

As a concrete example, consider the market leverage ratio $r_t^{\text{lev}} = D_t/(D_t + S_t P_t)$, where D_t is book debt, S_t is shares outstanding, and P_t is the share price. If the firm targets leverage $r_t^{\text{lev},*}$, the partial-adjustment dynamics are

$$r_t^{\text{lev}} = \lambda\, r_t^{\text{lev},*} + (1 - \lambda)\, r_{t-1}^{\text{lev}} + \varepsilon_t. \tag{34}$$

Connecting ratios to flows.

Equation (33) operates at the ratio level, but the simulator's accumulation equations (3) are driven by flows. To bridge this gap, we translate the ratio target into a stock target and then derive the required flow. Continuing the leverage example, the target debt stock implied by the target leverage and current equity value is

$$D_t^* = \frac{r_t^{\mathrm{lev},*}}{1 - r_t^{\mathrm{lev},*}}\,(S_t P_t). \tag{35}$$

The net debt issuance (a flow) that partially adjusts toward this target is

$$I_t = \lambda\,(D_t^* - D_{t-1}) + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0,\, \sigma^2), \tag{36}$$

and the debt stock updates via the standard accumulation rule:

$$D_t = D_{t-1} + I_t = \lambda\, D_t^* + (1-\lambda)\, D_{t-1} + \varepsilon_t. \tag{37}$$

The structure of (37) mirrors (33) at the stock level, confirming internal consistency. Importantly, this formulation respects the DAG ordering because D_t^* is computed from $r_t^{\mathrm{lev},*}$ and the equity value at time t, both of which are known before the flow I_t is determined.

4.4.3 Multi-Objective Policy

Motivation.

The partial-adjustment policy of Section 4.4.2 targets a single financial ratio at a time. In practice, firms simultaneously manage multiple, potentially conflicting objectives. A firm may wish to maintain a low debt-to-equity ratio (reducing financial risk) while also maintaining a high interest coverage ratio (signalling creditworthiness). These two objectives may conflict. Reducing debt improves coverage but raises the equity required to fund investment, while issuing debt to fund growth improves return on equity but worsens coverage. When objectives conflict, no single policy can minimise all deviations simultaneously, and the firm must navigate a trade-off.

This is the domain of *multi-objective optimisation*. Rather than producing a single optimal policy, it characterises the set of *Pareto-efficient* policies: those for which no objective can be improved without worsening at least one other. Incorporating multi-objective awareness into the policy layer prevents the simulator from implicitly prioritising one ratio over another. It instead exposes the full trade-off surface, allowing the user to select an operating point that reflects the firm's risk appetite and strategic priorities.

Formulation.

Let $\mathbf{p}_t = (p_t^{(1)}, \ldots, p_t^{(m)})^\top$ denote the vector of policy (decision) variables at time t, e.g., debt issued, debt repaid, and dividend payout ratio. Let $\mathbf{r}(\mathbf{p}_t) = (r_t^{(1)}, \ldots, r_t^{(n)})^\top$ denote the resulting financial ratios, computed from the updated stocks after applying $\mathbf{p}_t$ through the accumulation equations, and let $\mathbf{r}^* = (r^{(1),*}, \ldots, r^{(n),*})^\top$ denote the corresponding targets. The multi-objective problem is

$$\min_{\mathbf{p}_t \in \mathcal{P}} \left(\left| r_t^{(1)}(\mathbf{p}_t) - r^{(1),*} \right|, \; \ldots, \; \left| r_t^{(n)}(\mathbf{p}_t) - r^{(n),*} \right| \right), \tag{38}$$

where $\mathcal{P}$ is the feasible set of policies (e.g. $p_t^{(\text{issued})} \geq 0$, $0 \leq p_t^{(\text{d})} \leq 1$), and the minimisation is understood in the Pareto sense. A policy $\mathbf{p}^A$ dominates $\mathbf{p}^B$ if it achieves a weakly smaller deviation on every objective and a strictly smaller deviation on at least one.

As a concrete example, consider a firm with three policy variables, net debt issuance $p_t^{(\text{issued})}$, debt repayment $p_t^{(\text{repaid})}$, and dividend payout ratio $p_t^{(\text{d})}$, and two ratio targets, debt-to-equity

$$r_t^{(\text{dte})} = \frac{D_0 + p_t^{(\text{issued})} - p_t^{(\text{repaid})}}{E_0 + \text{EBIT}\,(1 - p_t^{(\text{d})})} \tag{39}$$

and interest coverage

$$r_t^{(\text{icr})} = \frac{\text{EBIT}}{r_{\text{int}} \times (D_0 + p_t^{(\text{issued})} - p_t^{(\text{repaid})})}, \tag{40}$$

where D_0 and E_0 are beginning-of-period debt and equity, EBIT is earnings before interest and taxes, and r_{int} is the interest rate. The numerator of (39) increases with net issuance, while the denominator of (40) also increases with net issuance. Issuing more debt worsens coverage but may improve return on equity. The Pareto frontier of (38) traces out the achievable combinations of $|r_t^{(\text{dte})} - r^{(\text{dte}),*}|$ and $|r_t^{(\text{icr})} - r^{(\text{icr}),*}|$ under the feasibility constraints, making the conflict between the two objectives explicit and quantifiable.

Integration with the simulator.

In practice, the multi-objective formulation serves as a *policy-generation* step. Given the state $\mathbf{s}_t$, one solves (38) to obtain a policy vector $\mathbf{p}_t$ on or near the Pareto frontier, which then enters the flow equations and accumulation rules as before. The user selects a point on the frontier, via a preference vector or a weighted-sum scalarisation, corresponding to the firm's relative priority among the objectives. The partial-adjustment framework of Section 4.4.2 can be layered on top, rather than jumping to the Pareto-optimal policy in one step. The firm adjusts gradually, with the Pareto solution serving as the target toward which the partial-adjustment dynamics converge.

5 Financial Dynamics Modelling

5.1 Working Capital

Working capital is a measure of a company's short-term liquidity and operational efficiency. It represents the difference between a company's current assets and current liabilities.

$$\text{Working Capital} = \text{Current Assets} - \text{Current Liabilities} \tag{41}$$

Common examples of current assets

- Cash and cash equivalents
- Inventory
- Trade receivables
- Prepayments (excluded)
- Assets held for sale (excluded)
- Other (excluded)

Common examples of current liabilities

- Trade payables
- Short-term debt (excluded)
- Accrued expenses (excluded)
- Current portion of long-term debt (excluded)
- Other (excluded)

This work will focus on the working capital components that are directly related to the company's operations, such as inventory, trade receivables, and trade payables.

5.2 Terminology

Net realisable value The estimated selling price in the ordinary course of business less the estimated costs of completion and the estimated costs necessary to make the sale

Fair Value The price that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date.

Net realisable value refers to the net amount that an entity expects to realise from the sale of inventory in the ordinary course of business. Fair value reflects the price at which an orderly transaction to sell the same inventory in the principal (or most advantageous) market for that inventory would take place between market participants at the measurement date. The former is an entity-specific value; the latter is not. Net realisable value for inventories may not equal fair value less costs to sell

5.3 Literature

5.3.1 Cost of Inventory

Inventories are assets (1) held for sale in the ordinary course of business, (2) in the process of production for such sale, or (3) in the form of materials or supplies to be consumed in the production process or in the rendering of services.

The cost of inventories comprises: (1) cost of purchase, (2) cost of conversion, and (3) other costs incurred in bringing the inventories to their present location and condition.

Cost of purchase

The cost of purchase is the purchase price plus non-recoverable taxes and directly attributable acquisition costs such as transport and handling, net of trade discounts, rebates and other similar items.

Cost of conversion

It comprises direct costs directly attributable to units of production, such as direct labour, together with a systematic allocation of production overheads incurred in transforming materials into finished goods. Production overheads include fixed and variable components.

Overhead classification.

Fixed production overheads are indirect costs that remain relatively constant regardless of production volume, for example depreciation and maintenance of factory buildings, equipment and right-of-use assets used in the production process, and the cost of factory management and administration. Variable production overheads are indirect costs that vary directly, or nearly directly, with production volume, such as indirect materials and indirect labour.

Allocation of overheads.

Fixed production overheads are allocated based on the normal capacity of the production facilities. Normal capacity is the average production expected over multiple periods under normal circumstances, taking into account planned maintenance. The actual level of production may be used if it approximates normal capacity. The fixed overhead allocated to each unit is not increased in periods of low production or idle plant; any unallocated overhead is recognised as an expense in the period. In periods of abnormally high production, the fixed overhead allocated per unit is decreased so that inventories are not measured above cost. Variable production overheads are allocated to each unit based on the actual use of the production facilities.

Multiple products.

When a process yields more than one product and conversion costs for each product are not separately identifiable, total conversion costs are allocated on a rational and consistent basis, commonly by relative sales value at the split-off point (when products become separately identifiable) or at completion.

By-products

Most by-products are immaterial. They are often measured at net realisable value, and this amount is deducted from the cost of the main product so that the carrying amount of the main product is not materially different from its cost.

Other costs

Other costs are included in inventory cost only when they are necessary to bring inventories to their present location and condition; examples include directly attributable non-production overheads and essential customer-specific design costs. Costs that do not contribute to readiness are expensed as incurred, including abnormal waste, storage costs not required before a further production stage, administrative overheads that do not ready inventories, and selling costs. Borrowing costs are included in inventory cost only when inventories meet the qualifying asset criteria in IAS 23; otherwise they are recognised as an expense. When inventories are purchased on deferred settlement terms with a financing element, measure inventories at the cash price equivalent and recognise the financing component separately as interest over the deferral period.

Techniques for measurement of cost (optional)

Standard cost and retail methods may be used if they approximate actual cost. Standard costs reflect normal levels of materials, labour, efficiency and capacity utilisation and are reviewed regularly. The retail method derives cost by reducing selling price by the appropriate percentage gross margin.

5.3.2 Cost formulas

Use specific identification for items that are not ordinarily interchangeable or are segregated for specific projects. For interchangeable inventories, use FIFO or weighted average and apply the same cost formula to inventories with similar nature and use. Differences in geographical location or tax regime alone do not justify different formulas. FIFO assumes the earliest purchases or production are sold first, leaving the most recent costs in ending inventory. Weighted average smooths unit costs and may be calculated periodically or as a moving average.

Notes: Discussion of net realisable value and write-downs is deferred to a later section.

5.3.3 Trade receivables

Initial recognition and measurement

Trade receivables are initially measured at the transaction price in accordance with IFRS 15 when they do not contain a significant financing component or when the IFRS 15 practical expedient is applied. If a significant financing component exists, initial measurement follows the general IFRS 9 approach at fair value plus or minus transaction costs, as applicable.

Subsequent measurement

For trade receivables without a significant financing component, IFRS 9 requires the simplified approach, under which the loss allowance is measured at lifetime expected credit losses. For trade receivables with a significant financing component, an entity may elect an accounting policy to apply lifetime expected credit losses; if elected, this policy is applied consistently to those receivables. Trade receivables are typically measured at amortised cost.

Entities may use the practical expedient of a provision matrix to estimate expected credit losses, using historical loss experience adjusted for current and forward-looking information. Applying fixed provision rates by days past due is common, and receivables should be segmented when loss patterns differ across customer groups, such as by region, product, customer rating, collateral or trade credit insurance, and customer type.

Notes: we deferred the discussion of significant increase in credit risk assessments, expected credit loss mechanics, write-off criteria, and related disclosure requirements to future work.

5.3.4 Trade payables

Initial recognition and measurement

Trade payables are financial liabilities within the scope of IFRS 9. Recognise a trade payable when, and only when, the entity becomes party to the contractual provisions of the instrument. At initial recognition, measure the trade payable at its fair value plus transaction costs if it is not designated at fair value through profit or loss. For typical short-term, non-interest-bearing supplier invoices, fair value generally equals the invoiced amount net of discounts and rebates.

Subsequent measurement

After initial recognition, measure trade payables at amortised cost in accordance with IFRS 9 unless they meet the criteria to be, or are irrevocably designated as, measured at fair value through profit or loss. Interest expense is recognised using the effective interest method when a financing element exists. For short-term, non-interest-bearing payables, the amortised cost typically approximates the invoiced amount.

Modification and exchange of terms

When the terms of a trade payable are substantially modified, account for the change as an extinguishment of the original liability and the recognition of a new financial liability. A common indicator of a substantial modification is when the discounted present value of the revised cash flows differs by at least 10 per cent from that of the remaining original cash flows, using the original effective interest rate. If the modification is not substantial, adjust the carrying amount to the present value of the modified cash flows discounted at the original effective interest rate and recognise the resulting gain or loss in profit or loss.

Derecognition

Derecognise a trade payable when, and only when, the obligation is extinguished, that is, when it is discharged, cancelled, or expires. On derecognition, recognise in profit or loss the difference between the carrying amount of the liability (or part thereof) and the consideration paid, including any non-cash assets transferred or new liabilities assumed. If only a part of a payable is repurchased or settled, allocate the previous carrying amount between the part derecognised and the part retained on a relative fair value basis at the date of the transaction.

Gains and losses

For trade payables measured at amortised cost and not part of a hedging relationship, recognise gains and losses in profit or loss through the amortisation process and upon derecognition. If a trade payable is designated at fair value through profit or loss, present the effects of changes in the liability's credit risk in other comprehensive income unless doing so creates or enlarges an accounting mismatch in profit or loss, recognise the remaining fair value changes in profit or loss.

Notes: We deferred the discussion of FX gains and losses to our future work.

5.3.5 Cash Conversion Cycle

It is a conceptual framework that helps to understand the time it takes for a company to convert its investments in inventory and other resources into cash flows from sales. It consists of three main components: Days Inventory Outstanding (DIO), Days Sales Outstanding (DSO), and Days Payables Outstanding (DPO). The formula for calculating the Cash Conversion Cycle is as follows:

$$\text{Cash Conversion Cycle} = \text{DIO} + \text{DSO} - \text{DPO} \tag{42}$$

5.4 Modelling

We borrow the concept of the cash conversion cycle to model the working capital components to help us understand the timing of cash flows related to inventory, trade receivables, and trade payables.

Assumptions: Annual setting, inventory is only a block (excluding the FIFO mechanism), no detailed inventory costing,no prepayment, no assets held for sale, no other current assets, no short-term debt, no accrued expenses, no current portion of long-term debt, and no other current liabilities.

Variables

$g_t \geq 0:$ gross inventory at end of year t

$v_t \geq 0:$ inventory valuation allowance at end of year t

$inv_t \geq 0:$ net inventory at end of year t $(= g_t - v_t)$

$AR_t \geq 0:$ trade receivables at end of year t

$AP_t \geq 0:$ trade payables at end of year t

Drivers

$R_t \geq 0:$ revenue at year t

$C_t \geq 0:$ cost of sales at cost at year t

$DIO_t \geq 0:$ days inventory outstanding at year t

$DSO_t \geq 0:$ days sales outstanding at year t

$DPO_t \geq 0:$ days payables outstanding at year t

$a_t^S \in [0, 1]:$ credit sales share at year t

$a_t^P \in [0, 1]:$ credit purchase share at year t

$q_t \in [0, 1]:$ NRV factor indicator at year t

5.4.1 System Dynamics

Split revenue into cash vs credit

We split revenue R_t into cash and credit portions based on the credit sales share a_t^S:

$$S_t^C = a_t^S\, R_t, \quad S_t^{Cash} = (1 - a_t^S)\, R_t$$

where S_t^C is credit sales and S_t^{Cash} is cash sales.

Inventory purchases, inventory roll-forward, and valuation (block inventory)

We model inventory in gross cost terms with a separate valuation allowance. Let g_{t-1} and v_{t-1} denote beginning-of-year gross inventory and allowance. Given annual cost of sales at cost C_t

and the inverted policy driver DIO_t, the target end-of-year *gross* inventory at cost is

$$g_t^\star = \frac{DIO_t}{365}\, C_t. \tag{43}$$

Planned purchases are then the amount required to (i) cover current-period cost of sales and (ii) move the inventory position toward the target:

$$P_t^{raw} = C_t + (g_t^\star - g_{t-1})\,, \tag{44}$$

and actual purchases are constrained to be non-negative:

$$P_t = \max(0, P_t^{raw}). \tag{45}$$

Define the post-activity (pre-valuation) gross inventory balance as the standard roll-forward:

$$g_t^{post} = g_{t-1} + P_t - C_t. \tag{46}$$

When $P_t = 0$, the model mechanically implies an inventory drawdown, $g_t^{post} = g_{t-1} - C_t$.

Split purchases into cash vs credit

Given total purchases P_t, we split them into cash and credit portions based on the credit purchase share a_t^P:

$$U_t^C = a_t^P\, P_t, \quad U_t^{Cash} = (1 - a_t^P)\, P_t$$

where U_t^C is credit purchases (creating trade payables) and U_t^{Cash} is cash purchases (immediate cash outflow).

Allowance roll-forward and cost of sales (net of allowance release)

To connect inventory valuation effects to both the balance sheet and the income statement, we release a proportion of the opening allowance as the opening inventory is consumed. Let the opening impairment ratio be

$$\phi_{t-1} = \begin{cases} \dfrac{v_{t-1}}{g_{t-1}}, & g_{t-1} > 0, \\ 0, & g_{t-1} = 0. \end{cases} \tag{47}$$

We approximate the portion of current-period cost of sales sourced from opening inventory as

$$C_t^{old} = \min(g_{t-1}, C_t). \tag{48}$$

The allowance released with the sale of opening inventory is

$$Rel_t = \phi_{t-1}\, C_t^{old}, \tag{49}$$

so the allowance remaining after this release (but before the year-end valuation test) is

$$v_t^{pre} = v_{t-1} - Rel_t. \tag{50}$$

Accordingly, cost of sales attributable to inventory, net of allowance release, is

$$COS_t^{inv} = C_t - Rel_t. \tag{51}$$

Year-end net realisable value (NRV) test and valuation effects

We implement a reduced-form NRV mechanism using an indicator/discount factor $q_t \in [0, 1]$. The implied NRV of ending *gross* inventory is

$$NRV_t = q_t \, g_t^{post}. \tag{52}$$

The year-end valuation allowance is then

$$v_t = \max(0, g_t^{post} - NRV_t) = \max\bigl(0, (1 - q_t)g_t^{post}\bigr), \tag{53}$$

and net inventory carried on the balance sheet is

$$inv_t = g_t^{post} - v_t = \min(g_t^{post}, NRV_t). \tag{54}$$

We decompose the change in allowance into write-downs and reversals:

$$WD_t = \max(0, v_t - v_t^{pre}), \tag{55}$$

$$Rev_t = \max(0, v_t^{pre} - v_t), \tag{56}$$

so that the net valuation expense recognised in profit or loss is

$$ValExp_t = WD_t - Rev_t = v_t - v_t^{pre}. \tag{57}$$

Total inventory-related expense in the period is therefore

$$InvExp_t = COS_t^{inv} + ValExp_t. \tag{58}$$

Trade receivables and payables targets, constraints, and implied cash flows

Credit sales and credit purchases are S_t^C and U_t^C as defined above. Using DSO_t and DPO_t, the unconstrained end-of-year targets are

$$AR_t^\star = \frac{DSO_t}{365} \, S_t^C, \qquad AP_t^\star = \frac{DPO_t}{365} \, U_t^C. \tag{59}$$

To avoid infeasible states (i.e., receivables or payables exceeding the available "stock"), we impose the budget constraints

$$AR_t \leq AR_{t-1} + S_t^C, \qquad AP_t \leq AP_{t-1} + U_t^C, \tag{60}$$

and implement them with clipping:

$$AR_t = \min(AR_t^\star, AR_{t-1} + S_t^C), \qquad AP_t = \min(AP_t^\star, AP_{t-1} + U_t^C). \tag{61}$$

These stock-flow identities imply customer collections and supplier payments:

$$Col_t = AR_{t-1} + S_t^C - AR_t \geq 0, \qquad Pay_t = AP_{t-1} + U_t^C - AP_t \geq 0. \tag{62}$$

End-of-period state vector and financial statement links

The end-of-year gross inventory state is

$$g_t \equiv g_t^{post} = g_{t-1} + P_t - C_t, \tag{63}$$

and the working-capital balance sheet items are

$$inv_t^{BS} = inv_t = g_t - v_t, \qquad AR_t^{BS} = AR_t, \qquad AP_t^{BS} = AP_t. \tag{64}$$

Under a direct-method operating cash flow view consistent with the cash/credit split, cash receipts from customers and cash paid to suppliers are

$$Cashin_t^{cust} = S_t^{Cash} + Col_t, \qquad Cashout_t^{sup} = U_t^{Cash} + Pay_t. \tag{65}$$

6 Inventory Modelling

6.1 Scope

Goal: Model the inventory age structure and its evolution over time under FIFO mechanics, using a distributional approach.

Note: we skip the driver discussion (inventory management decision, managerial accounting, financial shenanigans, etc.)

6.2 Accounting Framework

Inventories are assets (i) held for sale in the ordinary course of business, (ii) in the process of production for such sale, or (iii) in the form of materials or supplies to be consumed in the production process or in the rendering of services. Under IFRS standards, the cost of inventory includes the cost of purchase, cost of conversion, and other costs incurred in bringing the inventory to its present location and condition.

Once the cost is recognized, entities must choose a cost formula (e.g., FIFO, weighted average)[2] to allocate costs between COGS and ending inventory. With different cost formula assumptions, the COGS and ending inventory valuation can differ. For example, under inflation, FIFO policy records the COGS at older, lower costs and ending inventory at newer, higher costs, resulting in higher reported profits and inventory values compared to LIFO or weighted average. Therefore, the choice of cost formula has implications for financial reporting, tax liabilities, and business decision-making. It is important to model the inventory to support adjustments for different cost formula assumptions.

Let us begin with FIFO, which is the most common cost formula. Under FIFO, the oldest inventory items are sold first, and the ending inventory consists of the most recently purchased items. This means that the age structure of inventory is crucial and the simplest way to model it is to create a vintage structure, where inventory is categorized into discrete age buckets (e.g., 0–1 year, 1–2 years, etc.). **However, this approach can lead to a high-dimensional state space if there are many age buckets.**

6.3 Literature

There are many approaches to model inventory dynamics in the field of inventory control.

Fluid models

Inventory control is trying to understand and determine (i) when to order/produce, (ii) how much to order/produce, (iii) how to adjust the price (dynamic pricing), etc. to optimize some objective (e.g., profit, service level, etc.). Fluid models are a common approach to model inventory dynamics, where the inventory level is treated as a continuous variable that evolves over time according to some deterministic or stochastic dynamics. The fluid model can capture

[2] LIFO exists but is not permitted under IFRS.

the aggregate behavior of inventory and can be used to derive optimal policies for inventory control.

By modeling inventory as a fluid, it avoids the curse of dimensionality associated with discrete inventory levels. The common fluid model is the Economic Order Quantity (EOQ) model, and its extensions—stochastic fluid model and queueing model for inventory. However, these models typically do not capture the age structure of inventory, which is crucial for FIFO mechanics.

Perishable inventory control

Depending on the nature of the business, inventory can be classified as perishable (e.g., food, pharmaceuticals) or non-perishable (e.g., electronics, clothing). Perishable inventory has a limited shelf life and can become obsolete or unsellable after a certain period. Knowing the age structure of perishable inventory is crucial for effective inventory management, as it allows businesses to prioritize the sale of older inventory and minimize waste. This setting is closely related to the FIFO mechanics, as both require tracking the age structure of inventory.

In the early literature for perishable inventory control (Nahmias, 1982), the age structure is modelled as a state vector of discrete age buckets representing the remaining-life. However, this approach can lead to a high-dimensional state space if there are many age buckets when they are trying to solve the optimal control problem. The curse of dimensionality commonly arises in perishable inventory control problems (Nomura et al., 2025)

Some tricks to solve the curse of dimensionality include (i) approximate dynamic programming (Powell, 2007), (ii) approximate algorithm (Zhang et al., 2016) which avoids iterating over the state space, (iii) heuristics approach to reduce the search dimension (Gutierrez-Alcoba et al., 2017). However, most of these approaches are designed to reduce the search space and do not reduce the age-structure dimension.

6.4 Terminology

FIFO

FIFO operates on physical units (oldest units sold first), so the simulator tracks physical quantities. Cost values for financial reporting are computed by integrating the quantity distribution against the price index.

Weighted Average Cost (WAC)

WAC operates on a cost basis, where the cost of goods sold and ending inventory are calculated based on the average cost of all units available for sale during the period. This method smooths out price fluctuations over time.

6.5 FIFO Age Structure

Under FIFO, one must track the age composition of inventory, because older units are sold first and their historical purchase costs determine COGS. The natural modeling idea is to represent the inventory state as a distribution over discrete age buckets and to estimate that distribution from observed financial data. This section explains why that natural approach fails, both statistically and structurally, and previews the resolution adopted in Section 6.6.

6.5.1 Distributional Age Tracking

To formalise the age-tracking problem, we introduce the following notation used throughout this section. Let $t = 1, \dots, T$ index reporting periods (e.g. fiscal years). Fix an integer $N \geq 2$ denoting the number of age buckets. Bucket i corresponds to inventory of age $[i-1, i)$ years for $i = 1, \dots, N-1$, while bucket N is an absorbing bucket containing inventory of age $\geq N-1$ periods.

For each bucket i at the end of period t, we maintain two quantities:

- $p_{t,i} \geq 0$: the fraction of total physical inventory in bucket i (probability mass), satisfying $\sum_{i=1}^{N} p_{t,i} = 1$;
- $c_{t,i} \geq 0$: the average unit cost (\$/unit) of inventory in bucket i.

Let $Q_t \geq 0$ denote the total physical quantity of inventory on hand at the end of period t (units). The ending inventory value (balance-sheet figure) and cost of goods sold (income-statement figure) are then

$$\mathrm{Inv}_t = Q_t \cdot \sum_{i=1}^{N} p_{t,i}\, c_{t,i}, \qquad \mathrm{COGS}_t = \sum_{i=1}^{N} r_{t,i}\, c_{t,i}, \tag{66}$$

where $r_{t,i} \geq 0$ is the physical quantity sold from bucket i during period t under the FIFO allocation rule (oldest buckets depleted first).

Two distributional representations were investigated.

Gamma model.

The age composition is represented as a continuous density on $\mathbb{R}_+$, parameterised by a Gamma distribution $\Gamma(k, \theta)$ with shape $k > 0$ and scale $\theta > 0$. Each period, the density undergoes deterministic aging, mixing with new purchases, and FIFO tail removal (selling the oldest units). Because the post-sale density is not itself Gamma, it is projected back onto the Gamma family via moment matching. The latent state is two-dimensional: (k, θ).

Multinomial model.

The age composition is discretised into the N buckets defined above. At each period, aging shifts mass from bucket i to bucket $i+1$; bucket N absorbs all inventory from bucket $N-1$ and retains its own. New purchases of P_t units at average purchase price π_t^P enter bucket 1. FIFO sales of S_t units deplete buckets from oldest ($i = N$) to newest ($i = 1$). The latent state at the initial period $t = 0$ is $(2N+1)$-dimensional:

$$\theta = (p_{0,1}, \ldots, p_{0,N},\ c_{0,1}, \ldots, c_{0,N},\ Q_0).$$

Both models produce model-implied trajectories of Inv_t and COGS_t that can be compared to observed financial data, enabling estimation of the initial latent state θ via forward simulation and loss minimisation. Full technical details of both models are provided in Section 6.9.

6.5.2 Identifiability

Why the age distribution is non-identifiable.

Recall from the notation above that both accounting observables—ending inventory value Inv_t and cost of goods sold COGS_t—are weighted averages of bucket costs $c_{t,i}$, with weights determined by the age distribution $p_{t,i}$ and the FIFO sales allocation $r_{t,i}$. When the cost gap between adjacent buckets is small—as is typical under moderate inflation—redistributing probability mass across buckets barely changes these weighted averages. The loss surface is therefore effectively flat along directions that permute the age distribution while preserving the weighted-average cost. In the limiting case of flat prices ($c_{t,i} = c$ for all i), the observables are completely independent of the distribution: any age profile yields the same Inv and COGS.

Synthetic experiment.

We evaluate the recoverability of the multinomial initial state by generating synthetic data from the FIFO simulator with $N = 5$ age buckets. Let $\mathbf{p}_0 = (p_{0,1}, \ldots, p_{0,N})$ denote the true initial bucket probabilities, $\mathbf{c}_0 = (c_{0,1}, \ldots, c_{0,N})$ the true initial bucket costs, and Q_0 the true initial total quantity. The ground-truth state $(\mathbf{p}_0, \mathbf{c}_0, Q_0)$ is obtained by running the simulator for 40 burn-in periods from an initial condition (uniform probabilities, constant costs, small quantity), allowing the age structure to reach a realistic distribution shaped by each scenario's turnover and inflation dynamics. The state at the burn-in boundary is recorded as the true initial condition.

The subsequent $L = 20$ periods form the estimation window. Over this window we observe

$$\{\mathrm{Inv}_s^{\mathrm{obs}}\}_{s=0}^{L} \quad \text{and} \quad \{\mathrm{COGS}_s^{\mathrm{obs}}\}_{s=1}^{L},$$

both corrupted by additive Gaussian observation noise with standard deviation $\sigma_{\mathrm{obs}} = 0.02$. Bucket costs for ages 1 through $N-1$ are treated as known from the price history; only the absorbing-bucket cost $c_{0,N}$ is trainable. The estimator minimises a normalised mean-squared-error loss

$$\mathcal{L}(\theta) = \sum_{s=0}^{L} \left(\frac{\mathrm{Inv}_s^{\mathrm{obs}} - \mathrm{Inv}_s(\theta)}{\mathrm{Inv}_s^{\mathrm{obs}}} \right)^2 + \sum_{s=1}^{L} \left(\frac{\mathrm{COGS}_s^{\mathrm{obs}} - \mathrm{COGS}_s(\theta)}{\mathrm{COGS}_s^{\mathrm{obs}}} \right)^2,$$

where $\theta = (\mathbf{p}_0, c_{0,N}, Q_0)$ is the parameter vector and $\mathrm{Inv}_s(\theta)$, $\mathrm{COGS}_s(\theta)$ are the model-implied values obtained by forward-simulating the FIFO dynamics from θ. Optimisation uses Adam with cosine learning-rate decay. We test six scenarios spanning high, low, and very low turnover rates under inflation, deflation, and flat price regimes (Table 8).

Table 8: Scenario Configurations (Simulation Period = 20)

Scenario	**Inflation**	**Turnover**
High Turnover / Inflation	+4%	0.9
High Turnover / Deflation	−3%	0.9
Low Turnover / Inflation	+4%	0.3
Low Turnover / Deflation	−3%	0.3
Very Low Turnover / Inflation	+4%	0.1
Flat Price	0%	0.7

Results.

Let $\hat{\mathbf{p}}_0 = (\hat{p}_{0,1}, \ldots, \hat{p}_{0,N})$ denote the recovered bucket probabilities from the optimiser. Across all six scenarios, the optimiser converges to near-zero loss ($\mathcal{L} < 0.001$ in five of six cases). Physical quantity Q_0 is consistently recovered with high accuracy (Table 9). However, the recovered probabilities $\hat{\mathbf{p}}_0$ diverge substantially from the ground truth $\mathbf{p}_0$ in every scenario (Figure 6). The high-turnover-inflation case is particularly striking: the true distribution is concentrated entirely in bucket 1 ($\mathbf{p}_0 = [1, 0, 0, 0, 0]$), yet the estimator spreads mass across all buckets. The flat-price scenario confirms the theoretical prediction—with identical costs across buckets, the distribution is completely unidentifiable, and $\hat{\mathbf{p}}_0$ drifts toward a near-uniform allocation. The absorbing-bucket cost $c_{0,N}$ is poorly recovered across all scenarios: except in the very-low-turnover case, FIFO sales never reach the absorbing bucket, so $c_{0,N}$ does not enter the COGS calculation and receives no gradient signal during optimisation.

Table 9: Recovery Accuracy — Scalar Metrics

		Q (Total Quantity)			c_{last} (Oldest-Layer Cost)		
Scenario	**Loss**	True	Recovered	$\lvert\Delta\rvert$	True	Recovered	$\lvert\Delta\rvert$
High TO / Inflation	3.84e-4	11.40	11.40	0.00	0.00	47.57	47.57
High TO / Deflation	3.90e-4	11.40	11.40	0.00	0.00	2.39	2.39
Low TO / Inflation	4.63e-4	233.33	231.49	1.84	0.00	47.45	47.45
Low TO / Deflation	4.02e-4	233.33	233.12	0.20	0.00	2.81	2.81
Very Low TO / Inflation	2.81e-2	891.16	718.69	172.47	34.75	45.14	10.39
Flat Price	4.00e-4	43.39	43.40	0.01	0.00	9.07	9.07

Summary.

The poor recovery of $\hat{\mathbf{p}}_0$ despite near-zero loss $\mathcal{L}$ confirms that the multinomial model is practically non-identifiable from aggregate financial-statement data. The optimiser can find many distinct probability vectors $\hat{\mathbf{p}}_0$ that produce nearly identical Inv_t and COGS_t trajectories, because the loss surface is effectively flat along the age-distribution directions.

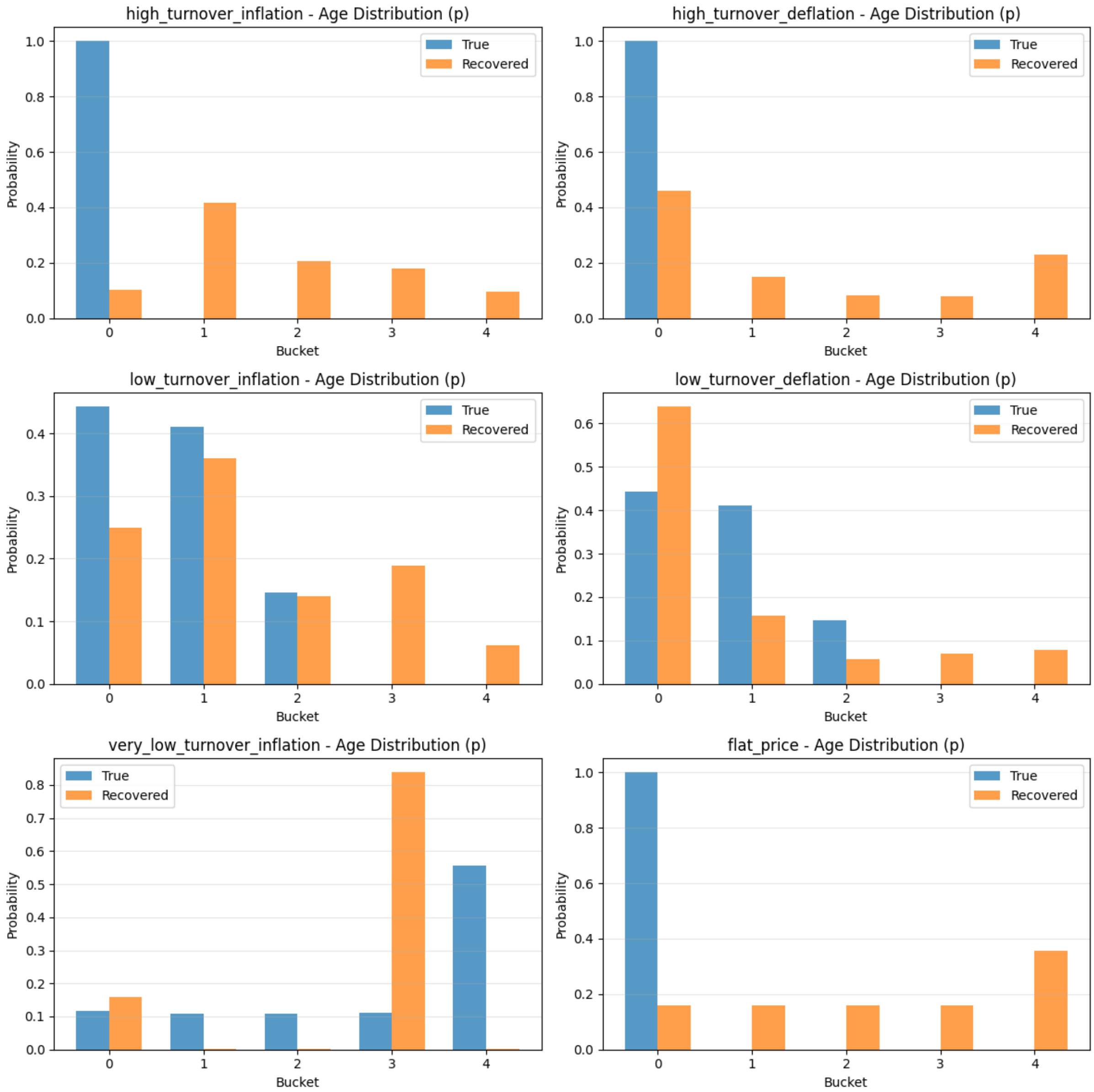


Figure 6: Recovered vs. true age distribution ($\mathbf{p}$) across scenarios. Blue bars represent the true initial bucket probabilities; orange bars represent the recovered probabilities from estimation.

6.5.3 Industry Evidence

We revisit the motivation for the bucket system. Under FIFO, COGS is determined by the cost of the oldest inventory on hand. When purchase prices change over time—for instance, under persistent inflation—firms accumulate inventory layers at different historical costs. The age distribution therefore governs the gap between COGS (old costs) and ending inventory (recent costs), affecting reported profitability, tax liabilities, and balance-sheet valuation. Furthermore, when firms switch accounting methods (e.g., from FIFO to weighted average), the age distribution determines the magnitude of the resulting restatement. This motivated our attempt to estimate the full age structure from observed financial data.

However, the practical relevance of the age distribution depends on two factors: (1) how long inventory is typically held, and (2) how much unit costs change over that holding period. If both are small, the cost layers are nearly identical regardless of the distribution shape, and a simpler representation suffices.

We proxy inventory age using Days Inventory Outstanding (DIO):

$$\text{DIO} = \frac{\text{Average Inventory}}{\text{COGS}/365}. \tag{67}$$

Industry benchmarks suggest that most sectors have short holding periods: grocery retailers typically report 10–15 days, electronics manufacturers 30–50 days, luxury goods 120–180 days, and pharmaceuticals 150–200 days. Over such horizons, unit-cost changes are generally modest—even at 4% annual inflation, the cost drift over 50 days is roughly 0.5%.

Extreme cases do exist. Diageo plc, the world's largest Scotch whisky producer, reports an inventory turnover ratio of 0.79 for 2025, equivalent to approximately 486 days of inventory on hand[3]. In such industries, the age structure may carry meaningful cost variation. However, these cases are the exception rather than the norm.

Notably, we found no existing literature that explicitly models or estimates the inventory age distribution from financial data—the concept appears absent from both the accounting and operations research literatures. This itself suggests that the practical demand for such estimation is limited.

Taken together, the evidence suggests that for the majority of firms and industries, the cost differential across inventory layers is too small to justify a distributional bucket system.

6.5.4 State Coherence

Beyond statistical non-identifiability, the earlier distributional approaches suffer from a deeper structural deficiency in their state representation itself.

The natural state specification for a stochastic bucket model is

$$x_t = (q_t,\ a_t,\ c_t),$$

where $q_t \in \mathbb{R}_{\geq 0}$ is total physical quantity, $a_t = (a_{t,1}, \dots, a_{t,N}) \in \Delta^{N-1}$ is the age-share vector on the $(N-1)$-simplex (i.e. $a_{t,i} \geq 0$, $\sum_i a_{t,i} = 1$), and $c_t = (c_{t,1}, \dots, c_{t,N}) \in \mathbb{R}^N_{\geq 0}$ are per-bucket unit costs. To build a stochastic state-space model, one naturally introduces noise on the age shares—for instance via logit-normal perturbations. Let a_t^{det} denote the deterministic age-share vector produced by the aging-and-purchase dynamics before noise is applied, and let $\varepsilon_t \in \mathbb{R}^N$ be a noise vector with covariance $\Sigma_\varepsilon \in \mathbb{R}^{N \times N}$. The perturbed age shares are

$$\tilde{a}_t = \operatorname{softmax}\bigl(\log(a_t^{\text{det}}) + \varepsilon_t\bigr), \qquad \varepsilon_t \sim \mathcal{N}(0, \Sigma_\varepsilon).$$

[3] https://www.macrotrends.net/stocks/charts/DEO/diageo/financial-ratios

This construction is **not generally coherent** under FIFO layer accounting. The reason is that perturbing age shares while carrying forward costs separately can reassign physical mass across buckets *without* reassigning the corresponding carrying values. Under FIFO, each unit of inventory has both a physical identity (which bucket it belongs to) and a dollar identity (the carrying value from its original purchase). If a perturbation moves mass from bucket 2 to bucket 4, but the carrying values remain unchanged, then the resulting state no longer corresponds to any valid FIFO inventory configuration. The accounting identity linking Inv_t to the sum of layer values is broken.

To see this concretely, define the carrying value in bucket i as $v_{t,i} = q_t\, a_{t,i}\, c_{t,i}$, so that $\mathrm{Inv}_t = \sum_{i=1}^{N} v_{t,i}$. If the age shares are perturbed from a_t to $\tilde{a}_t$ but the costs c_t are held fixed, then the implied carrying values $\tilde{v}_{t,i} = q_t\, \tilde{a}_{t,i}\, c_{t,i}$ will differ from the original $v_{t,i}$. In other words, the perturbation has moved dollar value between buckets without any corresponding physical transaction. The resulting state is internally inconsistent: it does not correspond to any realisation of FIFO purchase, aging, and sales operations.

6.5.5 Resolution: (b, v) Layers

The state-coherence problem is resolved by abandoning the (q, a, c) decomposition and instead tracking inventory layers directly as joint quantity–value pairs. The corrected latent state is

$$x_t = \begin{pmatrix} b_t \\ v_t \end{pmatrix} \in \mathbb{R}_+^{2N}, \tag{68}$$

where

- $b_t = (b_{t,1}, \ldots, b_{t,N})^\top$, with $b_{t,i} \geq 0$ denoting the physical quantity (units) in bucket i at the end of period t;
- $v_t = (v_{t,1}, \ldots, v_{t,N})^\top$, with $v_{t,i} \geq 0$ denoting the total carrying value ($) in bucket i at the end of period t.

In this representation:

- Age shares and unit costs are *derived* quantities: $a_{t,i} = b_{t,i} / \sum_j b_{t,j}$ and $c_{t,i} = v_{t,i}/b_{t,i}$ (when $b_{t,i} > 0$). They are not primitive state variables.
- Any perturbation applied to $b_{t,i}$ has a natural counterpart in $v_{t,i}$, because both are carried as a pair. There is no mechanism by which physical mass can be reassigned without also moving the associated carrying value.
- The ending inventory value is an exact sum of state variables: $\mathrm{Inv}_t = \sum_{i=1}^{N} v_{t,i}$. Similarly, COGS is computed as $\mathrm{COGS}_t = \sum_{i=1}^{N} s_{t,i}^v$, where $s_{t,i}^v = r_{t,i} \cdot (v_{t,i}^{\mathrm{pre}} / b_{t,i}^{\mathrm{pre}})$ is the dollar amount relieved from bucket i during FIFO sales, with $r_{t,i}$ denoting the quantity sold from bucket i and $(b_{t,i}^{\mathrm{pre}}, v_{t,i}^{\mathrm{pre}})$ denoting the pre-sale quantity and carrying value in bucket i. There is no ratio or weighted-average indirection in the measurement equations.

This layer-based representation is both mechanically correct and structurally compatible with stochastic perturbations. It forms the basis of the FIFO state-space model developed in Section 6.6.

6.6 FIFO Layer SSM

This section develops the full nonlinear state-space formulation for FIFO inventory dynamics using the joint quantity–value layer representation (b_t, v_t) motivated in Section 6.5.5. We specify the latent state, observed inputs, within-period mechanics, state transition, measurement equation, initial-state parameterisation, and the canonical SSM form.

Objective

We model the hidden age structure of inventory under FIFO accounting as a nonlinear state-space system. The key latent object is the joint collection of inventory quantities and carrying values across age buckets. This is required because under FIFO, physical layers and cost layers evolve jointly: when units age, their associated carrying values age with them; when units are sold, both quantity and value are relieved from the oldest surviving layers.

Time and Buckets

Let $t = 1, \dots, T$ index reporting periods (e.g. fiscal years). Fix an integer $N \geq 2$ denoting the number of age buckets. Bucket i corresponds to inventory of age

$$[i-1,\ i), \quad i = 1, \dots, N-1,$$

while bucket N is an absorbing bucket containing inventory of age at least $N-1$ periods. This bucket structure was introduced informally in Section 6.5.5; here we develop its full dynamic implications.

Inputs and Outputs

Let the observed exogenous input vector be

$$u_t = \begin{pmatrix} P_t \\ \pi_t^P \\ S_t \\ \pi_t^S \end{pmatrix} \in \mathbb{R}_+^4, \tag{69}$$

where

- $P_t \geq 0$ is the purchase quantity (units) during period t,
- $\pi_t^P \geq 0$ is the average purchase price (\$/unit) during period t,
- $S_t \geq 0$ is the sales quantity (units) during period t,

- $\pi_t^S \geq 0$ is the average selling price (\$/unit) during period t.

For FIFO inventory valuation, the state transition depends on (P_t, π_t^P, S_t); the selling price π_t^S is not required unless one also wishes to model revenue or gross profit.

Let the accounting observation vector be

$$y_t = \begin{pmatrix} \mathrm{Inv}_t \\ \mathrm{COGS}_t \end{pmatrix} \in \mathbb{R}_+^2, \tag{70}$$

where Inv_t is the ending inventory value (\$, balance sheet) and COGS_t is the cost of goods sold (\$, income statement) during period t.

Latent State

The hidden state at the end of period t is

$$x_t = \begin{pmatrix} b_t \\ v_t \end{pmatrix}, \quad b_t = (b_{t,1}, \dots, b_{t,N})^\top, \quad v_t = (v_{t,1}, \dots, v_{t,N})^\top, \tag{71}$$

with domain

$$b_t \in \mathbb{R}_+^N, \quad v_t \in \mathbb{R}_+^N, \quad x_t \in \mathbb{R}_+^{2N}. \tag{72}$$

Here

- $b_{t,i} \geq 0$ is the physical quantity (units) in bucket i at the end of period t,
- $v_{t,i} \geq 0$ is the total carrying value (\$) in bucket i at the end of period t.

Thus the latent state tracks inventory layers directly. Bucket quantities and bucket values must evolve jointly—this is the central design requirement established in Section 6.5.4.

Derived Quantities

The total ending quantity is

$$q_t = \sum_{i=1}^{N} b_{t,i}. \tag{73}$$

If $q_t > 0$, the age-share vector is the derived quantity

$$a_{t,i} = \frac{b_{t,i}}{q_t}, \quad i = 1, \dots, N. \tag{74}$$

Hence $a_t = (a_{t,1}, \dots, a_{t,N})^\top$ lies on the $(N-1)$-simplex Δ^{N-1}.

If $b_{t,i} > 0$, the bucket unit cost is the derived quantity

$$c_{t,i} = \frac{v_{t,i}}{b_{t,i}}, \quad i = 1, \dots, N. \tag{75}$$

Therefore age shares and unit costs are not primitive state variables; they are fully determined by (b_t, v_t).

Transition Mechanics

Each period transition proceeds in three stages:

1. aging and new purchases,
2. FIFO sales allocation,
3. post-sale layer update.

Pre-sale state after aging and purchases.

Let $(b_t^{\text{pre}}, v_t^{\text{pre}}) \in \mathbb{R}_+^N \times \mathbb{R}_+^N$ denote the pre-sale quantity and value vectors for period t, obtained by aging the previous end-of-period state x_{t-1} by one period and inserting new purchases into bucket 1. The pre-sale quantities are given by

$$b_{t,1}^{\text{pre}} = P_t, \tag{76}$$
$$b_{t,i}^{\text{pre}} = b_{t-1,i-1}, \quad i = 2, \dots, N-1, \tag{77}$$
$$b_{t,N}^{\text{pre}} = b_{t-1,N-1} + b_{t-1,N}. \tag{78}$$

The corresponding pre-sale carrying values are

$$v_{t,1}^{\text{pre}} = P_t \, \pi_t^P, \tag{79}$$
$$v_{t,i}^{\text{pre}} = v_{t-1,i-1}, \quad i = 2, \dots, N-1, \tag{80}$$
$$v_{t,N}^{\text{pre}} = v_{t-1,N-1} + v_{t-1,N}. \tag{81}$$

Equations (76)–(78) and (79)–(81) ensure that quantity layers and value layers age together. This is the central accounting-consistency requirement under FIFO.

Define the total pre-sale quantity

$$B_t = \sum_{i=1}^{N} b_{t,i}^{\text{pre}}. \tag{82}$$

A natural feasibility condition is

$$S_t \leq B_t, \tag{83}$$

so that requested sales do not exceed available stock.

FIFO sales allocation.

Let

$$r_t = (r_{t,1}, \dots, r_{t,N})^\top \in \mathbb{R}_+^N \tag{84}$$

denote the quantity sold from each bucket during period t.

FIFO means that the oldest surviving inventory is sold first. Define the recursive remaining sales demand by

$$R_t^{(N)} = S_t, \tag{85}$$

and for $i = N, N-1, \ldots, 1$,

$$r_{t,i} = \min\left\{R_t^{(i)},\ b_{t,i}^{\text{pre}}\right\}, \tag{86}$$

$$R_t^{(i-1)} = R_t^{(i)} - r_{t,i}. \tag{87}$$

This defines a deterministic FIFO allocation map

$$r_t = k(b_t^{\text{pre}},\ S_t). \tag{88}$$

Value relief and post-sale state.

To compute the cost of goods sold, define the dollar amount relieved from each bucket:

$$s_{t,i}^{v} = \begin{cases} r_{t,i}\,\dfrac{v_{t,i}^{\text{pre}}}{b_{t,i}^{\text{pre}}}, & \text{if } b_{t,i}^{\text{pre}} > 0, \\ 0, & \text{if } b_{t,i}^{\text{pre}} = 0, \end{cases} \qquad i = 1, \ldots, N. \tag{89}$$

Equivalently, if one defines pre-sale bucket unit costs

$$c_{t,i}^{\text{pre}} = \begin{cases} \dfrac{v_{t,i}^{\text{pre}}}{b_{t,i}^{\text{pre}}}, & \text{if } b_{t,i}^{\text{pre}} > 0, \\ 0, & \text{if } b_{t,i}^{\text{pre}} = 0, \end{cases} \tag{90}$$

then

$$s_{t,i}^{v} = r_{t,i}\, c_{t,i}^{\text{pre}}. \tag{91}$$

The post-sale state is then

$$b_{t,i} = b_{t,i}^{\text{pre}} - r_{t,i}, \tag{92}$$

$$v_{t,i} = v_{t,i}^{\text{pre}} - s_{t,i}^{v}, \tag{93}$$

for $i = 1, \ldots, N$.

Thus sales remove both quantity and carrying value from the oldest layers first, exactly as FIFO requires.

Transition Equation

Collecting the previous steps, the latent state evolves according to the nonlinear transition equation

$$x_t = f(x_{t-1},\ u_t), \tag{94}$$

where $f\colon \mathbb{R}_+^{2N} \times \mathbb{R}_+^4 \to \mathbb{R}_+^{2N}$ is defined implicitly by the pre-sale update (76)–(81), the FIFO allocation rule (86)–(87), and the post-sale update (92)–(93).

The nonlinearity arises from two sources: the min operator in the FIFO allocation (86) and the ratio in the value-relief formula (89).

Observation Equation

The accounting observables are deterministic functions of the updated inventory layers and the value relieved during sales.

Ending inventory value is

$$\mathrm{Inv}_t = \sum_{i=1}^{N} v_{t,i}. \tag{95}$$

Cost of goods sold is

$$\mathrm{COGS}_t = \sum_{i=1}^{N} s_{t,i}^{v}. \tag{96}$$

Hence the measurement equation is

$$y_t = h(x_{t-1},\, u_t) = \begin{pmatrix} \sum_{i=1}^{N} v_{t,i} \\ \sum_{i=1}^{N} s_{t,i}^{v} \end{pmatrix}, \tag{97}$$

where $h\colon \mathbb{R}_+^{2N} \times \mathbb{R}_+^4 \to \mathbb{R}_+^2$ is the observation map. Although written as a function of (x_{t-1}, u_t), the same measurement map can equivalently be written using (x_t, r_t) once the within-period FIFO allocation has been computed.

Initial State

Let the initial latent state satisfy

$$x_0 = \begin{pmatrix} b_0 \\ v_0 \end{pmatrix} \in \mathbb{R}_+^{2N}. \tag{98}$$

A convenient parameterisation decomposes x_0 into interpretable components. Let $q_0 \geq 0$ denote the total initial physical quantity, $a_0 = (a_{0,1}, \ldots, a_{0,N})^\top \in \Delta^{N-1}$ the initial age distribution (with $\sum_{i=1}^{N} a_{0,i} = 1$, $a_{0,i} \geq 0$), and $c_0 = (c_{0,1}, \ldots, c_{0,N})^\top \in \mathbb{R}_+^N$ the initial bucket unit costs. Then

$$b_{0,i} = q_0\, a_{0,i}, \quad i = 1, \ldots, N, \tag{99}$$

together with

$$v_{0,i} = b_{0,i}\, c_{0,i} = q_0\, a_{0,i}\, c_{0,i}, \quad i = 1, \ldots, N. \tag{100}$$

Thus the initial state may be parameterised by the triple (q_0, a_0, c_0). This parameterisation is used for estimation and for specifying prior distributions, while the actual state carried forward in the transition is always the joint pair (b_0, v_0).

Observation Noise

If reported accounting quantities are measured with noise or are contaminated by aggregation, rounding, or non-FIFO adjustments, one may write

$$y_t^{\mathrm{obs}} = y_t + \eta_t, \quad \eta_t \sim \mathcal{N}(0,\, \Sigma_y), \tag{101}$$

where $y_t = h(x_{t-1}, u_t)$ is the latent (noise-free) accounting identity, $y_t^{\mathrm{obs}} \in \mathbb{R}^2$ is the observed report, $\eta_t \in \mathbb{R}^2$ is the observation noise, and $\Sigma_y \in \mathbb{R}^{2\times 2}$ is a symmetric positive-semidefinite noise covariance matrix. In the noise-free idealisation, simply set $\eta_t \equiv 0$.

Canonical Form

The corrected FIFO inventory-layer model can therefore be summarised as the canonical nonlinear state-space system:

$$\boxed{\begin{array}{rl} \text{Transition:} & x_t = f(x_{t-1},\, u_t), \\ \text{Observation:} & y_t^{\mathrm{obs}} = h(x_{t-1},\, u_t) + \eta_t, \quad \eta_t \sim \mathcal{N}(0,\, \Sigma_y), \end{array}} \tag{102}$$

with latent state $x_t = (b_t^\top, v_t^\top)^\top \in \mathbb{R}_+^{2N}$, exogenous inputs $u_t \in \mathbb{R}_+^4$, and observations $y_t^{\mathrm{obs}} \in \mathbb{R}^2$. In the noise-free case, $y_t^{\mathrm{obs}} = y_t$.

Necessity of This Representation

A state specification based only on total quantity, age shares, and per-bucket unit costs,

$$x_t = (q_t,\, a_t,\, c_t),$$

is not generally coherent if stochastic perturbations are applied directly to age shares while costs are carried forward separately. Such a construction can reassign physical mass across buckets without reassigning the corresponding carrying values, thereby breaking FIFO layer accounting (see the detailed discussion in Section 6.5.4).

By contrast, the layer-based state (b_t, v_t) preserves the joint evolution of physical units and dollar basis, and therefore yields internally consistent expressions for both ending inventory and COGS.

Identification Remark

The formulation above provides a mechanically correct state-space representation of FIFO inventory dynamics. It does not by itself guarantee statistical identification of the latent layer state from annual financial reports. If only $(\mathrm{Inv}_t, \mathrm{COGS}_t)$ are observed each period, the inverse problem may remain weakly identified unless one imposes additional structure, such as small N, informative priors on x_0, observed purchase and sales flows, or extra disclosures on inventory composition. The following section develops the *washout* strategy, which exploits the finite-memory property of FIFO dynamics to achieve forward identification of the latent state.

6.7 Washout Identification

Overview

Under FIFO, the oldest surviving inventory is consumed first. This creates a natural *washout* mechanism: if cumulative sales are sufficiently large relative to the initial stock, then all units present at time 0 are eventually sold. After that point, the latent inventory state depends only on observed post-0 purchase and sales flows, not on the unknown initial state x_0.

This observation is useful, but its scope must be stated carefully. Washout yields a *forward determinacy* result: after the washout horizon, the state no longer depends on the unknown initial state. It does not, in general, imply exact backward recovery of x_0 from annual financial reports, because the FIFO transition f (defined in equation (94)) is not globally invertible and information is lost through bucket exhaustion and through the absorbing oldest bucket.

The goal of this section is therefore to formalise the valid implication of washout—namely, reduced sensitivity to initial conditions and forward identification of post-washout states—while clarifying the limits of what can be inferred about the initial latent state.

Required Inputs

To operationalise the FIFO state transition, one needs period-by-period measures of purchase quantity, purchase unit cost, and sales quantity. Recall the input vector u_t from equation (69). For the washout analysis, the relevant components are

$$u_t^{\text{trans}} = \begin{pmatrix} P_t \\ \pi_t^P \\ S_t \end{pmatrix}, \tag{103}$$

where P_t is purchase quantity (units), π_t^P is purchase unit cost (\$/unit), and S_t is sales quantity (units), as defined in Section 6.6.

In practice these inputs may not be directly observed in public financial reports and may need to be estimated using auxiliary assumptions. For example:

- Dollar purchases may be approximated using the inventory rollforward identity

  $$\text{Purchases}_t^{\$} \approx \text{Inv}_t - \text{Inv}_{t-1} + \text{COGS}_t, \tag{104}$$

  but this identity is exact only in a stylised setting without write-downs, shrinkage, FX effects, acquisitions, or other inventory adjustments.

- Purchase quantity P_t requires converting dollar purchases into units using an estimate of π_t^P.

- Sales quantity S_t may be obtained from operational disclosures, or approximated using revenue divided by an average selling price when such an approximation is economically meaningful.

Accordingly, the washout argument below should be interpreted as a property of the model conditional on the input sequence $(u_1, \ldots, u_T)$, rather than as a claim that public financial statements alone always reveal those inputs exactly.

Washout Condition

The state-space model in Section 6.6 adopts a period-aggregated convention: in each period, purchases are first added to inventory, the age buckets are updated via equations (76)–(81), and then period-t sales are allocated by FIFO to the resulting pre-sale stock. Under this convention, one can define washout at the reporting-period level.

Definition 1 (Cumulative sales)**.** For any integer $T' \geq 1$, define cumulative sales through period T' by

$$\bar{S}_{T'} = \sum_{t=1}^{T'} S_t. \tag{105}$$

Let $q_0 = \sum_{i=1}^{N} b_{0,i}$ denote the total initial inventory quantity, as in equation (73).

Definition 2 (Washout horizon)**.** The washout horizon is the smallest integer $T^* \geq 1$ such that, under the model's FIFO allocation rule (86)–(87), all units present at time 0 have been sold by the end of period T^*. A sufficient condition under the period-aggregated model is

$$\bar{S}_{T^*} \geq q_0. \tag{106}$$

If no such T^* exists within the sample, washout does not occur in-sample.

Remark 1. Condition (106) should be interpreted as a model-based sufficient condition, not as a purely accounting fact recoverable from annual reports without further assumptions. Its meaning depends on the period-aggregation convention of the state-space model.

DIO Lower Bound

While the formal washout horizon T^* depends on the full sequence of sales and initial stock, a simple lower bound can be obtained from a standard accounting ratio. Recall the Days Inventory Outstanding (DIO) from Section 6.5.3:

$$\text{DIO} = \frac{\text{Average Inventory}}{\text{COGS}/365}.$$

DIO measures the average number of days a unit of inventory is held before being sold. Converting to annual periods gives the heuristic lower bound

$$T^*_{\text{lb}} = \left\lceil \frac{\text{DIO}}{365} \right\rceil. \tag{107}$$

The intuition is straightforward: if average inventory turns over every DIO days, then after T^*_{lb} annual periods the "typical" initial unit has been sold. For a firm with DIO $= 50$ days, $T^*_{\mathrm{lb}} = 1$ year; for DIO $= 486$ days (Diageo-type), $T^*_{\mathrm{lb}} = 2$ years.

This is a *lower* bound rather than an exact estimate because DIO reflects the *mean* holding period, not the maximum. The true washout horizon could be longer if the age distribution has a heavy right tail—i.e. if a small fraction of initial inventory is very old and slow-moving. A tight upper bound cannot be established without knowledge of the full age distribution, which is precisely the unobservable quantity. Accordingly, T^*_{lb} should be treated as a practical heuristic: if the available historical window is substantially longer than T^*_{lb}, washout is likely; if it is close to or shorter than T^*_{lb}, the user should exercise caution and run the sensitivity analysis described in Section 6.8.

Forward Determinacy

The central implication of washout is that the state eventually ceases to depend on the unknown initial condition.

Proposition 1 (Loss of dependence on the initial state after washout)**.** Suppose the period-by-period inputs $(u_1, \ldots, u_{T^*})$ are known, and suppose washout occurs by time T^* (Definition 2). Then the latent state $x_{T^*} = (b_{T^*}, v_{T^*})$ is uniquely determined by the observed input sequence $(u_1, \ldots, u_{T^*})$ and is independent of the unknown initial state x_0.

Proof. Under FIFO, sales are allocated to the oldest surviving inventory first via equations (86)–(87). By assumption, washout occurs by time T^*, meaning that every unit present at time 0 has been sold by the end of period T^*.

Consider any unit still present in inventory at time T^*. Since all time-0 units have been exhausted, that surviving unit must have originated from a purchase in some period $t \in \{1, \ldots, T^*\}$. Therefore every surviving inventory layer at time T^* is generated by a purchase cohort observed within the sample window.

Given the known input sequence $(u_1, \ldots, u_{T^*})$, one can run the FIFO transition f (equation (94)) forward from an empty pre-sample state for the purchase cohorts created in periods $1, \ldots, T^*$. Since all time-0 units have been removed by time T^*, the surviving quantities and values of those cohorts are exactly the same as in the true system. Hence x_{T^*} is a deterministic function of $(u_1, \ldots, u_{T^*})$ alone and does not depend on x_0. $\square$

Remark 2. Proposition 1 is a forward result. It states that the post-washout state is free of initial-condition dependence. It does not imply that the initial state itself can be uniquely recovered by running the dynamics backward.

Cohort Representation

A convenient way to compute the post-washout state is to track the survival of purchase cohorts.

For each purchase cohort $t \in \{1, \ldots, T^*\}$, let $Q_t^{(\ell)} \geq 0$ denote the quantity from cohort t surviving at the end of period ℓ, for $\ell \geq t$. Initialise

$$Q_t^{(t-1)} = P_t. \tag{108}$$

Then, within each period ℓ, sales are allocated by FIFO from the oldest surviving cohort to the youngest. Applying the period-ℓ sales quantity S_ℓ to the ordered list of surviving cohorts yields the residual cohort quantities $Q_t^{(\ell)}$ after period ℓ. This defines a deterministic recursion

$$\left(Q_1^{(\ell)}, \ldots, Q_\ell^{(\ell)}\right) = \Phi_\ell\Big(\left(Q_1^{(\ell-1)}, \ldots, Q_{\ell-1}^{(\ell-1)}, P_\ell\right),\ S_\ell\Big), \tag{109}$$

where $\Phi_\ell \colon \mathbb{R}_+^\ell \times \mathbb{R}_+ \to \mathbb{R}_+^\ell$ is the FIFO depletion operator that removes S_ℓ units starting from the oldest cohort.

At the end of period T^*, cohort t has age $T^* - t$ periods and therefore belongs to bucket

$$i(t, T^*) = \min(T^* - t + 1,\ N). \tag{110}$$

Hence the post-washout state is

$$b_{T^*,i} = \sum_{\substack{t=1,\ldots,T^* \\ i(t,T^*)=i}} Q_t^{(T^*)}, \tag{111}$$

$$v_{T^*,i} = \sum_{\substack{t=1,\ldots,T^* \\ i(t,T^*)=i}} Q_t^{(T^*)}\, \pi_t^P, \tag{112}$$

for $i = 1, \ldots, N$, where π_t^P is the purchase price of cohort t (from equation (69)).

Thus, once washout has occurred, the entire latent state is a deterministic function of the known purchase cohorts and the realised sales sequence.

Non-Identification of Initial State

It is tempting to conclude from Proposition 1 that one can recover the unknown initial state x_0 by inverting the transition equation backward from x_{T^*}. In general, this conclusion is false.

The reason is that the FIFO transition f is not globally invertible. Information is lost in at least three ways:

1. **Bucket exhaustion.** If a bucket is fully depleted during a period (i.e. $r_{t,i} = b_{t,i}^{\text{pre}}$ in equation (86)), then its pre-sale quantity and value cannot be reconstructed uniquely from the post-sale state alone, because the post-sale values $b_{t,i} = 0$ and $v_{t,i} = 0$ are consistent with any pre-sale quantity that was fully consumed.

2. **Absorbing oldest bucket.** The transition rules

$$b_{t,N}^{\text{pre}} = b_{t-1,N-1} + b_{t-1,N}, \qquad v_{t,N}^{\text{pre}} = v_{t-1,N-1} + v_{t-1,N} \tag{113}$$

(from equations (78) and (81)) merge two previous buckets into one current bucket. This map is many-to-one and therefore not invertible without additional restrictions.

3. **Aggregation of accounting observables.** The observed quantities Inv_t and COGS_t (equation (97)) are low-dimensional aggregates ($\mathbb{R}^2$) of the $2N$-dimensional latent layers and generally do not contain enough information to reconstruct the full pre-sale layer composition.

Accordingly, washout should be understood as a *forgetting* or *truncation* result, not as a general identification theorem for x_0.

Remark 3 (Absorbing bucket as an identifiability sink)**.** The absorbing bucket N is particularly important. Even if all other steps of the transition were invertible, the merging map in equation (113) destroys information by pooling old vintages. This makes exact backward recovery of deep initial history impossible unless one introduces additional structure, such as a parametric prior on the contents of the oldest bucket or a sufficiently large choice of N so that the absorbing bucket is empirically empty.

Partial Washout

Full washout is not the only useful case. Even before all initial inventory is exhausted, cumulative sales reduce the extent to which the current state depends on the oldest parts of the initial condition.

Suppose that by the end of period $T' < T^*$, FIFO sales have completely consumed the oldest portion of the initial inventory queue—i.e. the initial buckets $b_{0,N}, b_{0,N-1}, \ldots, b_{0,j}$ for some j have been fully depleted. Then those consumed components of x_0 no longer affect the current state $x_{T'}$. In that sense, partial washout reduces the dimensional importance of the unknown initial state.

However, this should not be confused with exact identification of the corresponding initial buckets from aggregate accounting data. In general, partial washout does not imply that the quantities and values of the consumed initial buckets are separately recoverable from the observed $(\mathrm{Inv}_t, \mathrm{COGS}_t)$ sequence. What it does imply is weaker but still useful: the admissible set of initial states consistent with the data becomes smaller as more of the initial inventory is forced out of the system by FIFO depletion.

Post-Washout Validation

Washout has an important empirical implication. Once the washout horizon T^* is reached, the latent state is pinned down by observed inputs alone (Proposition 1). Therefore, subsequent accounting observations become model-implied restrictions rather than inputs needed to infer the hidden state.

More precisely, for any $t > T^*$, one can compute the model-implied state recursively from x_{T^*} and $(u_{T^*+1}, \ldots, u_t)$ via the transition (94), and then compute the model-implied accounting observables

$$\hat{y}_t = h(x_{t-1},\, u_t). \tag{114}$$

Comparing $\hat{y}_t$ with the observed y_t^{obs} yields out-of-sample specification checks:

$$y_t^{\text{obs}} - \hat{y}_t = \eta_t. \tag{115}$$

In the noise-free idealisation, these residuals should be zero. Under the observation-noise formulation (101), the residuals can be used for model validation, specification testing, or estimation of the noise covariance Σ_y.

Estimation Implications

The washout property suggests three practical uses.

Finite-horizon inference.

It justifies focusing inference on a finite recent history when the sample is long enough and inventory turnover is sufficiently fast. After washout, the effect of the unknown initial state x_0 disappears entirely.

Principled initialisation.

It provides a principled way to initialise filtering or smoothing algorithms. Rather than attempting to estimate a deep pre-sample inventory history, one may begin at or after a washout horizon where the state is determined by observed flows.

Natural validation regime.

Periods after washout provide out-of-sample restrictions (equation (115)) that can be used to assess whether the assumed FIFO layer model is consistent with observed accounting data.

Summary

The correct role of washout in this framework is the following:

1. **Forward determinacy.** Once washout occurs, the latent state x_{T^*} is a deterministic function of observed post-0 purchase and sales flows and no longer depends on the unknown initial state x_0.

2. **Reduced sensitivity to initialisation.** Even before full washout, FIFO depletion progressively weakens the influence of the oldest components of the initial inventory distribution.

3. **No general backward identification.** Washout does not, by itself, imply exact recovery of the initial latent state x_0, because the FIFO transition is not globally invertible and annual accounting observables are too aggregated. (See Section 6.7 for details.)

4. **Validation after washout.** Observations after the washout horizon T^* provide model-implied restrictions (Section 6.7) that can be used for specification testing and robustness analysis.

Thus washout should be viewed not as a full identification strategy for the initial latent state, but as a structural property of FIFO dynamics that yields finite-memory behaviour and a useful source of post-washout restrictions.

6.8 Forward Simulation

Overview

For many practical users, the primary goal of the model is not forensic recovery of the firm's latent inventory composition in the distant past, but rather forward simulation of inventory dynamics from the present onward. In that setting, the washout property (Section 6.7) has an important operational implication: when a sufficiently long history of financial reports is available, the current latent inventory state can often be approximated without precise knowledge of the firm's ancient pre-sample inventory composition.

Put differently, the relevant question for forward simulation is usually not whether the exact initial latent state x_0 at the beginning of the sample can be identified, but whether the available historical window is long enough that the current state x_T depends mainly on observed flows within the sample rather than on unknown inventory layers from long ago.

State Construction

Suppose the user has access to more than ten years of financial reports for a firm and wishes to simulate future inventory states under alternative scenarios for purchases, costs, and sales. In such a setting, the main role of the historical data is to construct a credible estimate of the current latent state

$$x_T = (b_T, v_T),$$

rather than to exactly recover the latent state x_0 at the beginning of the historical sample.

The washout logic developed in Section 6.7 implies that, if inventory turnover is sufficiently high relative to the sample length, then the current state x_T will be largely determined by observed flows over the recent historical window. The influence of arbitrary assumptions about very old inventory layers will either disappear entirely (if washout has occurred) or become concentrated in the oldest absorbing bucket N.

This makes the model substantially more useful for forward simulation than for exact retrospective identification.

Why Long Histories Help

A long sequence of historical reports provides three distinct benefits for forward simulation.

Reduced dependence on initialisation.

If the sample begins only a short time before the simulation start date, then the estimated current latent state x_T may depend heavily on arbitrary assumptions about the initial bucket composition (b_0, v_0). By contrast, when ten or more years of history are available, FIFO depletion often removes most of the effect of these arbitrary starting assumptions. The resulting current state is therefore more stable and more suitable as the starting point for scenario analysis.

Improved reconstruction of the current age-cost structure.

The future path of FIFO inventory value and cost of goods sold depends on the age distribution and carrying values of the inventory layers that survive today. A longer history allows the model to reconstruct this current age-cost structure from many years of observed purchase and sales flows (P_t, π_t^P, S_t), rather than relying on a short recent window that may omit important older layers.

More credible forward scenarios.

Once the current latent state has been credibly constructed, the user can simulate future inventory dynamics under alternative assumptions about demand, procurement, inflation, or clearance of slow-moving stock. These forward scenarios are meaningful only to the extent that the starting latent state x_T is credible. The main value of a long historical window is precisely that it makes this starting state less sensitive to arbitrary initialisation choices.

Operational Interpretation

From the user's perspective, the availability of a long financial history shifts the modelling objective. The model is best viewed as a *state-construction-and-simulation tool* rather than as a complete identification device for the entire historical latent path.

Concretely, the practical workflow is:

1. Use the long historical sequence of observed or estimated inputs $(u_1, \ldots, u_T)$ to run the FIFO state transition f (equation (94)) through history.

2. Obtain a current latent state $x_T = (b_T, v_T)$ that reflects the cumulative effect of those historical flows.

3. Treat x_T as the starting point for forward simulation under user-specified future scenarios $(u_{T+1}, u_{T+2}, \ldots)$.

If the model has effectively washed out the dependence on the beginning-of-sample state x_0, then the quality of forward simulation depends mainly on two factors:

1. the quality of the estimated current state x_T;

2. the quality of the future scenario assumptions $(P_{T+1}, \pi^P_{T+1}, S_{T+1}),\ (P_{T+2}, \pi^P_{T+2}, S_{T+2}), \ldots$

Turnover Regimes

The practical value of a long sample depends on the turnover characteristics of the firm. This discussion connects back to the DIO-based industry evidence in Section 6.5.3.

High-turnover firms.

For retailers, distributors, food producers, and other fast-turning businesses, ten years of annual reports are often far more than sufficient for effective washout (Definition 2). In such cases, even a crude assumption on the initial latent state x_0 at the start of the sample is unlikely to materially affect the reconstructed current state x_T. The model can therefore be used confidently for forward simulation.

Medium-turnover firms.

For many industrial and manufacturing firms, a ten-year window may be sufficient but should not be assumed to be so automatically. In these settings, the user should test whether different plausible initial-state assumptions lead to materially different reconstructed current states after running the model through the historical sample. If the resulting current states are close, then the historical window is effectively long enough for forward simulation purposes.

Low-turnover firms.

For firms with slow-moving, specialised, or lumpy inventory—for example heavy equipment, aerospace, shipbuilding, or certain real-estate-related businesses—even ten years of reports may not fully remove dependence on old inventory. In such cases, the remaining uncertainty is typically concentrated in the oldest absorbing bucket N, and forward simulation remains possible but should be accompanied by sensitivity analysis with respect to that bucket's composition.

Practical Procedure

The washout logic suggests the following practical procedure for users with long report histories.

1. **Use the full available history.** Even if early years are measured imperfectly, a long historical window helps reduce sensitivity to arbitrary initialisation of x_0.

2. **Run multiple initial-state assumptions.** For example, initialise the model under several plausible age profiles for (b_0, v_0), such as young-inventory-heavy, old-inventory-heavy, and approximately uniform age composition.

3. **Compare the reconstructed current states.** If these different initialisations produce similar current latent states x_T after the model is run through the full historical window, then the system has effectively forgotten its starting condition.

4. **Locate the remaining uncertainty.** If differences persist, they will often be concentrated in the oldest absorbing bucket N (see the absorbing-bucket remark in Section 6.7). This residual uncertainty should be reported explicitly.

5. **Simulate forward from the current state.** Once a stable current state x_T has been obtained, simulate the future under alternative scenarios for purchase quantities P_t, purchase costs π_t^P, and sales quantities S_t.

Sensitivity Analysis

The key robustness question for forward simulation is not whether the initial latent state x_0 at the start of the historical sample is uniquely identified, but whether uncertainty about that initial state materially affects the simulated future outcomes of interest.

Accordingly, the relevant robustness criterion is:

> Do different plausible historical initialisations of x_0 produce materially different present states x_T or future simulations after conditioning on the full historical record?

If the answer is no, then the model is practically well initialised for simulation, even though the distant historical latent state is not exactly identified. If the answer is yes, then the user should retain a distribution over possible current states and propagate that uncertainty through the forward scenarios.

Conclusion

The central implication of the washout property for forward simulation is therefore the following:

> When a sufficiently long history of financial reports is available, the model can often construct a credible current latent inventory state $x_T = (b_T, v_T)$ without precise knowledge of the distant historical inventory composition. This is usually enough for forward simulation, even when exact identification of the ancient latent state x_0 is impossible.

Thus, for a user interested in simulating the future inventory dynamics of a firm, the primary value of a long historical reporting window is not exact historical recovery, but rather stabilisation of the present-state estimate that serves as the starting point for scenario analysis.

Limitation

This conclusion depends not only on the washout property, but also on the quality of the estimated historical inputs (P_t, π_t^P, S_t). If purchase quantities, purchase costs, or sales quantities are measured with substantial error, then a long historical window may propagate those errors rather than eliminate them. Long history helps only when combined with a sufficiently reliable reconstruction of the historical flow inputs (see the input-estimation discussion in Section 6.7).

Summary

For users with more than ten years of financial reports, the model should be interpreted primarily as a tool for:

1. constructing a credible current latent inventory state $x_T = (b_T, v_T)$;
2. checking whether that current state is robust to alternative initialisations of x_0;
3. simulating future inventory dynamics under counterfactual scenarios.

The practical message is that long history usually matters because it reduces initialisation error, not because it fully reveals the latent inventory composition in the distant past.

6.9 Alternative Approaches

This section documents several earlier or simplified approaches to inventory modelling that were explored before we settled on the joint (b_t, v_t)-layer SSM of Section 6.6. Each approach shares a common set of accounting variables and identities; we collect these once below and reference them throughout.

6.9.1 Common Notation

Exogenous drivers.

At each annual period t the model receives four scalar drivers, all non-negative:

- $P_t \geq 0$: purchase quantity (units), as in Section 6.6.
- $\pi_t^P \geq 0$: average purchase price (\$/unit).
- $S_t \geq 0$: sales quantity (units).
- $\pi_t^S \geq 0$: average selling price (\$/unit).

The derived dollar flows are $\text{Purchase}_t = P_t \cdot \pi_t^P$ and $\text{Sales}_t = S_t \cdot \pi_t^S$.

Aggregate state variables.

All approaches track at least the following aggregate quantities at the end of period t:

- $Q_t \geq 0$: total physical inventory on hand (units).
- $\text{Inv}_t \geq 0$: ending inventory value (\$, balance sheet).
- Cash_t: cash balance (\$).
- RE_t: retained earnings (\$).

The cost of goods sold $\text{COGS}_t \geq 0$ is an annual flow variable whose computation differs across models.

Accounting identities.

The following four identities hold in every model:

$$
\begin{aligned}
Q_t &= Q_{t-1} + P_t - S_t, && (116)\\
\mathrm{Inv}_t &= \mathrm{Inv}_{t-1} + \mathrm{Purchase}_t - \mathrm{COGS}_t, && (117)\\
\mathrm{Cash}_t &= \mathrm{Cash}_{t-1} + \mathrm{Sales}_t - \mathrm{Purchase}_t, && (118)\\
\mathrm{RE}_t &= \mathrm{RE}_{t-1} + \mathrm{Sales}_t - \mathrm{COGS}_t. && (119)
\end{aligned}
$$

Feasibility constraints.

Every model enforces

$$Q_{t-1} \geq 0, \quad Q_t \geq 0, \quad S_t \geq 0, \quad P_t \geq 0, \quad S_t \leq Q_{t-1} + P_t. \tag{120}$$

Pre-sale quantity and sold fraction.

After purchases arrive but before sales are executed, the total on-hand quantity is

$$B_t = Q_{t-1} + P_t, \tag{121}$$

and the fraction of pre-sale inventory that is sold is

$$\pi_t = \frac{S_t}{B_t} \in [0, 1] \qquad (\text{set } \pi_t = 0 \text{ if } B_t = 0). \tag{122}$$

Observation model.

In every approach the observed data from financial statements are

$$y_t = (\mathrm{Inv}_t,\ \mathrm{COGS}_t), \tag{123}$$

and the estimation problem is to choose model parameters so that the forward-simulated $\mathrm{Inv}_t(\theta)$ and $\mathrm{COGS}_t(\theta)$ track these observations.

6.9.2 WAC Model

The simplest aggregated model tracks only total physical quantity and a single average unit cost, without explicit age buckets or FIFO assumptions. Under the hood, this is a weighted-average cost model.

State.

$$x_t = (Q_t,\ c_t) \in \mathbb{R}^2_{\geq 0}, \tag{124}$$

where $c_t \geq 0$ is the average unit cost of ending inventory (\$/unit) at the end of period t.

Annual transition.

1. **Pre-sale pooling.** Purchases are added and a pooled average cost is computed:

$$Q_t^{\text{pre}} = Q_{t-1} + P_t, \qquad c_t^{\text{pre}} = \frac{Q_{t-1}\, c_{t-1} + P_t\, \pi_t^P}{Q_t^{\text{pre}}}. \tag{125}$$

2. **Sales.** COGS is valued at the pre-sale average cost:

$$\text{COGS}_t = S_t \cdot c_t^{\text{pre}}. \tag{126}$$

3. **Post-sale state.**

$$Q_t = Q_t^{\text{pre}} - S_t, \qquad c_t = c_t^{\text{pre}}. \tag{127}$$

4. **Accounting.**

$$\text{Inv}_t = Q_t \cdot c_t, \qquad \text{Cash}_t,\ \text{RE}_t \text{ via (118)–(119)}. \tag{128}$$

Estimation.

The parameter vector is $\theta = (Q_0, c_0)$ (two scalars). Given observed series $\{\text{Inv}_s^{\text{obs}}\}_{s=0}^{L}$ and $\{\text{COGS}_s^{\text{obs}}\}_{s=1}^{L}$, the estimator solves

$$\theta^* = \arg\min_{\theta \in \mathbb{R}^2_{\geq 0}} \sum_{s=0}^{L} \big(\text{Inv}_s^{\text{obs}} - \text{Inv}_s(\theta)\big)^2 + \sum_{s=1}^{L} \big(\text{COGS}_s^{\text{obs}} - \text{COGS}_s(\theta)\big)^2, \tag{129}$$

where $\text{Inv}_s(\theta)$ and $\text{COGS}_s(\theta)$ are obtained by forward-simulating the transition from θ using the observed drivers. With only two parameters the problem is well-conditioned and typically converges to a unique solution.

Remark 4. The WAC model does not attempt to represent FIFO dynamics. It is appropriate as a fallback for firms where the cost differential across inventory layers is negligible (i.e. short holding periods or near-constant purchase prices), making the distinction between FIFO and weighted-average immaterial.

6.9.3 Gamma Model

The Gamma model represents inventory age structure as a continuous probability density $f_t(a)$ over age $a \in \mathbb{R}_+$ (years since purchase), parameterised within the Gamma family. This keeps the state two-dimensional at the cost of a moment-matching projection at each step.

State and Parameterisation

The state is $(Q_t,\ k_t,\ \theta_t)$, where $Q_t \geq 0$ is the total physical inventory on hand (units) and the age density is

$$f_t(a) \approx g(a;\, k_t, \theta_t) = \frac{a^{k_t-1}\, e^{-a/\theta_t}}{\Gamma(k_t)\, \theta_t^{k_t}}, \quad a \geq 0, \quad k_t, \theta_t > 0. \tag{130}$$

The first two moments are $\mathbb{E}[A] = k_t\theta_t$ and $\mathrm{Var}(A) = k_t\theta_t^2$.

Annual Transition

Within each fiscal year t the distribution undergoes four operations in sequence.

1. Aging.

Every on-hand unit ages by one year:

$$f_t^{\mathrm{old}}(a) = \begin{cases} f_{t-1}(a-1), & a \geq 1, \\ 0, & a < 1. \end{cases} \tag{131}$$

2. Purchase arrival.

Purchases are modelled as arriving uniformly throughout the year, so their age at year-end is $\mathrm{Unif}(0,1)$:

$$f_t^{\mathrm{new}}(a) = \begin{cases} 1, & 0 \leq a \leq 1, \\ 0, & \text{otherwise.} \end{cases} \tag{132}$$

3. Pre-sale mixing.

The pre-sale density is the quantity-weighted mixture

$$f_t^{\mathrm{mix}}(a) = \omega_t\, f_t^{\mathrm{old}}(a) + (1-\omega_t)\, f_t^{\mathrm{new}}(a), \qquad \omega_t = \frac{Q_{t-1}}{B_t}, \tag{133}$$

where $B_t = Q_{t-1} + P_t$ is the pre-sale physical quantity.

4. FIFO tail removal and projection.

FIFO sales remove the oldest π_t-fraction of f_t^{mix}:

$$f_t^{\mathrm{post}}(a) = \frac{f_t^{\mathrm{mix}}(a)\, \mathbf{1}_{[0,\, a_t^*]}(a)}{1-\pi_t}, \tag{134}$$

where the FIFO cutoff a_t^* satisfies $\int_{a_t^*}^{\infty} f_t^{\mathrm{mix}}(a)\, da = \pi_t$ (solved numerically). Because f_t^{post} is generally not Gamma, we project back via moment matching:

$$\mu_t = \mathbb{E}_{f_t^{\mathrm{post}}}[A], \quad \nu_t = \mathrm{Var}_{f_t^{\mathrm{post}}}(A), \qquad k_t = \frac{\mu_t^2}{\nu_t}, \quad \theta_t = \frac{\nu_t}{\mu_t}. \tag{135}$$

Cost Mapping via Price Index

The simulator tracks physical quantities; cost values are recovered by integrating against a price index $p(\tau)$ that maps calendar time τ to the historical unit purchase price.

Ending inventory value.

$$\mathrm{Inv}_t = Q_t \int_0^\infty p(t-a)\, f_t^{\mathrm{post}}(a)\, da. \tag{136}$$

COGS.

$$\mathrm{COGS}_t = S_t \cdot \bar{p}_t^{\mathrm{sold}}, \qquad \bar{p}_t^{\mathrm{sold}} = \frac{\int_{a_t^*}^\infty p(t-a)\, f_t^{\mathrm{mix}}(a)\, da}{\int_{a_t^*}^\infty f_t^{\mathrm{mix}}(a)\, da}, \tag{137}$$

where $\bar{p}_t^{\mathrm{sold}}$ is the quantity-weighted average purchase price of the sold tail.

Estimation

Transition operator.

Define the one-step operator

$$\mathcal{T}_s : (k_{s-1}, \theta_{s-1}) \mapsto (k_s, \theta_s), \tag{138}$$

which, given the observed flows (Q_{s-1}, P_s, S_s), applies aging $\to$ purchase mixing $\to$ FIFO tail removal $\to$ moment-matching projection. The operator is deterministic; the only free parameters are the initial condition $(k_0, \theta_0) \in \mathbb{R}^2_{>0}$.

Loss and optimisation.

Given observed $\{\mathrm{COGS}_s^{\mathrm{obs}}\}_{s=1}^L$, forward-simulate from (k_0, θ_0) and solve

$$(k_0^*, \theta_0^*) = \arg\min_{k_0, \theta_0 > 0} \sum_{s=1}^{L} \big(\mathrm{COGS}_s^{\mathrm{obs}} - \widehat{\mathrm{COGS}}_s(k_0, \theta_0)\big)^2, \tag{139}$$

where $\widehat{\mathrm{COGS}}_s$ is computed via (137) using the forward-simulated density at step s. The terminal estimate is $(k_L^*, \theta_L^*) = (\mathcal{T}_L \circ \cdots \circ \mathcal{T}_1)(k_0^*, \theta_0^*)$.

Limitations

- **Projection error.** Moment matching onto the Gamma family at every step accumulates approximation error, causing the simulated distribution to drift from the true post-sale shape over long horizons.
- **Unbounded support.** After projection the Gamma density has unbounded support on $\mathbb{R}_+$, even though FIFO truncation imposes a hard upper age a_t^*. This reintroduces

spurious mass above the cutoff.

- **Flat-price degeneracy.** When $p(\tau)$ is nearly constant, all age distributions produce similar COGS values and (k_0, θ_0) is poorly identified—the same flat-likelihood pathology discussed in Section 6.5.2.
- **Price index requirement.** The model requires a product-level price index $p(\tau)$; in practice only category-level proxies are available.

6.9.4 Multinomial Model

The multinomial model discretises the age space into N buckets and tracks, for each bucket, the probability mass (fraction of total physical inventory) and the average unit cost. This eliminates the Gamma model's moment-matching approximation but increases the state dimension.

Bucket Definition

- Bucket $i = 1$: inventory aged $[0, 1)$ years (most recent purchases).
- Bucket $i = 2, \ldots, N-1$: inventory aged $[i-1,\, i)$ years.
- Bucket $i = N$: inventory aged $N-1$ years *or older* (absorbing bucket).

For each bucket i at the end of year t we maintain

$$p_{t,i} \geq 0, \quad c_{t,i} \geq 0, \qquad \sum_{i=1}^{N} p_{t,i} = 1, \tag{140}$$

where $p_{t,i}$ is the probability mass and $c_{t,i}$ is the average unit cost (\$/unit). The inventory cost valuation is

$$\mathrm{Inv}_t = Q_t \sum_{i=1}^{N} p_{t,i}\, c_{t,i} = Q_t\, \bar{c}_t, \tag{141}$$

where $\bar{c}_t = \sum_i p_{t,i}\, c_{t,i}$ is the weighted-average unit cost.

System Dynamics

Aging and purchases.

At the start of year t the on-hand state is $(p_{t-1,i},\, c_{t-1,i})_{i=1}^{N}$ with total quantity Q_{t-1}. Aging shifts each bucket one position older; new purchases of P_t units at price π_t^P enter bucket 1. With $B_t = Q_{t-1} + P_t$:

$$p_{t,i}^{\mathrm{pre}} = \begin{cases} P_t / B_t, & i = 1, \\ p_{t-1,i-1}\, Q_{t-1} / B_t, & i = 2, \ldots, N-1, \\ (p_{t-1,N-1} + p_{t-1,N})\, Q_{t-1} / B_t, & i = N. \end{cases} \tag{142}$$

$$
c^{\mathrm{pre}}_{t,i} = \begin{cases} \pi^P_t, & i = 1, \\ c_{t-1,i-1}, & i = 2, \dots, N-1, \\ \dfrac{p_{t-1,N-1}\, c_{t-1,N-1} + p_{t-1,N}\, c_{t-1,N}}{p_{t-1,N-1} + p_{t-1,N}}, & i = N. \end{cases} \tag{143}
$$

FIFO sales.

Sales of S_t units are allocated from oldest to newest. Let $q_{t,i}$ denote the quantity sold from bucket i; initialise $R^{(N)} = S_t$ and iterate for $i = N, N-1, \dots, 1$:

$$
q_{t,i} = \min\bigl(R^{(i)},\ p^{\mathrm{pre}}_{t,i}\, B_t\bigr), \qquad R^{(i-1)} = R^{(i)} - q_{t,i}. \tag{144}
$$

COGS is the cost-weighted sum $\mathrm{COGS}_t = \sum_{i=1}^{N} q_{t,i}\, c^{\mathrm{pre}}_{t,i}$.

Post-sale update.

$$
Q_t = B_t - S_t, \qquad p^{\mathrm{post}}_{t,i} = \frac{p^{\mathrm{pre}}_{t,i}\, B_t - q_{t,i}}{Q_t}, \qquad c_{t,i} = c^{\mathrm{pre}}_{t,i}. \tag{145}
$$

Estimation

Parameters.

The initial state at $s = 0$ has $2N + 1$ unknowns:

$$
\theta = \bigl(p_{0,1}, \dots, p_{0,N},\ c_{0,1}, \dots, c_{0,N},\ Q_0\bigr). \tag{146}
$$

Simulator operator.

$$
\mathcal{M}_s : (p_{s-1,i},\, c_{s-1,i},\, Q_{s-1}) \ \mapsto\ (p^{\mathrm{post}}_{s,i},\, c_{s,i},\, Q_s), \tag{147}
$$

deterministic given the drivers $(P_s,\, \pi^P_s,\, S_s,\, \pi^S_s)$.

Loss function.

$$
\mathcal{L}(\theta) = w_{\mathrm{inv}} \sum_{s=0}^{L} \bigl(\mathrm{Inv}^{\mathrm{obs}}_s - \mathrm{Inv}_s(\theta)\bigr)^2 + w_{\mathrm{cogs}} \sum_{s=1}^{L} \bigl(\mathrm{COGS}^{\mathrm{obs}}_s - \mathrm{COGS}_s(\theta)\bigr)^2, \tag{148}
$$

subject to simplex and non-negativity constraints on θ. The weights $w_{\mathrm{inv}}, w_{\mathrm{cogs}} > 0$ balance the two terms.

Comparison with Gamma.

- **Gamma:** 2 parameters, computationally light, but accumulates projection error.
- **Multinomial:** $2N+1$ parameters, exact discrete age tracking, no projection error, but higher-dimensional optimisation that may require regularisation for large N.

Limitations

- **State dimension.** With $2N+1$ parameters the optimisation landscape is high-dimensional; for $N \geq 5$ multiple local minima are common.
- **Absorbing bucket.** Bucket N accumulates all inventory aged $\geq N-1$ years, blending distinct cost layers. The cost $c_{t,N}$ is a weighted average that becomes increasingly approximate for very old stock.
- **Identifiability.** As shown in Section 6.5.2, the bucket probabilities $p_{0,i}$ are poorly recovered from aggregate Inv and COGS observations because different probability vectors yield nearly identical financial outputs when inter-bucket cost differentials are small.

6.9.5 SSM (q, a, c) with VI

This approach wraps the multinomial transition in a full state-space model with stochastic dynamics and performs posterior inference over the latent trajectory via variational methods.

State and Noise Structure

The state at time t is

$$x_t = (q_t,\ a_t,\ c_t), \qquad q_t \in \mathbb{R}_{>0},\ a_t \in \Delta^{N-1},\ c_t \in \mathbb{R}^N_{>0}, \tag{149}$$

where q_t is total physical quantity, a_t is the age distribution on the simplex, and c_t are bucket unit costs.

Transition function.

The transition $x_t, r_t = f(x_{t-1}, u_t, \varepsilon_t)$ proceeds in four steps:

1. **Aging and purchases.** Compute the deterministic pre-sale age distribution a_t^{det} and pre-sale costs c_t^{pre} via (142)–(143).
2. **Noise injection.** The pre-sale distribution is perturbed in logit space and mapped back to the simplex:
$$\tilde{a}_t = \operatorname{softmax}\big(\log a_t^{\text{det}} + \varepsilon_t\big), \qquad \varepsilon_t \sim \mathcal{N}(0,\ \Sigma_\varepsilon), \tag{150}$$
guaranteeing $\tilde{a}_t \in \Delta^{N-1}$ for any realisation of ε_t.
3. **FIFO sales.** Given $\tilde{a}_t$, c_t^{pre}, and B_t, allocate S_t units from oldest to newest, producing the FIFO allocation vector $r_t = (r_{t,1}, \ldots, r_{t,N})$.

4. **Post-sale update.**

$$q_t = B_t - S_t, \qquad a_{t,i}^{\text{post}} = \frac{\tilde{a}_{t,i}\, B_t - r_{t,i}}{q_t}, \qquad c_{t,i} = c_t^{\text{pre}}. \tag{151}$$

Observation function.

$$y_t = h(x_t,\, r_t) + \eta_t, \qquad h(x_t, r_t) = \begin{pmatrix} q_t \sum_i a_{t,i}^{\text{post}}\, c_{t,i} \\ \sum_i r_{t,i}\, c_{t,i}^{\text{pre}} \end{pmatrix}, \qquad \eta_t \sim \mathcal{N}(0,\, \Sigma_\eta). \tag{152}$$

Note that COGS_t requires the intermediate FIFO allocations r_t; it cannot be computed from x_t alone.

Complete SSM.

$$\boxed{\begin{aligned} \text{Transition:} &\quad x_t,\, r_t = f(x_{t-1},\, u_t,\, \varepsilon_t), &\quad \varepsilon_t \sim \mathcal{N}(0, \Sigma_\varepsilon), \\ \text{Observation:} &\quad y_t = h(x_t,\, r_t) + \eta_t, &\quad \eta_t \sim \mathcal{N}(0, \Sigma_\eta). \end{aligned}} \tag{153}$$

Covariance Parameterisation

Both covariance matrices are parameterised via their Cholesky factors $\Sigma = LL^T$ with diagonal elements passed through a softplus transform. For diagonal observation noise this simplifies to $\Sigma_\eta = \text{diag}(\sigma_{\text{inv}}^2,\, \sigma_{\text{cogs}}^2)$.

Decoupling Principle

An important design choice: the deterministic FIFO simulator (transition f and observation mapping h) enforces accounting identities *exactly*. The noise terms ε_t and η_t live *outside* the accounting logic. Process noise ε_t captures unmodelled stochastic variation in the age distribution; observation noise η_t bridges the gap between the simplified model output and reported financials (write-downs, NRV adjustments, rounding, overhead allocations). In particular, no accounting identity is violated by the noise.

Variational Inference

Prior.

$$a_0 \sim \text{Dirichlet}(\mathbf{1}_N), \tag{154}$$
$$\log c_{0,i} \sim \mathcal{N}\big(\log \pi_0^P,\ \sigma_c^2\big), \tag{155}$$
$$\log q_0 \sim \mathcal{N}\big(\log(\text{Inv}_0/\pi_0^P),\ \sigma_q^2\big). \tag{156}$$

Joint distribution.

$$p(y_{1:T}, x_{0:T} \mid u_{1:T}) = p(x_0) \prod_{t=1}^{T} p(x_t \mid x_{t-1}, u_t) \prod_{t=1}^{T} p(y_t \mid x_t, r_t), \tag{157}$$

where r_t is a deterministic byproduct of the transition and is not an additional latent variable.

Mean-field approximation.

The posterior is approximated by

$$q(x_{0:T}) = \prod_{t=0}^{T} q_t(a_t)\, q_t(c_t)\, q_t(q_t), \tag{158}$$

with $q_t(a_t)$ logistic-normal ($2(N-1)$ parameters), $q_t(c_t)$ log-normal per bucket ($2N$ parameters), and $q_t(q_t)$ log-normal (2 parameters), giving $\mathcal{O}(6N)$ variational parameters per timestep.

ELBO.

$$\boxed{\mathrm{ELBO} = \sum_{t=1}^{T} \mathbb{E}_{q_t}\big[\log p(y_t \mid x_t, r_t)\big] + \sum_{t=1}^{T} \mathbb{E}_{q_t\, q_{t-1}}\big[\log p(x_t \mid x_{t-1}, u_t)\big] - \sum_{t=0}^{T} \mathrm{KL}\big(q_t \,\|\, p_t\big).} \tag{159}$$

Expectations are estimated via Monte Carlo sampling from q_t.

Remark 5 (Structured VI)**.** The mean-field factorisation $q(x_{0:T}) = \prod_t q_t(x_t)$ ignores temporal correlations and is known to underestimate posterior uncertainty. A natural extension is structured VI $q(x_{0:T}) = q(x_0) \prod_{t=1}^{T} q(x_t \mid x_{t-1})$, which respects the Markov structure but requires amortised inference networks or autoregressive flows. We regard this as a future direction.

Why This Approach Was Superseded

The (q, a, c)-SSM was the most complete formulation prior to the (b, v)-layer model of Section 6.6. However, it inherits the fundamental state-coherence problem identified in Section 6.5.4: the three components (q_t, a_t, c_t) are estimated independently by the mean-field posterior, but must satisfy the nonlinear coupling $v_{t,i} = q_t\, a_{t,i}\, c_{t,i}$ for Inv_t and COGS_t to be consistent. The logit-space noise injection (150) adds further entanglement. In practice, we found that VI consistently recovers age distributions that minimise the ELBO without corresponding to the true latent state, precisely because of the flat likelihood landscape described in Section 6.5.2.

The (b, v)-layer model resolves these issues by parameterising the state so that financial-statement consistency is maintained by construction (Section 6.5.5), and by eliminating the need for stochastic inference entirely: washout identification (Section 6.7) and deterministic forward simulation (Section 6.8) replace the computationally expensive VI loop.

7 Property, Plant, Equipment Modelling

PPE represents tangible long-lived assets used in operations, such as buildings, machinery, and equipment. It treats them as long-term, tangible assets initially recorded at cost (purchase price + associated costs), systematically depreciated over their useful life (except land), tested for impairment, and derecognized (sold/disposed) for gains or losses, with primary focus on initial measurement, depreciation, revaluation, and derecognition to reflect their economic value and usage over time.

Unlike other line items in the income statement or balance sheet items like cash, which only represent a snapshot of the firm's financial position at a point in time, PPE represents a long-term investment that provides economic benefits over multiple periods. Part of the dynamics of PPE comes from the firm's historical investment decisions (Jennergren, 2010), which determine the age and condition of the assets, as well as their remaining useful life and residual value. Therefore, it is not sufficient to model PPE drivers or dynamics using simple ratios. To address this, we propose a **cohort-based model** that tracks the age and depreciation of assets over time, allowing for more accurate representation of PPE dynamics.

7.1 Scope

Recognition

- **Scope:** Additions, replacement, and major inspections
- **Deferred:** Spares, held for sale

Initial Costs

- **Scope:** Purchases, directly attributable costs (delivery, installation, testing)
- **Deferred:** Opening costs, training, advertising

Measurement of Cost

- **Scope:** Cash price equivalent (standard purchase: purchase price + directly attributable costs)
- **Deferred:** Non-monetary exchanges, deferred-payment interest

Depreciation

- **Scope:** Straight-line Depreciation
- **Deferred:** Other methods (diminishing, units of production), significant parts, changes in estimates, idle assets

Subsequent Cost

- **Scope:** Day-to-day servicing, part replacement, major inspections.
- **Deferred:** Training

Impairment

- **Scope:** Impairment losses
- **Deferred:** Reversals, compensation from third parties

Derecognition

- **Scope:** Derecognition of assets
- **Deferred:** Held for sales, sale and leaseback

Other

- **Excluded:** Assets under construction (AUC), Proceeds before intended use, grants, asset retirement obligation (ARO), borrowing costs, leases, FX, hyper inflation, investment property, self-constructed assets, tax effects.

7.2 Accounting Framework

This section summarizes the IFRS accounting treatment for each PPE component within scope. Modeling choices and empirical evidence are deferred to Section 7.3.

7.2.1 General Structure

Most companies present PPE information in a standardized format[4], typically divided into two main sections (see Table 10). Therefore, we structure our model around these two sections: (1) Gross PPE, and (2) Accumulated Depreciation and Impairment Losses. The net carrying value (NetPPE) is then derived as the difference between the closing balances of these two sections.

$$\text{NetPPE} = \text{GrossPPE} - \text{Accumulated Depreciation and Impairment Losses} \tag{160}$$

7.2.2 Capital Expenditure Classification

Lets begin with the addition of PPE. In capital investment literature, total additions to PPE are often split into **maintenance CapEx** versus **growth CapEx**. Maintenance CapEx is the portion of investment required to sustain existing operations, essentially replacing or refurbishing

[4] Firms do follow this structure: AstraZeneca (pharmaceutical), TESCO (retail), Shell (energy), Vodafone (telecommunications), Wise (fintech). Firms do not following this structure: British Land (real estate), Landsec (real estate), Made Tech (technology)

	Value
Gross PPE	
Opening balance	
Additions	
Disposals	
Closing balance	
Accumulated depreciation and impairment losses	
Opening balance	
Depreciation charge	
Impairment losses	
Disposals	
Closing balance	
Net carrying value	$xxx.xx

Table 10: Simplified PP&E Disclosure

assets that wear out without increasing the firm's capacity. In contrast, growth CapEx represents discretionary investment to expand capacity or enter new markets, intended to increase output or revenue beyond current levels.

$$\text{Addition (CAPEX)} = \text{Addition for Growth} + \text{Addition for Maintenance} \tag{161}$$

According to IFRS, the maintenance costs can be further classified into day-to-day servicing, part replacement, and major inspections as shown in Table 11.

Table 11: **Subsequent Cost Categories**

Type	**Example**	**Action**
Day-to-Day servicing	Routine repairs, maintenance and labour cost for service	Expense in P&L
Part Replacement	Replacing aircraft seats, and interior walls of a building	Capitalize and derecognize old part
Major Inspection	Mandatory major inspections	Capitalize and derecognize old inspection cost

Only part replacement and major inspections are capitalized, while day-to-day servicing is expensed in the profit and loss statement. We then define our CAPEX and OPEX for PPE as follows:

$$\text{CAPEX} = \text{Addition for Growth} + \text{Part Replacement} + \text{Major Inspections} \tag{162}$$

$$\text{OPEX} = \text{Day-to-Day Servicing} \tag{163}$$

Here, we assume that all the **CAPEX items are controlled variables**, where firms decide how much to invest in each period. On the other hand, **OPEX is treated as an uncontrolled variable**, determined by the firm's maintenance policy and the condition of the assets.

7.2.3 Depreciation

Depreciation is the systematic allocation of an asset's depreciable amount over its useful life. Under IFRS, multiple depreciation methods are permitted provided they reflect the pattern of consumption of economic benefits (straight-line, diminishing, and units of production method). Here we consider **straight-line depreciation**. Depreciation is an accounting policy, not an actual cash outflow, it should be modeled in a way that is consistent with accounting standards and that reflects the economic reality of asset usage.

7.2.4 Impairment

IAS 36 applies to PPE carried under the cost model. Entities assess at each reporting date whether there is any indicator of impairment. Indicators include external (market declines, adverse regulation, technological change) and internal (obsolescence, physical damage, under-performance).

Once the indicators are identified, an impairment test is performed. The test compares the carrying amount of the asset to its recoverable amount, where the recoverable amount is the higher of fair value less costs of disposal (FVLCD), representing the price that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date less costs of disposal, and value in use (VIU), representing the present value of future cash flows expected from the asset's continued use and eventual disposal[5].

$$\text{Impairment Loss} = \max\{0, \text{Carrying amount} - \text{Recoverable Amount}\}. \tag{164}$$

If an impairment loss is recognized, the carrying amount of the asset is reduced to its recoverable amount. The impairment loss is recognized in profit or loss immediately. In the subsequent periods, if there is an indication that the impairment loss may no longer exist or may have decreased, a reversal of the impairment loss is recognized. The reversal is limited to the carrying amount that would have been determined (net of depreciation) had no impairment loss been recognized in prior periods.

It is important to note that the IAS 36 defines two types of assets: (1) cash-generating units (CGUs), the smallest identifiable group of assets that generates cash inflows that are largely independent of the cash inflows from other assets or groups of assets, and (2) corporate assets, that do not generate independent cash inflows, such as headquarters buildings or shared services. They requires different impairment testing approaches.

Here we focus on the general idea of impairment testing, and leave the CGU and corporate assets classification for future research.

[5] If either of these amounts exceeds the asset's carrying amount, the asset is not impaired and it is not necessary to estimate the other amount.

7.2.5 Derecognition

When removing PPE from the balance sheet, firms must derecognize both the gross carrying amount and the related accumulated depreciation. The gain or loss on derecognition is calculated as the difference between the net carrying amount and the proceeds from disposal, and this value is recognized in profit or loss.

$$\text{Gain/Loss on Derecognition} = \text{Proceeds from Disposal} - \text{Net Carrying Amount} \tag{165}$$

Derecognition can occur due to disposal, or when no future economic benefits are expected from its use or disposal. This includes sales, scrapping, or abandonment of assets, etc.

*Note:*Derecognition related to held-for-sale assets, sale and leaseback transactions, financial leases are excluded from the current model scope.

7.3 Literature Review

This section collects the empirical and theoretical literature that motivates the modeling choices in Section 7.4.

7.3.1 Capital Expenditure

Macro Factor

We first examine the macro factors that influence the CAPEX. When the economy is expanding and demand is growing, firms tend to ramp up growth investments to increase capacity. This is the classic accelerator effect (Baddeley, 2003), where higher GDP or output growth leads to a proportionally larger rise in capital investment. In downturns, many companies postpone or cancel planned expansion projects, waiting for clearer signals of recovery. For example, during the 2008-09 financial crisis over half of surveyed CFOs reported canceling or deferring planned investments, especially at financially constrained firms (Campello et al., 2010). Expansion/improvement CAPEX rises with higher expected income growth and falls with higher volatility (Ambrose and Steiner, 2022).

On the other hand, maintenance CapEx is less sensitive to economic cycles, as firms must continue to replace worn-out assets to maintain operations. However, in recession, firms may delayed or reduce the maintenance expenditure (Kim and Ebdon, 2021). It points out the maintenance CapEx is comes form both capital and oeprating budget (major maintenance vs. routine maintenance). In details, it points out recession and funding shortages push maintenance spending down, increasing deferrals. Then the "deferred maintenance" accumulated that shows up as large renewal/replacement. The paper also points out the political enviroment, human resources, information systems, cost management, and funding mechanism will also affect the maintenance spending in different extend. These factor would be more abstract, and we leave it for future research.

Capital Utilization

For a capacity intensive industry, like manufacturing, mining, transportation, etc., higher capacity utilisation leads to a higher incentive to invest in the capital stock. For other industries, like service industry, higher capacity mostly signals labour tightness. The relationship between capacity utilisation and capital investment may be less direct (Lane et al., 2015).

Adjustment Cost

The cost of moving to a higher capital stock, including disruptions to business-as-usual operation (installation time, training, etc.). If the adjustment cost is high compared to the increase in capital stock, firms may delay or reduce their investment (Lane et al., 2015).

Others

While there are other factors that may affect CAPEX, such as financial constraints (Fazzari et al., 1987), short-term incentives (Almeida et al., 2024), operational policy (McGrattan and Schmitz Jr, 1999; Mullen and Williams, 2004), business Cycle (Goolsbee, 1998), etc. **We will revisit them in future**.

Summary

Growth CAPEX Drivers:

- (+) Expected income growth for the asset's market segment
- (-) Volatility of that growth
- (+) Capacity utilisation, if capacity-intensive industry
- (-) Adjustment cost

Maintenance CAPEX Drivers:

- (+) Asset age and condition
- (-) Economic downturns and funding shortages

7.3.2 Depreciation

The depreciation module is intentionally parsimonious in its literature demands: once the useful life and residual value are set, straight-line depreciation is a deterministic accounting formula. The review below therefore focuses on *why* a cohort-based implementation is needed rather than on the depreciation method itself.

A common shortcut is to approximate depreciation as a ratio of current PPE (e.g., $\text{Dep}_t \approx \delta \cdot NetPPE_t$ or $\delta \cdot GrossPPE_t$) based on a constant value or simple forecast. Alternatively, some work ties depreciation mechanically to maintenance under a steady-state assumption in the valuation literature (Vernimmen et al., 2022; Koller et al., 2010). These approaches are data-light and fast and do not require a detailed CAPEX forecast. However, they fail to capture the dynamics of depreciation driven by CAPEX timing and magnitude. PPE has long life and exists as overlapping "cohorts" (age classes). So current capital needs and depreciation depend on the firm's growth history, not just this year's sales (Jennergren, 2010). Taking ratio short cuts can lead to significant misstatements in simulated financials over time. Moreover, the steady-state assumption may also be invalid in practice, particularly during periods of growth, contraction, or technological change. Therefore, we avoid simplistic shortcuts and instead implement a cohort-based depreciation model that tracks asset vintages and their associated depreciation schedules over time.

7.3.3 Impairment

Financing tool

Goex and Wagenhofer (2009) show that impairment can function as a financing tool. Before borrowing, management has an internal estimate (a private, noisy appraisal) of what the pledged assets are worth. The firm can pre-commit to a reporting rule that maps this estimate to the book value shown to lenders. The optimal rule is one-sided: if the estimate falls below a preset threshold, record an impairment and write down to that estimate; otherwise keep the carrying amount at historical cost and do not recognize unrealized gains. Because lenders know this policy and value collateral at a discount in liquidation (a haircut), a "no impairment" report credibly reassures them: it implies the assets are at least at the threshold and, on average, sufficient to support the loan. If gains were also reported, a "no change" report would be ambiguous and funding would be less likely. The threshold should be stricter when financing conditions are tougher—projects are less profitable, less collateral is available, haircuts are larger, or incentive frictions are stronger. If managers can overstate their internal estimate slightly, lenders anticipate it, so the threshold should be stricter still. In short, impairment acts as a commitment to disclose bad news but not good news, turning "no impairment" into a reliable green light for lenders and giving us a clear one-sided trigger to model.

Accounting standard

Rennekamp et al. (2015) study how the accounting rule for impairment reversals (allowed under IFRS, prohibited under U.S. GAAP) alters managers' real decisions even when economics are held constant. In lab experiments, when the same manager is responsible for recording an impairment, reversibility triggers a behavioral response: managers allocate more future investment to the impaired division if reversals are allowed, and less if reversals are prohibited. The mechanism is cognitive dissonance: responsibility for a negative write-down conflicts with self-image as a capable manager; if the accounting outcome can still be "undone" (reversed), managers try to fix it via extra investment; if it cannot be undone, they rationalize the division as a lost cause and invest less. A second experiment shows that giving managers a chance to deny responsibility attenuates these effects. The takeaway is that the reversibility feature of impairment accounting can systematically push investment up or down in impaired units through psychology, not economics.

Macro perspective

Recent empirical work (Chang et al., 2025)shows that impairment recognition for long-lived assets reflects economy- and industry-wide adverse outcomes both contemporaneously and prospectively, while firm-specific adverse outcomes are incorporated only contemporaneously, indicating a timeliness asymmetry between external and firm-level drivers. Methodologically, the write-down indicator ($\mathrm{WO}_{i,t}$) is estimated via logit and the censored write-down amount ($\mathrm{WOTA}_{i,t}$) via tobit, using a linear index populated by macro ($\Delta\mathrm{GDP}_t$, $\Delta\mathrm{GDP}_{t+1}$), industry ($\Delta\mathrm{INDSALE}_t$, $\Delta\mathrm{INDSALE}_{t+1}$), firm-specific performance ($\mathrm{FIRM}_{i,t}$, $\mathrm{FIRM}_{i,t+1}$), and reporting-incentive controls. The asymmetry is motivated by attribution theory: users more readily attribute firm-specific adverse outcomes to managerial action, increasing incentives to delay recognition relative to external macro/industry shocks. The evidence is produced under U.S. GAAP for long-lived tangible and identifiable amortizing intangible assets, which informs but does not directly dictate IFRS PPE modeling choices.

Other (out of scope)

Acquired intangible assets (Krishnan et al., 2024). Goodwill under US. GAAP (Ramanna and Watts, 2008), IFRS (Glaum et al., 2015).

7.3.4 Derecognition

Economic replacement.

The retirement decision is not a purely mechanical function of age. It responds to economic condition (recessions, factor prices, cost of capital, firm performance) (Goolsbee, 1998). This motivates state-dependent retirement hazards, rather than constant disposal rates. (the replacement can be spiky)

Real options.

Investment and exit decisions are typically forward-looking because actions are partly irreversible and uncertainty creates an "option value of waiting." (Dixit and Pindyck, 1994) Firms may delay disposals in the hope of better future conditions, or accelerate them to avoid anticipated losses. We may consider disposal/replace probability increase when the value of keeping is low relative to selling/replacing.

Financial constraints.

Firms may sell assets to raise cash under financial stress, this is a distinct channel from replacement. This suggest we may consider a separate sales driver tied to liquidity or distress state (Campello et al., 2010).

Resales price.

Asset resales values depends on market liquidity and buyer constraints (Shleifer and Vishny, 1992). Distressed markets can produce fire-sales discounts. This suggests resale values may vary with macro liquidity conditions and firm distress.

Capex and disposal.

Selling an asset is a complex decision influenced by various factors. Ambrose and Steiner (2022) suggest that the option to invest in an asset and the option to sell it are interconnected, and empirically capital expenditure reduces subsequent sale likelihood.

7.4 System Dynamics

This section specifies the PPE cohort model in three layers: state-space definitions (§7.4.1), policy and probability models (§7.4.2), and period-by-period dynamics (§7.4.3). Financial-statement mappings are collected in §7.4.4.

7.4.1 State space and notation

Time is indexed by $t = 1, 2, \ldots$ (years). Reporting and disposal are end-of-year.

Useful life simplification.

To minimize mapping complexity across ledgers [6], we set

$$L_{\text{core}} = L_{\text{insp}} = L, \quad \rho_{\text{core}} = \rho_{\text{insp}} = \rho, \quad d = \frac{1-\rho}{L}.$$

where

- L is the useful life (in years),
- ρ is the residual fraction,
- d is the straight-line depreciation rate.
- Subscript "core" refers to the main PPE class, and "insp" refers to the major inspection component.

Cohort (age-bucket) structure.

To replicate the accounting treatment of PPE (see Table 10), we need to track the gross carrying amount G and accumulated depreciation A separately. The net book value (NetPPE) is then given by

$$\text{NetPPE}_t = G_t - A_t \tag{166}$$

[6] This constraint can be relaxed in future versions.

Moreover, as discussed earlier, the dynamics of PPE depends on the firms's (investment) history. We introduce a age bucket structure to capture the vintage effects of assets. Specifically, for each ledger ℓ, we track assets in age buckets $i = 0, 1, \dots, L-1$, where $i = 0$ represents assets added this year, $i = 1$ represents assets added last year, and so on.

Combining this two considerations, we can express the total gross carrying amount, accumulated depreciation, and net book value for each ledger ℓ at time t as follows:

$$G_t^\ell = \sum_{i=0}^{L-1} g_{i,t}^\ell, \quad \mathrm{A}_t^\ell = \sum_{i=0}^{L-1} \mathrm{a}_{i,t}^\ell, \quad \mathrm{NetPPE}_t^\ell = G_t^\ell - \mathrm{A}_t^\ell.$$

Total PPE:

$$G_t = G_t^{\text{core}} + G_t^{\text{insp}}, \quad \mathrm{A}_t = \mathrm{A}_t^{\text{core}} + \mathrm{A}_t^{\text{insp}}, \quad \mathrm{NetPPE}_t = \mathrm{NetPPE}_t^{\text{core}} + \mathrm{NetPPE}_t^{\text{insp}}.$$

Major Inspection Tracking

To model major inspections, we needs a inspection clock τ_t that tracks the number of years since the last completed major inspection. This clock resets to zero when a major inspection is completed, and increments by one each year otherwise.

$$\tau_t = \begin{cases} 0, & \text{if a major inspection was completed at the end of year } t-1, \\ \tau_{t-1} + 1, & \text{otherwise.} \end{cases} \tag{167}$$

where $\tau_t \in \{0, 1, \dots, L^{\text{insp}} - 1\}$, and L^{insp} is the deterministic interval (in years) between major inspections.

Stress Metric

We consider a stress metric $z_t = f_z(s_t)$ (e.g., liquidity shortfall risk, leverage, credit spread proxy). This metric captures the firm's financial health and influences decisions like distress sales and impairment testing.

For simplicity, we define this as a weighted sum of current ratio (liquidity) and debt-to-equity ratio (leverage) (lower is better):

$$z_t = -w_{\text{liq}} \cdot \text{Current Ratio}_{t-1} + w_{\text{lev}} \cdot \text{Debt-to-Equity Ratio}_{t-1} \tag{168}$$

Workflow summary

Within year t, the model applies the following steps in order:

1. **Beginning-of-year**

(a) **Capitalize core additions.** Add growth and replacement CapEx into age bucket 0.

(b) **Decide major inspection.** If triggered, derecognize all inspection cohorts (proceeds governed by recovery rate q_t^{insp}), then capitalize the new inspection cost into the inspection ledger's age bucket 0, and reset the inspection clock.

2. **Over the year**

(a) **Depreciation.** Apply straight-line depreciation to all cohorts, with the same rate across core and inspection ledgers.

3. **End-of-year**

(a) **Impairment test.** If triggered, apply impairment loss to all cohorts pro-rata based on their net book value.

(b) **Decide disposals.** Process distress sales and economic replacements, derecognize gross and accumulated depreciation by cohort, and record proceeds and gain or loss on derecognition.

(c) **Aging.** Move all remaining cohorts to the next age bucket, and remove fully depreciated assets from the last bucket.

We model only two PPE sub-ledgers $\ell \in \{\text{core}, \text{insp}\}$

- **core**: the main PPE class (single class baseline).
- **insp**: a pseudo-component ledger that represents the capitalized major inspection component. It is wiped when a new major inspection is completed, replicating IAS 16 inspection-component derecognition without full component tracking.

7.4.2 Policy and probability models

Each subsection below specifies a single stochastic or deterministic decision rule. The period dynamics in Section 7.4.3 invoke these rules in the workflow order defined above.

Core CapEx budget

The core capital expenditure (CapEx) budget B_t^{core} is defined as

$$B_t^{\text{core}} = \exp(\theta_B^\top s_t + \varepsilon_t^B), \quad \varepsilon_t^B \sim N(0, \sigma_B^2). \tag{169}$$

The state vector s_t encodes the following signal classes, motivated by the capital investment literature (Section 7.3):

Remark 6 (CapEx signal mapping)**.** Macro signals drive growth CapEx via the accelerator effect (Baddeley, 2003); GDP volatility dampens expansion investment (Ambrose and Steiner,

Signal Type	Proxy Variable
Macro (X_t^{macro})	Δ GDP, Δ Interest Rates, GDP Volatility
Industry (X_t^{ind})	Capacity Utilisation, Δ Sector Revenue
Firm (X_t^{firm})	Δ Revenue, Δ Operating Cash Flow
Asset (X_t^{asset})	Age Ratio, Maintenance Backlog

2022); industry capacity utilisation is relevant for capacity-intensive sectors (Lane et al., 2015); firm-level cash flow captures financial constraints that may defer both growth and maintenance spending (Campello et al., 2010); and asset-level signals (age ratio from the cohort structure, maintenance backlog) primarily influence the growth-vs-replacement split (Kim and Ebdon, 2021).

We then split the core budget into growth CapEx, and economic replacement CapEx using a multinomial regression.

$$U_t^{(g)} = \theta_g^\top s_t, \quad U_t^{(r)} = \theta_r^\top s_t.$$

where $U_t^{(g)}$ is for growth and $U_t^{(r)}$ is for replacement.

The portions of CapEx allocated to growth $\pi_t^{(g)}$ and replacement $\pi_t^{(r)}$ are given by

$$\pi_t^{(r)} = \frac{\exp(U_t^{(r)})}{\exp(U_t^{(g)}) + \exp(U_t^{(r)})}, \quad \pi_t^{(g)} = 1 - \pi_t^{(r)}. \tag{170}$$

where $\pi_t^{(g)} + \pi_t^{(r)} = 1$.

The actual growth CapEx $x_t^{(g)}$ and replacement CapEx $x_t^{(r)}$ are then given by

$$x_t^{\text{grow}} = \pi_t^{(g)} B_t^{\text{core}}, \quad x_t^{\text{rep}} = \pi_t^{(r)} B_t^{\text{core}}. \tag{171}$$

Major inspection hazard.

The major inspection event is modeled as a Bernoulli random variable, with probability increasing in time since last completed major inspection:

$$p_t^{\text{major}} = \sigma(\beta_0 + \beta_1 \tau_t), \quad I_t^{\text{major}} \sim \text{Bernoulli}(p_t^{\text{major}}). \tag{172}$$

where τ_t is the inspection clock, and $\beta_1 \geq 0$ enforces that the event is non-decreasing in time since last completed major inspection (details of τ_t in Section 7.4.1).

Additionally, the major inspection cost is modeled as a spiky cost that scales with PPE size:

$$x_t^{\text{major}} = \alpha \cdot I_t^{\text{major}} G_t, \qquad \alpha > 0. \tag{173}$$

where

- x_t^{major} is the major inspection cost at time t,

- I_t^{major} is the major inspection event indicator (1 if major inspection is due, 0 otherwise),
- G_t is the gross PPE at time t (measured at the start of t before any postings),
- α is a scaling constant factor representing the proportion of gross PPE cost incurred for major inspection.
- $\beta_1 > 0$ ensures that the probability of major inspection increases with time since last inspection.
- $p^{\text{major}}_{L^{\text{insp}}-1} = 1$ ensures that a major inspection is mandatory at the maximum interval.

Once a major inspection is completed ($I_t^{\text{major}} = 1$), the inspection clock τ_t resets to zero. Otherwise, the clock increments by one.

$$\tau_{t+1} = \begin{cases} 0, & \text{if } I_t^{\text{major}} = 1, \\ \tau_t + 1, & \text{otherwise.} \end{cases} \tag{174}$$

Once the major inspection is completed, the inspection ledger is derecognized (wiped out), according to IAS 16. We define a disposal factor $\delta_{i,t}^{\text{insp}}$ for the inspection ledger for age bucket i at time t:

$$\delta_{i,t}^{\text{insp}} = I_t^{\text{major}}, \quad \forall i. \tag{175}$$

This represent disposaling all ppe in the inspection ledger[7]. The usage of the disposal factor will be address in the later section.

Remark 7 (Inspection drivers)**.** The hazard model is intentionally parsimonious: τ_t is the dominant driver because major inspections are largely governed by regulatory or manufacturer-mandated intervals. Future extensions may condition on asset deterioration signals (e.g. unplanned maintenance frequency, operating hours) or allow deferral under financial stress (Kim and Ebdon, 2021).

Distress sales

Taking the stress metric $z_t = f_z(s_t)$ (e.g., liquidity shortfall risk, leverage, credit spread proxy) defined earlier, we model the distress sale decision as a Bernoulli random variable, with probability increasing in stress level:

$$p_t^{\text{dist}} = \sigma(\gamma_0^{\text{dist}} + \gamma_1^{\text{dist}} z_t + \varepsilon_t^{\text{dist}}), \quad \varepsilon_t^{\text{dist}} \sim N(0, \sigma^2_{\text{dist}}). \tag{176}$$

$$I_t^{\text{dist}} \sim \text{Bernoulli}(p_t^{\text{dist}}). \tag{177}$$

where

[7] Partial major inspection can be consider in the future

- I_t^{dist} is the distress sale indicator at time t,
- z_t is the scalar stress metric defined in Section 7.4.1,
- γ_0^{dist} is the intercept (baseline log-odds of distress sale),
- $\gamma_1^{\text{dist}} > 0$ controls sensitivity to the stress metric,
- $\varepsilon_t^{\text{dist}}$ is an unobserved shock.

The stress metric z_t aggregates the following proxy variables:

Signal Type	**Proxy Variable**
Liquidity (X_t^{liq})	Current Ratio, Cash / Total Assets
Leverage (X_t^{lev})	Debt-to-Equity Ratio, Interest Coverage Ratio
Market (X_t^{mkt})	Credit Spreads, Δ Equity Index

Remark 8 (Distress signal mapping)**.** Financially constrained firms are more likely to sell assets during downturns (Campello et al., 2010). Market-level signals (credit spreads) capture funding conditions that may amplify fire-sale dynamics, while asset-market liquidity influences achievable proceeds (Shleifer and Vishny, 1992). The current implementation uses a simplified two-factor form (Current Ratio and Debt-to-Equity); the full set above is available for future calibration.

When distressed, sell a fixed fraction $\psi_{\text{sale}} \in [0, 1]$ of core gross:

$$D_t^{\text{sale}} = I_t^{\text{dist}} \cdot \psi_{\text{sale}} G_t^{\text{core}}. \tag{178}$$

Here we assumes the distress sales only come from the core ledger, and all age buckets are sold proportionally:

$$\delta_{a,t}^{\text{sale}} = \begin{cases} D_t^{\text{sale}}/G_t^{\text{core}}, & G_t^{\text{core}} > 0, \\ 0, & G_t^{\text{core}} = 0, \end{cases} \quad \forall a. \tag{179}$$

where

- $\delta_{a,t}^{\text{sale}} \in [0, 1]$ is the disposal factor for age bucket a at time t due to distress sales.

Economic replacement (disposal + replacement CapEx)

Given the replacement CapEx budget x_t^{rep} (from section 7.4.2), here we model the derecognition (disposal) effect caused by the economic replacement decision. The disposal amount D_t^{rep} is determined by a scalling factor $\psi_{\text{rep}} \in [0, \infty)$:

$$D_t^{\text{rep}} = \min(\psi_{\text{rep}}\, x_t^{\text{rep}}, G_t^{\text{core}}). \tag{180}$$

where

- $\psi_{\text{rep}} = 1$ means full replacement, 1-to-1 derecognition (set this by default).
- $\psi_{\text{rep}} < 1$ means partial replacement.
- $\psi_{\text{rep}} > 1$ means over-replacement (derecognizing more than the replacement CapEx).

Here we also assume the economic replacement only comes from the core ledger, and all age buckets are sold proportionally[8]. Similarly to distress sales, the disposal factor for age bucket a at time t due to economic replacement is given by:

$$\delta^{\text{rep}}_{a,t} = \begin{cases} D^{\text{rep}}_t / G^{\text{core}}_t, & G^{\text{core}}_t > 0, \\ 0, & G^{\text{core}}_t = 0, \end{cases} \quad \forall a. \tag{181}$$

The replacement decision is driven by the following signals, channeled through the CapEx budget allocation (Section 7.4.2):

Signal Type	**Proxy Variable**
Macro (X_t^{macro})	Δ GDP, Δ Factor Prices, Cost of Capital
Firm (X_t^{firm})	Δ Revenue, Δ EBITDA
Asset (X_t^{asset})	Asset Age, Δ Operating Efficiency
Option Value (X_t^{opt})	Revenue Volatility, Δ Replacement Cost

Remark 9 (Replacement signal mapping)**.** The replacement CapEx share $\pi_t^{(r)}$ inherits its drivers from the state vector s_t defined in Section 7.4.2. Additionally, the retirement decision responds to economic conditions rather than age alone (Goolsbee, 1998), and the option value of waiting under uncertainty can delay replacement (Dixit and Pindyck, 1994).

Impairment trigger

The impairment test depends on the nature of the asset (CGU vs. corporate asset) and the availability of market data (FVLCD vs. VIU). Here, we propose a simplified impairment model that considers only fair value as the recoverable amount, with no reversal. The impairment loss is then given by:

$$\text{Impairment Loss}_t = \max\{0, \text{Carrying Amount}_t - \text{Fair Value}_t\}. \tag{182}$$

Referencing the empirical findings of Chang et al. (2025) on managerial bias in impairment recognition, we model the trigger probability as a function of multiple signal classes with heterogeneous sensitivity:

Remark 10 (Proxy variables)**.** The proxy variables above are illustrative and can be replaced with more suitable data sources. The firm-specific signal enters with a one-period lag, reflecting the empirical finding that firm-level adverse outcomes are incorporated into impairment

[8] I want to make it as a mixture of oldest-first and proportional disposal in the later version

Signal Type	Proxy Variable	Weight / Sensitivity
Macro (X_t^{macro})	Δ GDP, Δ Interest Rates	**High.** Managers are sensitive to "blame-free" external indicators.
Industry (X_t^{ind})	Δ Sector Rev. / Indices	**Medium.**
Firm (X_{t-1}^{firm})	Δ Sales, Δ CFO	**Low / Lagged.** Reflects bias to delay internal bad news.
Asset (X_t^{asset})	Δ Asset Price Indices	**Variable.** (e.g. Price Index from FRED).

recognition only contemporaneously, whereas macro and industry signals also have prospective predictive power.

The probability of triggering an impairment test is defined as:

$$p_t^{\text{trigger}} = \sigma\Big(\beta_0^{\text{imp}} + \beta_1^{\text{imp}} X_t^{macro} + \beta_2^{\text{imp}} X_t^{ind} + \beta_3^{\text{imp}} X_{t-1}^{firm} + \beta_4^{\text{imp}} X_t^{asset}\Big). \tag{183}$$

The decision to trigger is then a Bernoulli trial:

$$D_t^{\text{trigger}} \in \{0, 1\}, \quad D_t^{\text{trigger}} \sim \text{Bernoulli}(p_t^{\text{trigger}}). \tag{184}$$

Remark 11 (Deferred signal refinements)**.** The following statistically significant variables identified by Chang et al. (2025) require finer-grained data and are deferred to future iterations:

- Δ Industry Sales (requires an industry panel to compute median changes),
- Firm performance factor via PCA (over changes in sales, pre-write-down earnings, and CFO),
- Smoothing and Big Bath indicators (require cross-sectional earnings-change medians),
- Management change indicator (requires executive turnover data).

Recoverable amount

The fair value is derived from an asset-class price index (e.g. market index, industry index, producer price index). Let $P_{i,t}$ denote the index value when age bucket i was added (at time $t - i$). The unadjusted replacement-cost fair value is given by the relative index change $R_{i,t}$:

$$R_{i,t} = \frac{P_{0,t}}{P_{i,t}}, \qquad FV_t^{\text{raw}} = \sum_{\ell \in \{\text{core,insp}\}} \sum_{i=0}^{L-1} R_{i,t}\, g_{i,t}^{\ell}. \tag{185}$$

Remark 12 (Limitation of the unadjusted approach)**.** The unadjusted replacement-cost estimate produces an "as-new" fair value and therefore omits two adjustments required by IAS 36: the asset's current age and condition, and the costs of disposal. Because these adjustments are omitted, the estimated fair value often sits above Net PPE and suppresses impairment recognition. In simulations the price index typically must fall by roughly 50% before fair value drops below carrying amount, which suggests the unadjusted method understates realistic recoverable-value declines.

To address this, we introduce an obsolescence adjustment that incorporates multiple signals reflecting the asset's physical, economic, and technological condition:

Signal Type	**Proxy Variable**
Physical (X_t^{phy})	Asset Age, Δ Asset Condition, Δ Maintenance Costs
Economic (X_t^{econ})	Δ GDP, Δ Industry Rev, Δ Interest Rates, Market Demand
Technological (X_t^{tech})	Δ Industry Capex, Δ R&D Spending

The obsolescence ratio and adjusted fair value are defined as:

$$r_t^{obso} = \sigma\Big(\beta_0^{fv} + \beta_1^{fv} X_t^{phy} + \beta_2^{fv} X_t^{econ} + \beta_3^{fv} X_t^{tech} + \varepsilon_t^{fv}\Big), \quad \varepsilon_t^{fv} \sim N(0, \sigma_{fv}^2), \tag{186}$$

$$FV_t = (1 - r_t^{obso}) \times FV_t^{\text{raw}}. \tag{187}$$

Within the cohort structure, the weighted average asset age used as part of X_t^{phy} is approximated as:

$$\overline{\text{Age}}_t = \frac{\displaystyle\sum_{\ell} \sum_{i=0}^{L-1} g_{i,t}^{\ell} \cdot i}{\displaystyle\sum_{\ell} \sum_{i=0}^{L-1} g_{i,t}^{\ell}}, \qquad \text{Age Ratio}_t = \frac{\overline{\text{Age}}_t}{L}. \tag{188}$$

Remark 13 (Obsolescence proxy backing)**.** The physical, economic, and technological signal categories above are motivated by the external and internal impairment indicators listed in IAS 36.12 rather than a single empirical study. The cohort structure provides a natural source for the physical signal (weighted-average age). Economic and technological signals reuse macro and industry proxies that also drive the impairment trigger.

Remark 14 (Future extensions)**.** For future iterations, one may consider hedonic pricing models (Rosen, 1974; Sirmans et al., 2005) and fair value accounting for PPE (Jin et al., 2022) to further refine the recoverable-amount estimate.

Impairment loss allocation

The impairment loss is then given by:

$$L_t = \max\{0, \text{NetPPE}_t - FV_t\}, \tag{189}$$

where L_t is the total impairment loss at time t.

We apply the impairment loss proportionally across all age buckets and ledgers, using the weight

$w_{i,t}^{\ell}$:

$$w_{i,t}^{\ell} = \frac{g_{i,t}^{\ell} - a_{i,t}^{\ell}}{\mathrm{NetPPE}_t}, \quad \forall \ell, i, \qquad l_{i,t}^{\ell} = \min(w_{i,t}^{\ell} \cdot L_t,\ g_{i,t}^{\ell} - a_{i,t}^{\ell}) \quad \forall \ell, i. \tag{190}$$

Lastly, we update the accumulated depreciation for each age bucket and ledger:

$$a_{i,t}^{\ell} = a_{i,t}^{\ell} + l_{i,t}^{\ell}, \quad \forall \ell, i. \qquad \text{(gross remains unchanged)} \tag{191}$$

7.4.3 Period dynamics

This section walks through the within-year cohort updates in workflow order (see Section 7.4.1). Each step references the policy model that drives it.

At the beginning of year t, the agent samples the core CapEx budget B_t^{core} and determines the growth and replacement portions. Then the major inspection event is sampled, and the major inspection cost is determined. The total additions to PPE for year t is given by:

$$x_t^{\mathrm{core}} = x_t^{\mathrm{grow}} + x_t^{\mathrm{rep}} \tag{192}$$

$$x_t^{\mathrm{insp}} = x_t^{\mathrm{major}} \tag{193}$$

Inspection derecognition.

If a major inspection is completed ($I_t^{\mathrm{major}} = 1$), the inspection ledger is derecognized (wiped out) according to IAS 16. Define the inspection-ledger disposal factor as

$$\delta_{i,t}^{\mathrm{insp}} = I_t^{\mathrm{major}}, \quad \forall i. \tag{194}$$

Disposal flows (inspection ledger).

Define the disposed (gross) and disposed accumulated depreciation for each inspection cohort:

$$\mathrm{disp}_{i,t}^{\mathrm{insp}} = \delta_{i,t}^{\mathrm{insp}}\, g_{i,t}^{\mathrm{insp}}, \qquad \mathrm{adisp}_{i,t}^{\mathrm{insp}} = \delta_{i,t}^{\mathrm{insp}}\, a_{i,t}^{\mathrm{insp}}, \quad \forall i. \tag{195}$$

Aggregate these for reporting (PPE note disclosure):

$$\mathrm{Disp}_t^{\mathrm{insp}} = \sum_{i=0}^{L-1} \mathrm{disp}_{i,t}^{\mathrm{insp}}, \qquad \mathrm{ADisp}_t^{\mathrm{insp}} = \sum_{i=0}^{L-1} \mathrm{adisp}_{i,t}^{\mathrm{insp}}. \tag{196}$$

Proceeds from disposal (inspection ledger).

Let q_t^{insp} be the constant recovery rate [9](proceeds as a fraction of gross carrying amount). The proceeds from disposal for each age bucket is given by:

$$\text{Proceeds}_{i,t}^{\text{insp}} = q_t^{\text{insp}} \cdot \text{disp}_{i,t}^{\text{insp}}, \quad \forall i. \tag{197}$$

Apply derecognition to inspection cohorts.

$$g_{i,t}^{\text{insp}} = g_{i,t}^{\text{insp}} - \text{disp}_{i,t}^{\text{insp}}, \qquad a_{i,t}^{\text{insp}} = a_{i,t}^{\text{insp}} - \text{adisp}_{i,t}^{\text{insp}}. \tag{198}$$

Beginning-of-year additions.

The new additions are placed in service at the start of year t, and recorded in age bucket $a = 0$ for each ledger:

$$g_{0,t}^{\text{core}} = g_{0,t}^{\text{core}} + x_t^{\text{core}} \quad g_{0,t}^{\text{insp}} = g_{0,t}^{\text{insp}} + x_t^{\text{insp}}. \tag{199}$$

Depreciation.

Over the year, straight-line depreciation is applied to each age bucket. For each ledger ℓ and age bucket a, the depreciation dynamics is given by:

$$\text{dep}_{a,t}^{\ell} = min(d \cdot g_{a,t}^{\ell}, max(g_{a,t}^{\ell} - a_{a,t}^{\ell}, 0)), \quad a_{a,t}^{\ell} = a_{a,t}^{\ell} + \text{dep}_{a,t}^{\ell}. \tag{200}$$

where d is the straight-line depreciation rate, defined in Section 7.4.1.

Impairment.

At the end of year t, the impairment test is performed. If triggered ($D_t^{\text{trigger}} = 1$), the impairment loss is calculated and applied proportionally across all age buckets and ledgers, as described in the policy section (see equation 191 for the derivation of $l_{i,t}^{\ell}$).

$$a_{i,t}^{\ell} = a_{i,t}^{\ell} + l_{i,t}^{\ell}, \quad \forall \ell, i. \tag{201}$$

End-of-year disposals.

At the end of year t, the firm may derecognize PPE due to (i) distress sales and (ii) economic replacement. We combine these into a single disposal factor for the **core** ledger:

$$\delta_{i,t}^{\text{core}} = \min\{1, \delta_{i,t}^{\text{sale}} + \delta_{i,t}^{\text{rep}}\}, \quad \forall i. \tag{202}$$

[9]Ideally, it should be reflecting market conditions.

Disposal flows.

Define per-cohort disposal (gross) and disposal of accumulated depreciation:

$$\text{disp}^{\text{core}}_{i,t} = \delta^{\text{core}}_{i,t}\, g^{\text{core}}_{i,t}, \qquad \text{adisp}^{\text{core}}_{i,t} = \delta^{\text{core}}_{i,t}\, a^{\text{core}}_{i,t}, \quad \forall i. \tag{203}$$

Aggregate these for reporting (PPE note disclosure):

$$\text{Disp}^{\text{core}}_{t} = \sum_{i=0}^{L-1} \text{disp}^{\text{core}}_{i,t}, \qquad \text{ADisp}^{\text{core}}_{t} = \sum_{i=0}^{L-1} \text{adisp}^{\text{core}}_{i,t}. \tag{204}$$

Proceeds from disposal.

Let q_t be the constant recovery rate[10] (proceeds as a fraction of gross carrying amount). The proceeds from disposal for each age bucket is given by:

$$\text{Proceeds}^{\text{core}}_{i,t} = q_t \cdot \text{disp}^{\text{core}}_{i,t}, \quad \forall i. \tag{205}$$

Apply derecognition to core cohorts.

$$g^{\text{core}}_{i,t} = g^{\text{core}}_{i,t} - \text{disp}^{\text{core}}_{i,t}, \qquad a^{\text{core}}_{i,t} = a^{\text{core}}_{i,t} - \text{adisp}^{\text{core}}_{i,t}. \tag{206}$$

Aging.

At the end of year t, all assets age by one year. For each ledger ℓ, the assets in age bucket a move to age bucket $a+1$ for $i = 0, 1, \ldots, L-2$. Assets in age bucket $L-1$ are fully depreciated and removed from the balance sheet.

For $i = 0, 1, \ldots, L-2$ and each ledger ℓ:

$$g^{\ell}_{i+1,t+1} = g^{\ell}_{i,t}, \quad a^{\ell}_{i+1,t+1} = a^{\ell}_{i,t}, \quad i = 0, 1, \ldots, L-2. \tag{207}$$

$$g^{\ell}_{0,t+1} = 0, \quad a^{\ell}_{0,t+1} = 0. \tag{208}$$

where $g^{\ell}_{i,t} \geq 0$, $a^{\ell}_{i,t} \geq 0$, and $g^{\ell}_{i,t} - a^{\ell}_{i,t} \geq 0$ for all ℓ, i, t.

Cohorts that reach NBV $= 0$ before $L-1$ remain in the ledger and continue to age to the next bucket each year and are removed only at the end of life or if derecognized via the end-of-year disposals process

7.4.4 Links to Financial Statements

We map the simulated cohort flows/stocks to standard financial-statement line items. Below, we classify each output by statement.

[10]Ideally, it should be reflecting market conditions, expecially distress sales might have a lower recovery rate.

Balance Sheet

Net PPE (closing)

Closing net carrying amount of PPE (asset on the balance sheet):

$$\mathrm{NetPPE}_t = \sum_{\ell \in \{\mathrm{core,insp}\}} \sum_{i=0}^{L-1} \left(g_{i,t}^{\ell} - a_{i,t}^{\ell} \right). \tag{209}$$

Alternatively,

$$\begin{aligned}
\mathrm{NetPPE}_t = \mathrm{NetPPE}_{t-1} & \\
&+ (x_t^{\mathrm{core}} + x_t^{\mathrm{insp}}) \\
&- \mathrm{PPE\ Depreciation}_t \\
&- \mathrm{Impairment\ Loss}_t \\
&- (\mathrm{Disp}_t^{\mathrm{core}} - \mathrm{ADisp}_t^{\mathrm{core}}) \\
&- (\mathrm{Disp}_t^{\mathrm{insp}} - \mathrm{ADisp}_t^{\mathrm{insp}})
\end{aligned}$$

Cash Flow Statement

Payments to acquire PPE (investing outflow)

Cash outflows to acquire PPE (IAS 7: cash flows from investing activities). In this baseline module, we assume cash settlement occurs upon capitalization, so payments equal capitalized additions:

$$\mathrm{PaymentsForPPE}_t = x_t^{\mathrm{core}} + x_t^{\mathrm{insp}}. \tag{210}$$

Proceeds from disposal of PPE (investing inflow)

Cash inflows from the disposal of PPE (IAS 7: cash flows from investing activities):

$$\mathrm{ProceedsFromPPE}_t = \sum_{\ell \in \{\mathrm{core,insp}\}} \sum_{i=0}^{L-1} \mathrm{Proceeds}_{i,t}^{\ell}. \tag{211}$$

Income Statement

PPE Depreciation Expense

Non-cash depreciation expense recognized in year t:

$$\mathrm{PPE\ Depreciation}_t = \sum_{\ell \in \{\mathrm{core,insp}\}} \sum_{i=0}^{L-1} \mathrm{dep}_{i,t}^{\ell}. \tag{212}$$

PPE Impairment Loss

Impairment losses recognized in year t:

$$\text{Impairment Loss}_t = \max\{0, \text{NetPPE}_t^{\text{pre-impairment}} - FV_t\}. \tag{213}$$

PPE Derecognition Gain/Loss

Gain/loss on disposal equals proceeds minus disposed net book value (NBV):

$$\text{PPE Derecog.}_t = \sum_{\ell \in \{\text{core,insp}\}} \sum_{i=0}^{L-1} \left(\text{Proceeds}_{i,t}^{\ell} - \left(\text{disp}_{i,t}^{\ell} - \text{adisp}_{i,t}^{\ell} \right) \right). \tag{214}$$

8 Technical Documentation

This section documents the software architecture of `fs_sim`, a symbolic-to-compiled financial statement simulator. The design separates *what* the model computes (symbolic equations) from *how* it executes (TensorFlow graph), enabling rapid model iteration without manual pipeline changes.

8.1 Design Principles

Two core goals shape the architecture:

Automatic pipeline generation.

Adding or modifying a variable requires editing only the model specification layer (variables, parameters, equations). The symbolic graph, index registry, and TensorFlow execution pipeline are regenerated automatically. This eliminates a large class of "glue code" errors and makes the simulator scalable to models with many balance-sheet line items.

Symbolic pre-check system.

Before any numerical computation, the system validates the equation system symbolically:

- Variable completeness (every declared variable has exactly one equation and vice versa),
- DAG acyclicity and single connected component (no circular or orphaned dependencies),
- Balance-sheet constraints (BS items are leaf nodes; every BS item's lag term is used),
- Accounting identity (Assets = Liabilities + Equity) verified via recursive symbolic expansion,
- Parameter uniqueness (each trainable parameter appears in exactly one equation).

These checks catch specification errors at model-construction time, not at simulation runtime.

8.2 System Architecture Overview

The system follows a four-stage pipeline: **Declare** → **Validate** → **Index** → **Execute**. Algorithm 1 summarises how the four stages compose during model construction.

8.3 Model Specification Layer

The model specification layer provides a declarative, dataclass-based interface for defining the financial model. All components are SymPy-based and carry no TensorFlow dependency.

8.3.1 Variables (`variables.py`)

Every model variable is a `SimSymbol`, a subclass of `sympy.Symbol` that automatically appends the time suffix `_{t}`. Variables are organized into frozen dataclasses by financial statement role:

Algorithm 1 Model construction pipeline

Require: `ModelVariables`, `ModelEquations`, `ModelParameters`
Ensure: Compiled `FSSimulator` ready for simulation

Stage 1 (Declare): Collect all `Equation` / `ARXEquation` objects from equation dataclasses (including dynamically generated equations from `SymbolicSubmodules`)

Stage 2 (Validate): Build `SymbolicGraph`:
  Register each equation: build adjacency dict `lhs` $\rightarrow$ {rhs symbols}
  Build DAG G = `nx.DiGraph(adj_dict).reverse()`
  Run validation suite (Algorithm 3)
  Collapse ARX equations into `VARX_BLOCK`; remove exogenous nodes
  Compute execution order via lexicographic topological sort

Stage 3 (Index): Build `IndexRegistry`:
  Classify all graph nodes into `macro`, `state`, `lag_state`, `params`
  Assign contiguous indices within each bucket
  For each equation in execution order, emit a `LayerSpec` with read/write indices
  Precompute state $\rightarrow$ lag_state carryover mapping

Stage 4 (Execute): Build TF pipeline via `PipelineGenerator`:
for each `LayerSpec` ℓ **do**
  if ℓ is `DeterministicLayerSpec` **then**
    Create `SymPyDeterministicLayer`: `sp.lambdify`(rhs) $\rightarrow$ TF callable
  else if ℓ is `ARXLayerSpec` **then**
    Create `DiagonalVARXLayer` with trainable $\Phi, B, \log \sigma$
  end if
end for
Assemble layers into `FSSimulator` (a `tf.keras.Model`)

The unified container `ModelVariables` aggregates all groups and provides IDE auto-completion. Dynamically generated variable dataclasses from submodules (PPE cohorts, debt tranches) are also included.

8.3.2 Parameters (`parameters.py`)

Parameters are SymPy `Symbol` objects when trainable (e.g., depreciation rate, minimum cash threshold) or `sp.Float` when fixed. A key design constraint is *parameter uniqueness*: each trainable parameter symbol may appear in at most one equation. This is because each TensorFlow layer creates its own independent weight; sharing a parameter symbol across equations would result in separate weights with inconsistent gradient updates. Initial values are stored in a `PARAM_INITIAL_VALUES` dictionary and used to initialize Keras weights.

8.3.3 Equations (`equations/`)

Two equation types coexist:

- **Deterministic equations** (`Equation`, a subclass of `sympy.Eq`): express one state variable as a closed-form function of other variables and parameters. Used for all financial

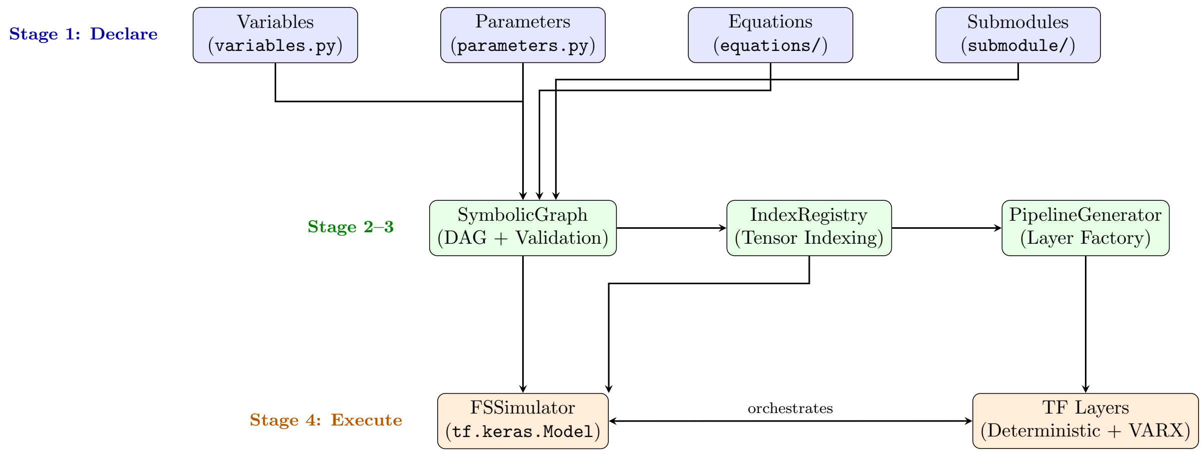


Figure 7: Four-stage architecture of `fs_sim`. The model specification layer (top) feeds into graph construction and validation (middle), which produces a compiled TensorFlow pipeline (bottom).

Dataclass	**Role**
`AssetVariables`, `LiabilityVariables`, `EquityVariables`	Balance-sheet stock variables
`IncomeStatementVariables`	Revenue, COGS, expenses, net income
`CashFlowStatementVariables`	Operating, investing, financing cash flows
`MacroVariables`	Exogenous macro inputs (GDP growth Δgdp, real risk-free rate r^f, nominal risk-free rate i^f, inflation π)
`FirmStateVariables`	Endogenous firm-level drivers (log-prices, log-quantities, etc.)
`PolicyVariables`	Firm decision/action variables
`AuxiliaryVariables`	Intermediate computational variables

Table 12: Variable categories in the model specification.

statement computations (e.g., $\texttt{cash}_t = \texttt{cash}_{t-1} + \texttt{cfo}_t + \texttt{cfi}_t + \texttt{cff}_t$).

- **ARX equations** (`ARXEquation`, subclass of `Equation`): represent autoregressive processes with exogenous inputs. The RHS is a symbolic function placeholder $f(Y_{t-1}, X_{t-1})$, mapped at execution time to a `DiagonalVARXLayer` with trainable Φ, B, and noise parameters.

Equations are organized by category: `FirmStateEquations` (ARX drivers), `PolicyEquations` (firm decisions), `FinancialStatementEquations` (IS, CFS, BS), and `AuxiliaryEquations` (intermediate computations).

8.3.4 Submodule System (`SymbolicSubmodule`)

Some balance-sheet items require a *dynamic* number of internal variables parameterized at runtime (e.g., n PPE age-buckets or M-year debt tranches). The `SymbolicSubmodule` abstract base class provides a contract for such subsystems:

- `var_dataclass()`: dynamically generates a frozen dataclass of `SimSymbols`.
- `eq_dataclass()`: dynamically generates a frozen dataclass of `Equations`.

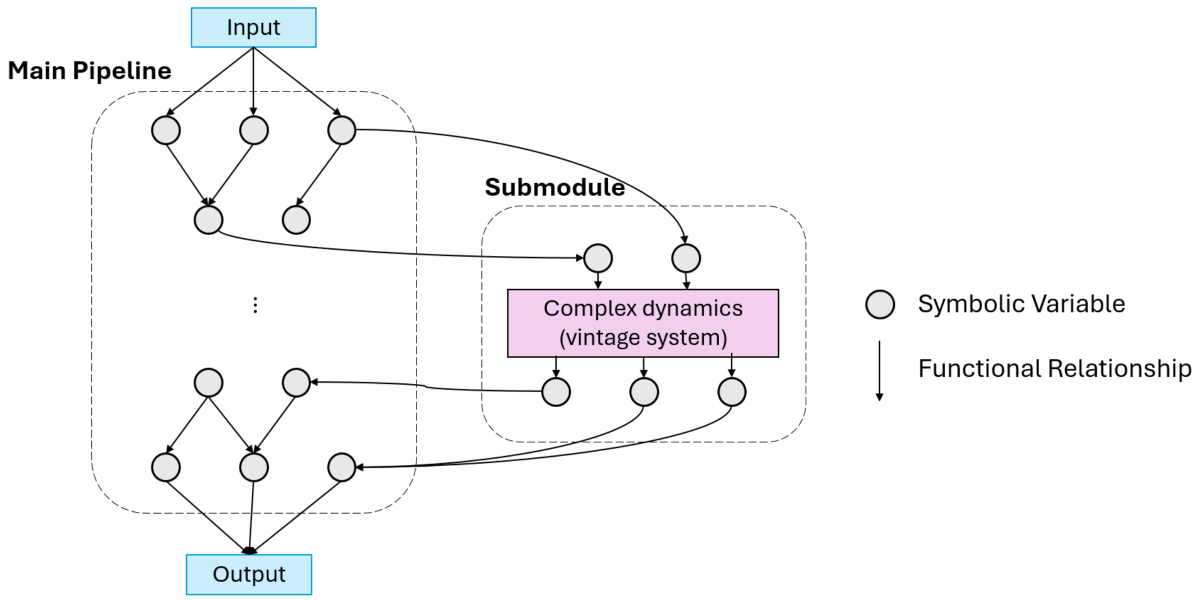
- `bs_identity_subs()`: declares substitutions mapping lag(`bs_item`) to internal bucket-level lags, enabling validators to resolve aggregate lagged balance-sheet terms.

Two concrete implementations exist:

- `PPECohortModule` — cohort-based PPE roll-forward with configurable `n_buckets` (see Section 7 for the accounting model).
- `DebtTermStructure` — maturity-based debt tranches with configurable `max_maturity` and amortized cost accounting.

Integration requires only two steps: add the generated variable dataclass to `ModelVariables` and the generated equation dataclass to `ModelEquations`. The rest of the system discovers submodule outputs automatically.

Figure 8.3.4 illustrates the conceptual relationship. The main pipeline DAG and each submodule DAG are separate graphs connected through a narrow interface: input variables flow from the main graph into the submodule, and aggregate output variables flow back, allowing the main graph to reference submodule results without knowledge of internal structure.



8.4 Symbolic Graph and Validation

8.4.1 Graph Construction

The `SymbolicGraph` class builds a directed acyclic graph (DAG) from the registered equations: each node is a variable symbol, each edge represents a dependency (RHS symbol → LHS symbol). Algorithm 2 shows how equations are registered and the adjacency structure is built.

All ARX driver equations are collapsed into a single `VARX_BLOCK` node in the aggregated DAG, reflecting their joint estimation as a vector process. Exogenous inputs (lagged variables and macro variables) have zero in-degree and are removed before computing the execution order.

Algorithm 2 Equation registration and DAG construction

Require: Equation dataclasses from all categories (firm state, policy, financial statement, auxiliary, submodules)

Ensure: Directed graph G, mapping `equations_by_lhs`

```
adj_dict ← {};    equations_by_lhs ← {}
for each equation section S in ModelEquations do
    for each equation e in S do
        Assert e.lhs ends with _{t} or _{t-1}
        Assert e.lhs ∉ adj_dict                                   ▷ no duplicate LHS
        rhs_vars ← {s.name | s ∈ e.rhs.free_symbols, s ∉ param_symbols}
        adj_dict[e.lhs.name] ← rhs_vars
        equations_by_lhs[e.lhs.name] ← e
    end for
end for
G ← nx.DiGraph(adj_dict).reverse()        ▷ edge direction: dependency → dependent
Run validation suite (Algorithm 3)
return G, equations_by_lhs
```

8.4.2 Validation Suite

Algorithm 3 details the six validation checks that run at construction time. The first three execute inside `_build_symbolic_graph` immediately after DAG construction; the remaining three run afterwards.

The recursive expansion in step (6) substitutes each RHS into the balance-sheet identity, but only for variables in the *expandable* set F—namely balance-sheet, income statement, cash flow statement, auxiliary, and changes-in-equity variables. Firm-state driver variables (whose RHS is an ARX placeholder $f(\cdot)$) and policy variables are treated as terminals and not expanded. This is essential because ARX equations do not have closed-form RHS expressions amenable to symbolic simplification. The result is memoised per symbol to avoid redundant expansion of shared sub-expressions.

8.4.3 Execution Order

A lexicographic topological sort (`nx.lexicographical_topological_sort`) of the pruned DAG produces a deterministic execution order. The `VARX_BLOCK` is placed first (it depends only on exogenous inputs), followed by all deterministic equations in dependency order. Lexicographic ordering ensures reproducibility: given the same equation set, the execution order is always identical regardless of Python dictionary ordering or registration order.

8.5 Index Registry

The `IndexRegistry` assigns contiguous integer indices to every variable, partitioned into four tensor buckets:

For each equation in the execution order, a `LayerSpec` is generated containing:

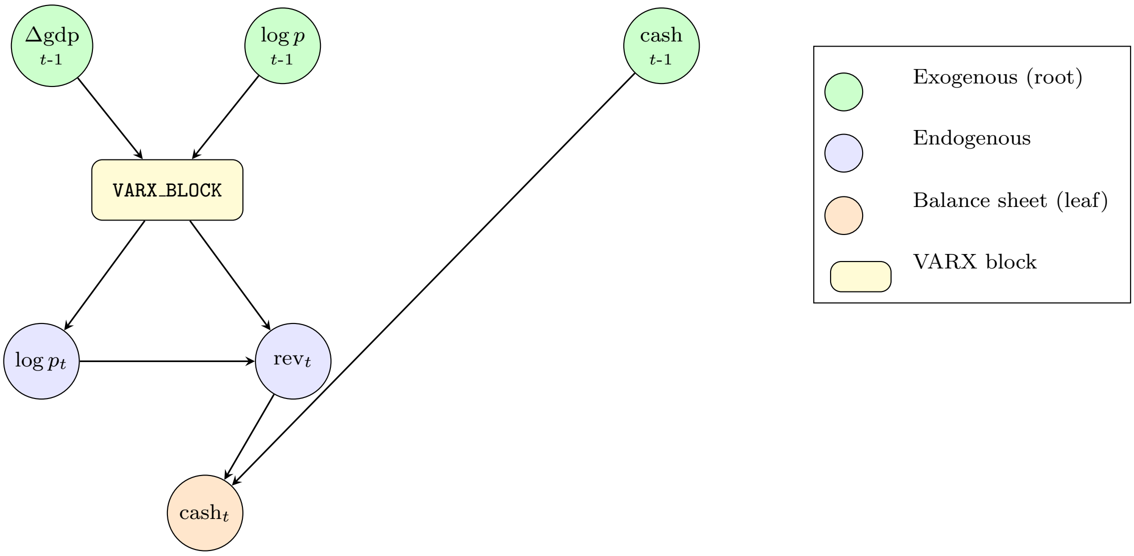


Figure 8: Simplified symbolic DAG. ARX driver equations are collapsed into a single `VARX_BLOCK`. Balance-sheet items (orange) are enforced as leaf nodes with zero out-degree. Exogenous inputs (green) are root nodes with zero in-degree.

Bucket	**Shape**	**Content**	**Description**
`macro`	$[\text{batch}, n_{\text{macro}}]$	$\Delta\text{gdp}_t, r^f_t, i^f_t, \pi_t$	Exogenous macro inputs (both $_t$ and $_{t-1}$ forms)
`lag_state`	$[\text{batch}, n_{\text{lag}}]$	$\text{cash}_{t-1}, \text{rev}_{t-1}, \ldots$	*Sparse*: only lagged variables actually referenced in equations
`state`	$[\text{batch}, n_{\text{state}}]$	$\text{cash}_t, \text{rev}_t, \ldots$	All endogenous variables (output of each step)
`params`	$[n_{\text{params}}]$	Trainable weights	One-to-one with trainable `Symbols`

Table 13: Tensor bucket layout. The sparse lag_state design avoids allocating indices for variables not used as lags.

- `read_from_macro`, `read_from_lag_state`, `read_from_state`, `read_from_params`: lists of `(var_name, index)` tuples specifying which tensor positions to read.
- `write_to_state`: the output position in the state tensor.

To support multi-year simulation, a *state-to-lag carryover mapping* precomputes which state indices should be extracted at the end of each period to form the next period's lag_state tensor. Concretely, for each lag variable `x_{t-1}` used in any equation, the mapping records its corresponding state index (for `x_{t}`) and lag index, enabling a single `tf.gather` to extract the next period's lag_state from the current state.

8.6 TensorFlow Execution Layer

8.6.1 Pipeline Generator

The `PipelineGenerator` is a factory that converts each `LayerSpec` into a concrete `tf.keras.layers.Layer`:

- `DeterministicLayerSpec` → `SymPyDeterministicLayer`
- `ARXLayerSpec` → `DiagonalVARXLayer`

8.6.2 SymPyDeterministicLayer

Each deterministic layer bridges SymPy and TensorFlow. Algorithm 4 shows its call-time execution.

Key implementation details:

- The lambdified function f is created at `__init__` time via `sp.lambdify(..., modules=[CUSTOM_MODULE, "tensorflow"])`, where `CUSTOM_MODULE` maps `exp` $\to$ `tf.exp`, `Max` $\to$ `tf.math.maximum`, etc.
- Symbols in the lambdified function are sorted to match the concatenation order (macro $\to$ lag $\to$ state $\to$ params), ensuring positional correctness.
- Each trainable parameter is a separate `add_weight` scalar with shape `(1,)`, initialized from `PARAM_INITIAL_VALUES` and concatenated via `tf.concat` (not `tf.stack`) for `tf.vectorized_map` compatibility.

8.6.3 DiagonalVARXLayer

The VARX(1) layer jointly models all firm-state driver variables:

$$Y_t = \Phi\, Y_{t-1} + B\, X_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \Sigma) \tag{215}$$

where $\Phi = \operatorname{diag}(\phi_1, \dots, \phi_k)$ encodes autoregressive persistence, $B \in \mathbb{R}^{n_{\text{exog}} \times n_{\text{endog}}}$ captures exogenous effects, and $\Sigma = \operatorname{diag}(\sigma_1^2, \dots, \sigma_k^2)$ is the noise covariance. Note that the exogenous inputs X_{t-1} are *lagged* macro variables (e.g., Δgdp_{t-1}), consistent with the ARX specification in `FirmStateEquations`. Both lagged endogenous and lagged macro variables reside in the `macro` tensor bucket (see Table 13), since the index registry classifies all macro-family variables—regardless of time suffix—into the macro bucket. All three parameter matrices are trainable Keras weights. Noise standard deviations are stored in log-space ($\log \sigma_i$) and exponentiated at call time to ensure positivity.

At call time:

1. Gather Y_{t-1} from `lag_state` and X_{t-1} from `macro` using precomputed indices.
2. Compute `output` $= \Phi \odot Y_{t-1} + X_{t-1} \cdot B + \exp(\log \sigma) \odot \mathcal{N}(0, I)$.
3. Scatter each output component into the corresponding `state` position.

8.6.4 FSSimulator (`tf.keras.Model`)

The top-level simulator inherits from `tf.keras.Model` and orchestrates the full pipeline. Algorithm 5 describes the multi-year simulation loop.

Key aspects:

- `call(inputs)`: executes all layers in sequence for a single period, producing the updated state tensor.

- `run_multi_year`: decorated with `@tf.function` for graph-mode execution and `@auto_build` to ensure `build()` runs in eager mode before tracing. The `@auto_build` decorator is necessary because TensorFlow variables (created in `build()`) can only be created once; calling `build()` inside `@tf.function` would fail on the second call.
- `run_multi_firm_multi_traj`: uses `tf.vectorized_map` to parallelize across firms, each running `run_multi_year` over shared macro scenarios. Output shape: $[\text{num\_firms}, \text{num\_traj}, T, n_{\text{state}}]$. Each firm's initial lag_state is tiled to cover all trajectories, so different firms can start from different initial conditions while sharing macro scenarios.

8.7 Post-Processing

The `to_dict()` method converts the output state tensor $[\text{num\_firms}, \text{num\_traj}, T, n_{\text{state}}]$ into nested dictionaries of named time series per firm per trajectory. Financial ratios (liquidity, profitability, leverage, efficiency) are computed symbolically using `sp.lambdify` on ratio expressions defined in `financial_ratio.py`. Ratios involving lagged terms (e.g., inventory turnover using average inventory) handle edge cases at $t = 0$ by inserting `None` values. A `_check_balances()` method verifies the accounting identity holds numerically across all firms, trajectories, and time steps.

Algorithm 3 Symbolic validation suite

Require: Graph G, equations map E, variables V, submodule substitutions $\mathcal{S}$

— *checks run inside `_build_symbolic_graph`, immediately after DAG construction* —

(1) Variable completeness:

$D \leftarrow \{v.\texttt{name} \mid v \in V,\ v \notin \text{excluded groups}\}$
$L \leftarrow \text{keys}(E)$ ▷ all equation LHS names
Assert $D = L$ ▷ bijection: every variable has exactly one equation

(2) DAG acyclicity and connectivity:

Assert `nx.find_cycle`(G) raises `NetworkXNoCycle`
Assert `nx.number_weakly_connected_components`$(G) = 1$

(3) Balance-sheet leaf-node constraint:

$B \leftarrow$ all asset $\cup$ liability $\cup$ equity symbols
for each $b \in B$ **do**
    Assert out-degree$_G(b.\texttt{name}) = 0$ ▷ BS items must be leaf nodes
end for

— *checks run after `_build_symbolic_graph` returns* —

(4) Balance-sheet lag-term presence:

for each $b \in B$ **do**
    Assert lag(b) appears in some equation RHS, or lag$(b) \in \text{dom}(\mathcal{S})$
end for

(5) Parameter uniqueness:

for each parameter p used in equations **do**
    Assert p appears in exactly one equation's RHS
end for

(6) Accounting identity:

$\Delta \leftarrow \sum_a a.\texttt{lhs} - \sum_l l.\texttt{lhs} - \sum_e e.\texttt{lhs}$ ▷ Assets − Liabilities − Equity
$F \leftarrow \{\text{BS, IS, CFS, Auxiliary, ChangesInEquity variable names}\}$ ▷ expandable set
$\Delta_{\text{exp}} \leftarrow \textsc{FullyExpand}(\Delta, E, F)$ ▷ substitute RHS only for $v \in F$; treat firm-state/policy as terminals
$\Delta_{\text{res}} \leftarrow \Delta_{\text{exp}} - (\text{lagged identity})$ ▷ subtract $\sum \text{lag(Assets)} - \cdots$
$\Delta_{\text{res}} \leftarrow \Delta_{\text{res}}.\text{subs}(\mathcal{S})$ ▷ apply submodule identity substitutions
Assert `simplify`$(\Delta_{\text{res}}) = 0$

Algorithm 4 `SymPyDeterministicLayer.call()` — single-equation execution

Require: Tensors: `macro` $[\mathrm{B}, n_m]$, `lag_state` $[\mathrm{B}, n_l]$, `state` $[\mathrm{B}, n_s]$
Require: Precomputed: gather indices I_m, I_l, I_s, I_p; write index w; lambdified function f
Ensure: Updated `state` tensor with position w overwritten

if equation is constant (RHS is a literal number c) **then**
    `output` $\leftarrow$ `tf.fill`([B], c) ▷ fast path
else
    $\vec{x}_m \leftarrow$ `tf.gather`(`macro`, I_m, axis $= 1$) ▷ macro inputs
    $\vec{x}_l \leftarrow$ `tf.gather`(`lag_state`, I_l, axis $= 1$) ▷ lag inputs
    $\vec{x}_s \leftarrow$ `tf.gather`(`state`, I_s, axis $= 1$) ▷ state inputs
    $\vec{x}_p \leftarrow$ `tf.broadcast_to`(`concat`(param_weights), $[\mathrm{B}, |I_p|]$) ▷ trainable params
    $\vec{x} \leftarrow$ `tf.concat`($[\vec{x}_m, \vec{x}_l, \vec{x}_s, \vec{x}_p]$, axis $= 1$)
    `output` $\leftarrow f$(`unstack`($\vec{x}$)) ▷ evaluate lambdified SymPy expression
end if
`state`$[:, w] \leftarrow$ `output` ▷ via `tf.tensor_scatter_nd_update`
return `state`

Algorithm 5 `run_multi_year` — multi-period simulation

Require: `macro_seq` $[\mathrm{B}, T, n_m]$, `lag_state`$_0$ $[\mathrm{B}, n_l]$
Ensure: `all_states` $[T, \mathrm{B}, n_s]$

`lag_state` $\leftarrow$ `lag_state`$_0$
`outputs` $\leftarrow$ `TensorArray`(size $= T$)
for $t = 0, 1, \ldots, T-1$ **do** ▷ inside `tf.while_loop`
    `macro`$_t \leftarrow$ `macro_seq`$[:, t, :]$ ▷ shape $[\mathrm{B}, n_m]$
    `state` $\leftarrow \vec{0}\ [\mathrm{B}, n_s]$
    for each layer ℓ in `tf_pipeline` **do** ▷ `call()`
        `state` $\leftarrow \ell$(`macro`$_t$, `lag_state`, `state`)
    end for
    `outputs`$[t] \leftarrow$ `state`
    `lag_state` $\leftarrow$ `tf.gather`(`state`, $I_{\text{carryover}}$, axis $= 1$) ▷ extract lag variables
end for
return `outputs.stack()`

9 Future Work

Our current model assumes that multiple firms can be trained jointly. In practice, even firms following the same accounting standards differ substantially in business models and reporting conventions. This section outlines the open issues and a proposed path forward.

9.1 Open Issues

Accounting Distribution Estimation.

Several distributional inputs required by the model are not directly observable from financial statements:

- Next-12-month debt maturities are often omitted from public filings.
- PP&E age distributions must be backed out from historical depreciation data.
- Both are critical for valuation; IBD currently performs these tasks manually.

Data Standardization.

No generic framework exists to convert an arbitrary financial statement into our model's input format. Key challenges include handling different reporting granularities (e.g. segment vs. consolidated), mapping line items to a common taxonomy, and dealing with missing or aggregated data. This is a major bottleneck for scaling the model across firms and industries.

9.2 Segment Reporting: Case Studies

Under IFRS 8 and ASC 280 (US GAAP), entities with publicly traded securities must disclose segment-level financial information when a segment meets quantitative thresholds ($\geq$ 10% of combined revenue, profit/loss, or assets). In practice, firms choose very different reporting granularities.

Company	**Standard**	**Disclosure Type**	**Reported Segments / Lines**
Apple	US GAAP	Product & Services	iPhone, Mac, iPad, Wearables/Home/Accessories, Services
Nvidia	US GAAP	End Market	Data Center, Gaming, Professional Visualization, Automotive, OEM/Other
Airbus	IFRS	Business Segment	Airbus, Airbus Helicopters, Airbus Defence & Space
Nestlé	IFRS	Product Category	Beverages, Water, Dairy/Ice Cream, Nutrition/Health Science, Prepared Dishes, Confectionery, PetCare, Other
Tesco	IFRS	Region	UK & ROI, Central Europe

Table 14: Segment disclosure formats across selected firms.

Three patterns emerge: *product-oriented* (Apple, Nvidia), *business-segment-oriented* (Airbus, Nestlé), and *region-oriented* (Tesco). Firms within the same industry tend to adopt similar formats, but cross-industry heterogeneity is significant. Our model must accommodate all three patterns.

9.3 Proposed Path Forward

We propose a phased approach:

1. **Single-company deep dive.** Start with a single-company time series using the detailed data already available. Extract and convert standardized line items from historical spreadsheets, then run end-to-end valuation.
2. **Cross-industry case studies.** Select firms across GICS sectors, perform in-depth modelling, and identify the minimal set of transformations required to fit each into the model.
3. **Sector-level extension.** Generalize the single-company pipeline to sector-level estimation, accounting for multiple subsidiaries under one parent entity.
4. **LLM-assisted data ingestion.** Use large language models to parse financial statements and produce a hierarchical knowledge graph aligned to IFRS taxonomies. This would serve as a reusable intermediate representation for downstream data transformation.
5. **Agentic workflow.** The long-term goal is an end-to-end agentic system that ingests filings, builds the knowledge graph, populates model inputs, and runs simulations. A current gap is that existing LLM tooling (e.g. OpenAI function calling) lacks a built-in simulation engine, which our framework would need to provide.

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